Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-23 18:37:13
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.2
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
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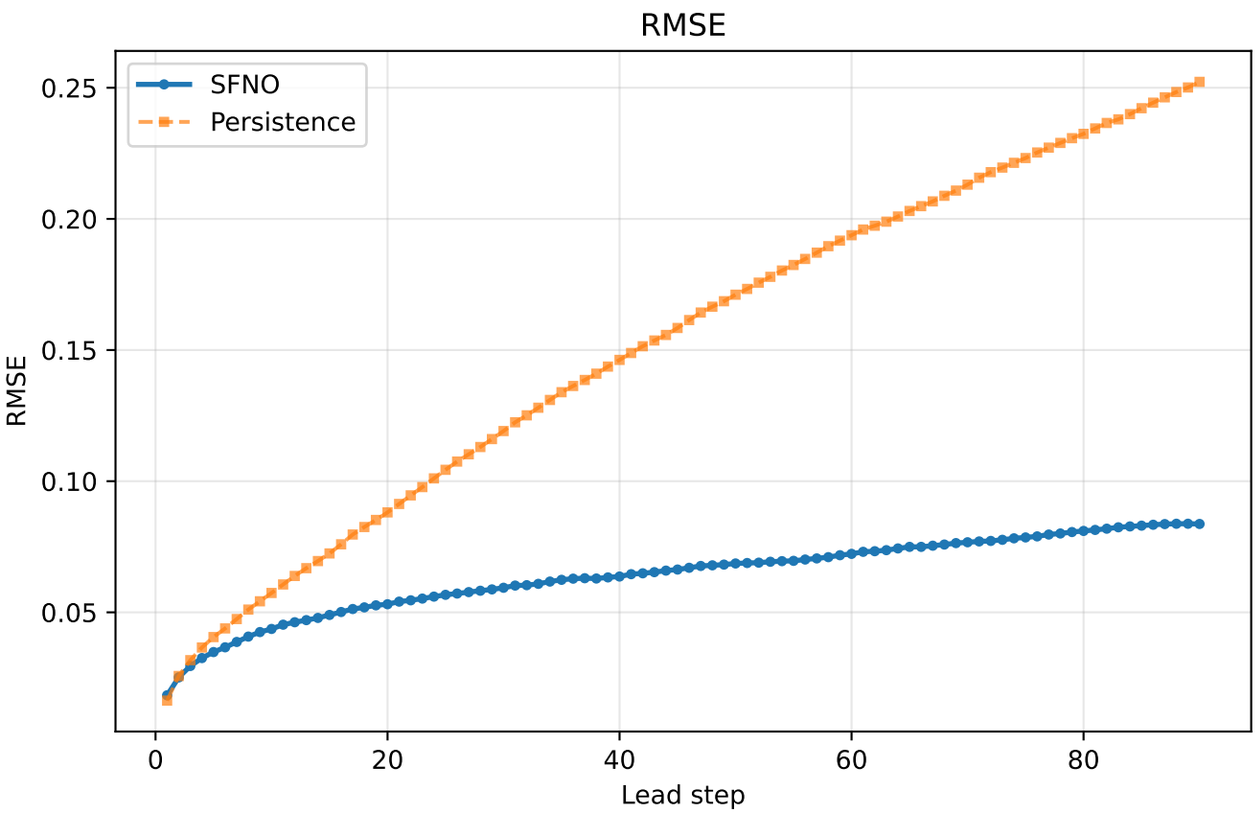
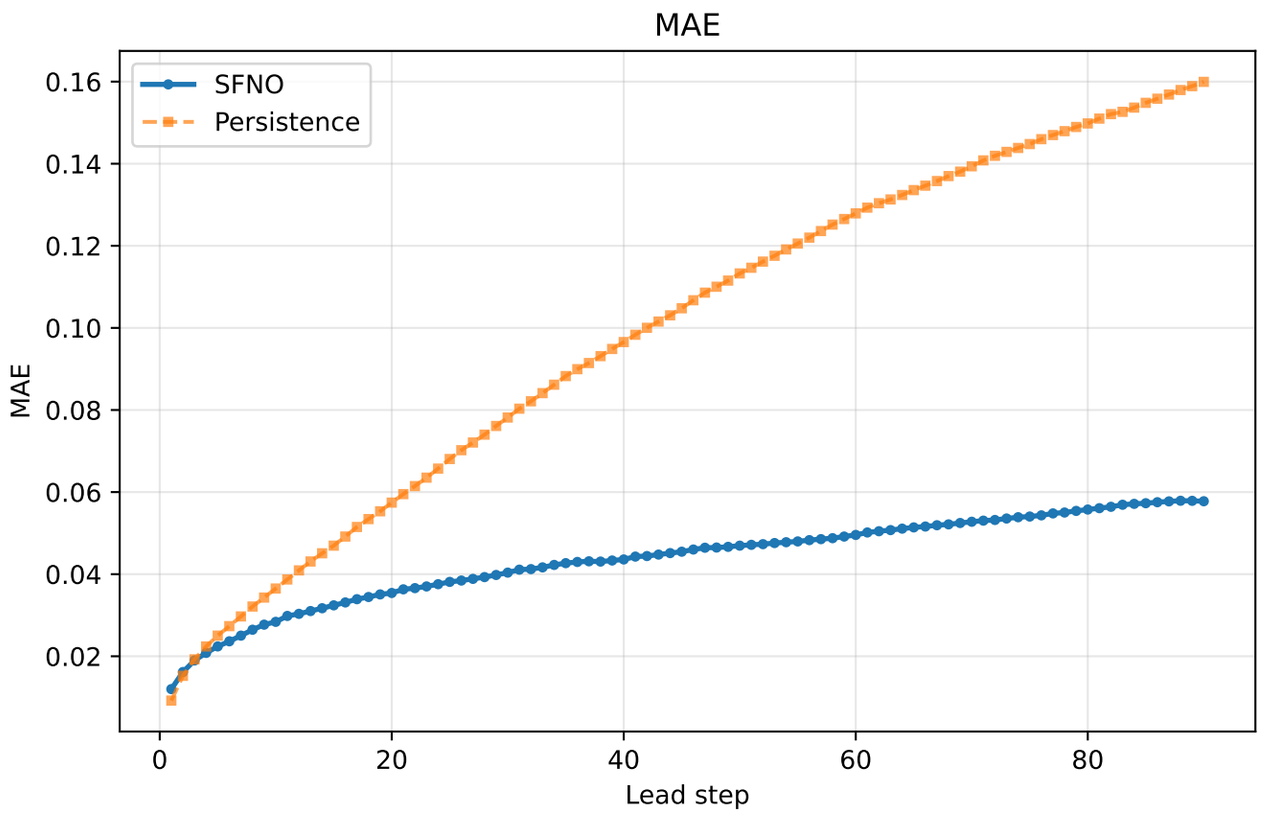
Metrics by Lead Step — Validation

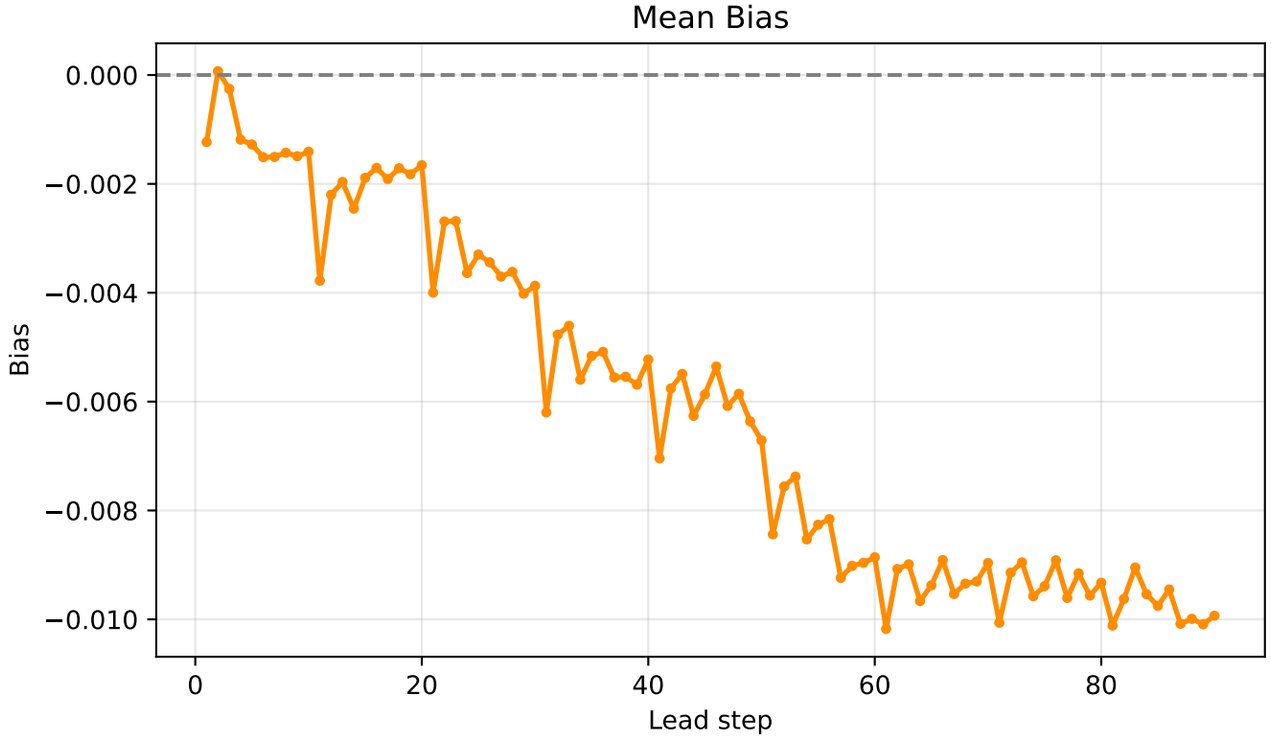
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0184	0.0164	-12.4%	0.0120	-0.0012	0.9521	0.9592	0.9997
2	0.0252	0.0257	2.1%	0.0162	0.0001	0.9045	0.8991	0.9994
3	0.0295	0.0318	7.0%	0.0190	-0.0003	0.8638	0.8481	0.9992
4	0.0326	0.0366	11.0%	0.0208	-0.0012	0.8315	0.8008	0.9990
5	0.0349	0.0406	14.1%	0.0224	-0.0013	0.8023	0.7574	0.9988
6	0.0367	0.0439	16.4%	0.0236	-0.0015	0.7773	0.7202	0.9987
7	0.0387	0.0474	18.4%	0.0250	-0.0015	0.7513	0.6815	0.9985
8	0.0407	0.0510	20.2%	0.0265	-0.0014	0.7251	0.6425	0.9984
9	0.0425	0.0542	21.6%	0.0277	-0.0015	0.7000	0.6072	0.9982
10	0.0437	0.0574	23.9%	0.0284	-0.0014	0.6791	0.5705	0.9981
11	0.0453	0.0606	25.3%	0.0298	-0.0038	0.6529	0.5345	0.9980
12	0.0462	0.0639	27.6%	0.0303	-0.0022	0.6339	0.4984	0.9979
13	0.0470	0.0668	29.6%	0.0310	-0.0020	0.6126	0.4642	0.9978
14	0.0479	0.0695	31.2%	0.0317	-0.0025	0.5917	0.4363	0.9978
15	0.0490	0.0725	32.4%	0.0324	-0.0019	0.5725	0.4112	0.9977
16	0.0501	0.0759	34.0%	0.0331	-0.0017	0.5541	0.3812	0.9976
17	0.0513	0.0797	35.6%	0.0339	-0.0019	0.5339	0.3482	0.9974
18	0.0519	0.0825	37.1%	0.0344	-0.0017	0.5123	0.3242	0.9974
19	0.0527	0.0852	38.2%	0.0351	-0.0018	0.4910	0.3058	0.9973
20	0.0531	0.0881	39.7%	0.0354	-0.0017	0.4720	0.2821	0.9973
21	0.0541	0.0913	40.8%	0.0363	-0.0040	0.4509	0.2589	0.9972
22	0.0546	0.0946	42.3%	0.0366	-0.0027	0.4408	0.2377	0.9971
23	0.0552	0.0978	43.5%	0.0370	-0.0027	0.4249	0.2157	0.9970
24	0.0560	0.1011	44.6%	0.0376	-0.0036	0.4053	0.1932	0.9969
25	0.0567	0.1043	45.7%	0.0381	-0.0033	0.3888	0.1725	0.9969
26	0.0572	0.1075	46.8%	0.0384	-0.0034	0.3731	0.1551	0.9968
27	0.0577	0.1103	47.7%	0.0389	-0.0037	0.3535	0.1418	0.9968
28	0.0582	0.1130	48.5%	0.0393	-0.0036	0.3379	0.1313	0.9967
29	0.0587	0.1161	49.4%	0.0398	-0.0040	0.3241	0.1164	0.9966
30	0.0594	0.1191	50.2%	0.0404	-0.0039	0.3092	0.1038	0.9966
31	0.0602	0.1225	50.8%	0.0411	-0.0062	0.2922	0.0889	0.9965
32	0.0604	0.1251	51.8%	0.0412	-0.0048	0.2872	0.0821	0.9965
33	0.0609	0.1280	52.5%	0.0417	-0.0046	0.2743	0.0694	0.9964
34	0.0617	0.1310	52.9%	0.0423	-0.0056	0.2602	0.0582	0.9963
35	0.0624	0.1339	53.4%	0.0427	-0.0052	0.2552	0.0464	0.9962
36	0.0628	0.1363	53.9%	0.0430	-0.0051	0.2457	0.0400	0.9962
37	0.0630	0.1386	54.5%	0.0431	-0.0056	0.2395	0.0378	0.9961
38	0.0629	0.1410	55.4%	0.0431	-0.0055	0.2334	0.0334	0.9961
39	0.0633	0.1437	55.9%	0.0433	-0.0057	0.2243	0.0265	0.9961
40	0.0637	0.1462	56.5%	0.0436	-0.0052	0.2130	0.0227	0.9960
41	0.0645	0.1489	56.6%	0.0443	-0.0070	0.1947	0.0178	0.9959
42	0.0649	0.1514	57.2%	0.0444	-0.0058	0.1906	0.0118	0.9959
43	0.0653	0.1536	57.5%	0.0448	-0.0055	0.1778	0.0084	0.9958
44	0.0659	0.1558	57.7%	0.0451	-0.0063	0.1643	0.0082	0.9958
45	0.0663	0.1584	58.1%	0.0455	-0.0059	0.1580	-0.0016	0.9957
46	0.0670	0.1614	58.5%	0.0460	-0.0054	0.1500	-0.0150	0.9956
47	0.0677	0.1643	58.8%	0.0464	-0.0061	0.1424	-0.0262	0.9955
48	0.0679	0.1665	59.2%	0.0465	-0.0059	0.1373	-0.0296	0.9955
49	0.0682	0.1686	59.6%	0.0467	-0.0064	0.1329	-0.0316	0.9955
50	0.0686	0.1711	59.9%	0.0469	-0.0067	0.1283	-0.0359	0.9954
51	0.0688	0.1733	60.3%	0.0471	-0.0084	0.1196	-0.0392	0.9954
52	0.0689	0.1757	60.8%	0.0473	-0.0076	0.1175	-0.0454	0.9954
53	0.0693	0.1779	61.1%	0.0476	-0.0074	0.1127	-0.0488	0.9953
54	0.0695	0.1803	61.5%	0.0478	-0.0085	0.1189	-0.0525	0.9953
55	0.0696	0.1824	61.8%	0.0480	-0.0083	0.1171	-0.0540	0.9953
56	0.0701	0.1847	62.0%	0.0483	-0.0082	0.1090	-0.0585	0.9952
57	0.0706	0.1871	62.3%	0.0485	-0.0092	0.1064	-0.0631	0.9952
58	0.0710	0.1896	62.5%	0.0488	-0.0090	0.1024	-0.0696	0.9951
59	0.0718	0.1917	62.6%	0.0492	-0.0090	0.0934	-0.0708	0.9950
60	0.0723	0.1938	62.7%	0.0496	-0.0089	0.0918	-0.0716	0.9949
61	0.0731	0.1959	62.7%	0.0502	-0.0102	0.0841	-0.0755	0.9948
62	0.0733	0.1974	62.9%	0.0505	-0.0091	0.0785	-0.0678	0.9948
63	0.0737	0.1989	63.0%	0.0507	-0.0090	0.0761	-0.0586	0.9947
64	0.0744	0.2009	63.0%	0.0511	-0.0097	0.0699	-0.0567	0.9946
65	0.0750	0.2030	63.1%	0.0514	-0.0094	0.0628	-0.0588	0.9945
66	0.0750	0.2049	63.4%	0.0516	-0.0089	0.0609	-0.0583	0.9945
67	0.0754	0.2067	63.5%	0.0519	-0.0095	0.0604	-0.0556	0.9945
68	0.0758	0.2088	63.7%	0.0521	-0.0093	0.0608	-0.0542	0.9944
69	0.0764	0.2108	63.8%	0.0525	-0.0093	0.0585	-0.0508	0.9943
70	0.0767	0.2131	64.0%	0.0528	-0.0090	0.0597	-0.0529	0.9943
71	0.0770	0.2157	64.3%	0.0530	-0.0101	0.0621	-0.0598	0.9942
72	0.0772	0.2178	64.5%	0.0532	-0.0091	0.0631	-0.0609	0.9942
73	0.0777	0.2196	64.6%	0.0536	-0.0090	0.0615	-0.0564	0.9941
74	0.0782	0.2214	64.7%	0.0539	-0.0096	0.0606	-0.0508	0.9940
75	0.0786	0.2232	64.8%	0.0540	-0.0094	0.0547	-0.0483	0.9940
76	0.0790	0.2253	65.0%	0.0543	-0.0089	0.0454	-0.0507	0.9939
77	0.0797	0.2272	64.9%	0.0548	-0.0096	0.0374	-0.0482	0.9938
78	0.0801	0.2290	65.0%	0.0550	-0.0092	0.0300	-0.0477	0.9937
79	0.0806	0.2308	65.1%	0.0554	-0.0096	0.0257	-0.0476	0.9937
80	0.0810	0.2324	65.1%	0.0557	-0.0093	0.0220	-0.0445	0.9936
81	0.0814	0.2345	65.3%	0.0561	-0.0101	0.0226	-0.0448	0.9935
82	0.0819	0.2366	65.4%	0.0564	-0.0096	0.0197	-0.0443	0.9935
83	0.0824	0.2380	65.4%	0.0569	-0.0090	0.0114	-0.0426	0.9934
84	0.0828	0.2399	65.5%	0.0571	-0.0095	0.0119	-0.0439	0.9933
85	0.0830	0.2422	65.7%	0.0573	-0.0098	0.0181	-0.0456	0.9933
86	0.0834	0.2443	65.9%	0.0575	-0.0095	0.0193	-0.0445	0.9932
87	0.0836	0.2464	66.0%	0.0577	-0.0101	0.0213	-0.0446	0.9932
88	0.0838	0.2484	66.3%	0.0579	-0.0100	0.0212	-0.0447	0.9932
89	0.0838	0.2501	66.5%	0.0579	-0.0101	0.0219	-0.0431	0.9932
90	0.0837	0.2522	66.8%	0.0578	-0.0099	0.0266	-0.0441	0.9932

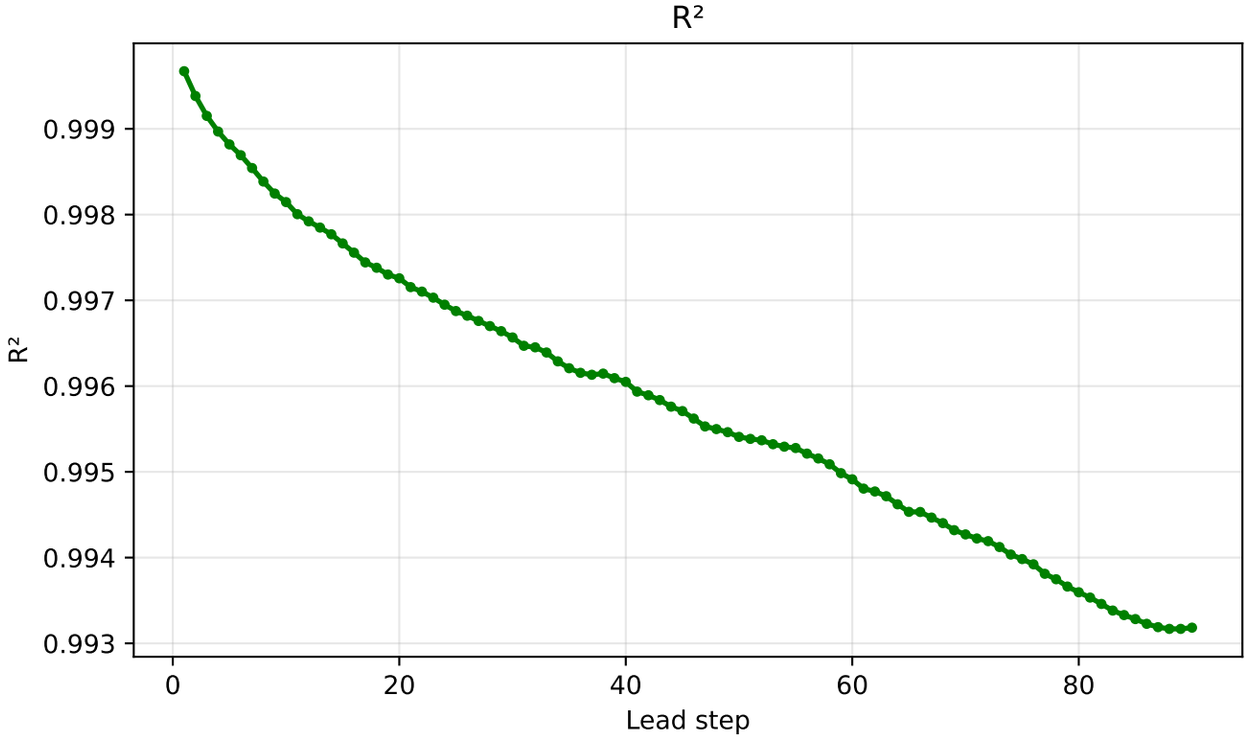
Forecast Skill vs Lead Step — Validation





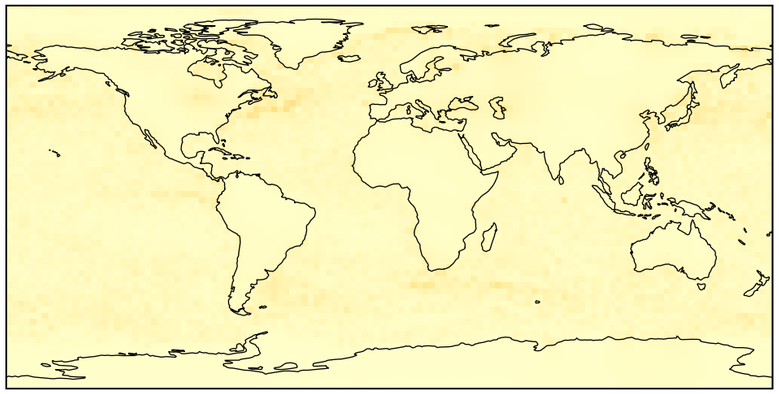
Bias and R² — Validation



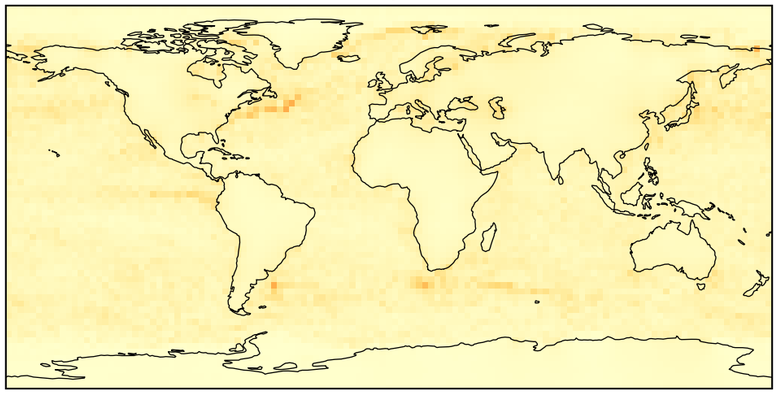


Spatial RMSE (all grid) — Validation

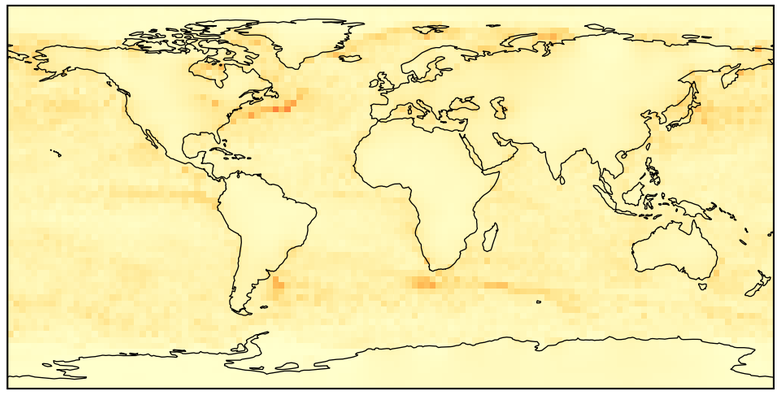
Step 1



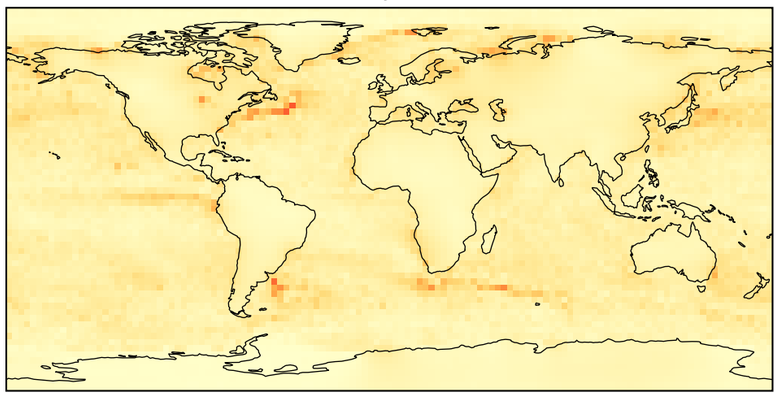
Step 3



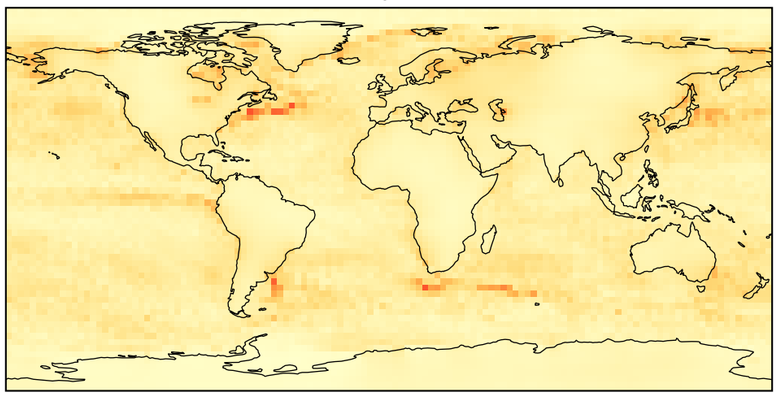
Step 7



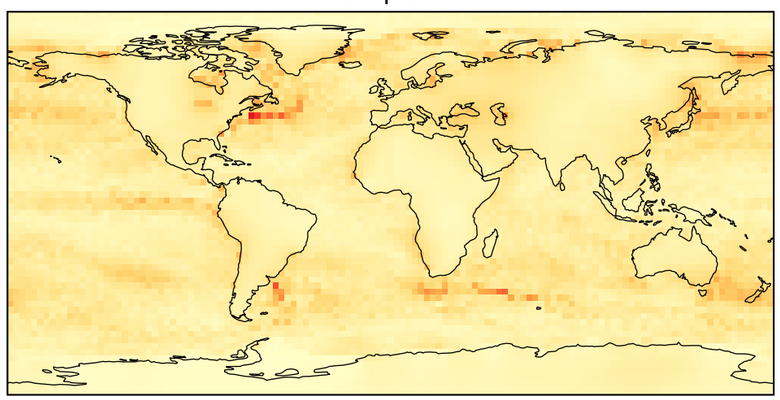
Step 14



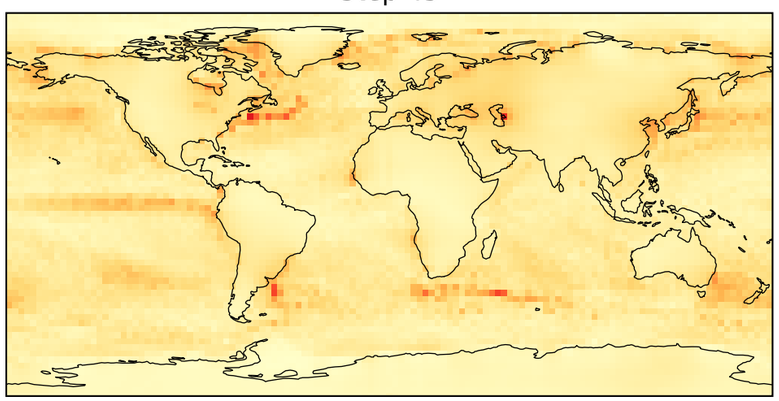
Step 21



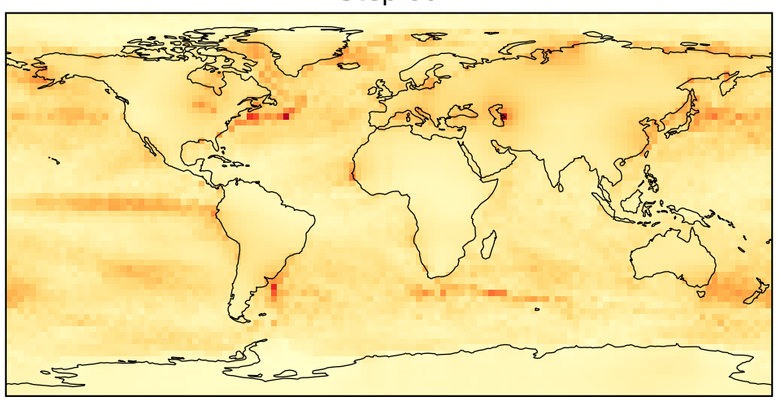
Step 30



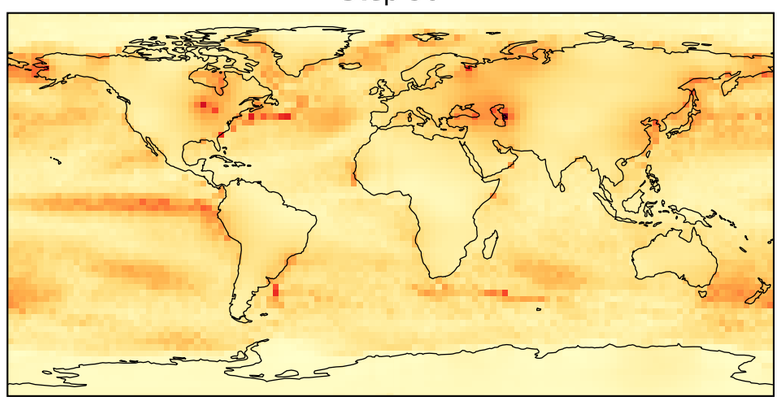
Step 45

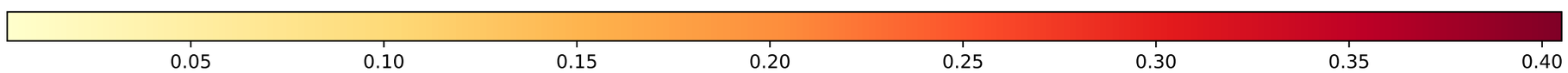


Step 60



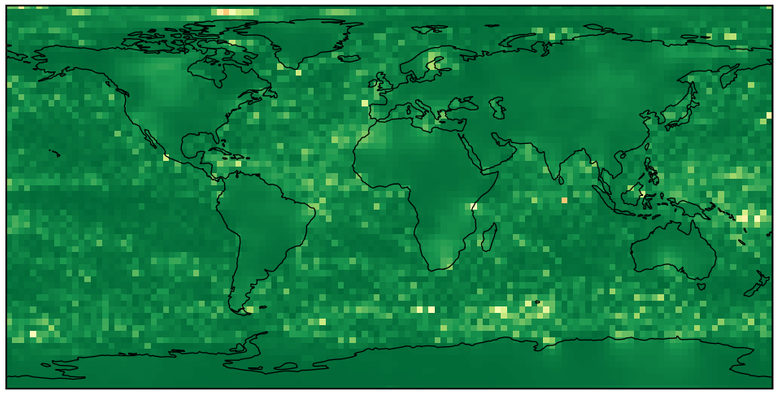
Step 90



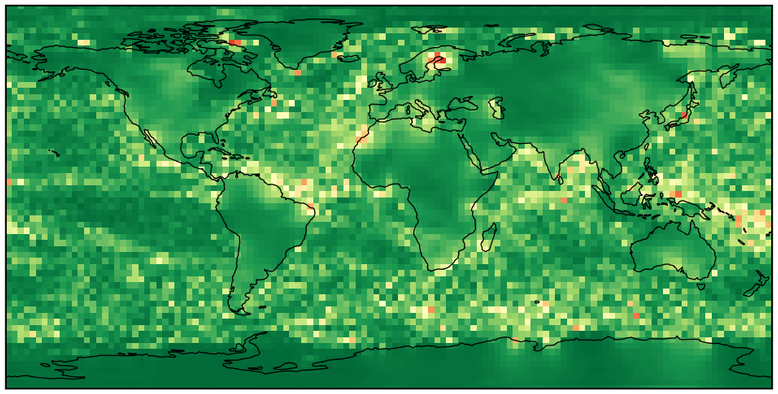


Spatial ACC (all grid) — Validation

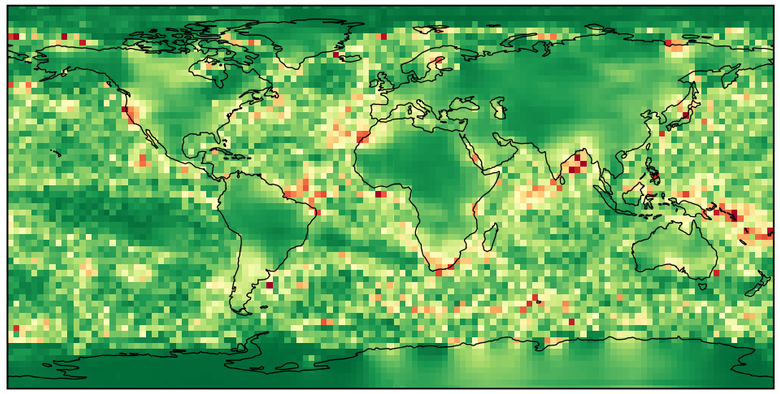
Step 1



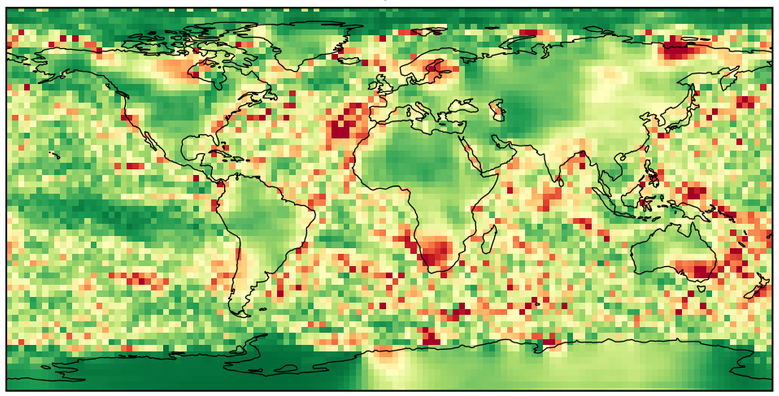
Step 3



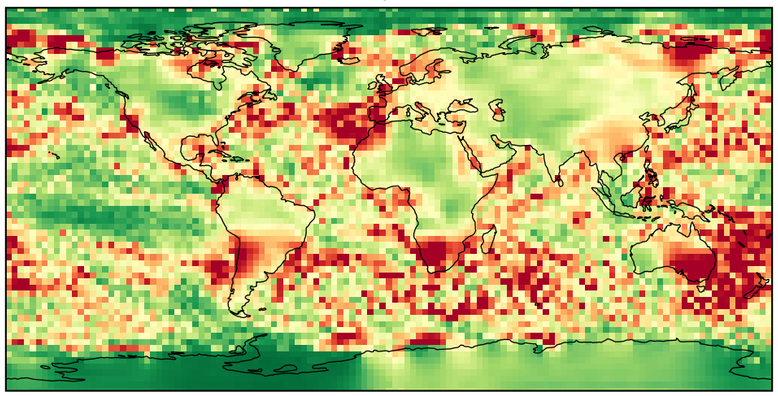
Step 7



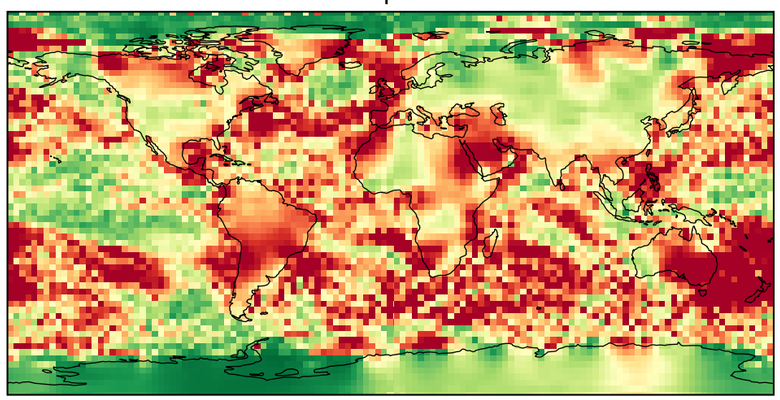
Step 14



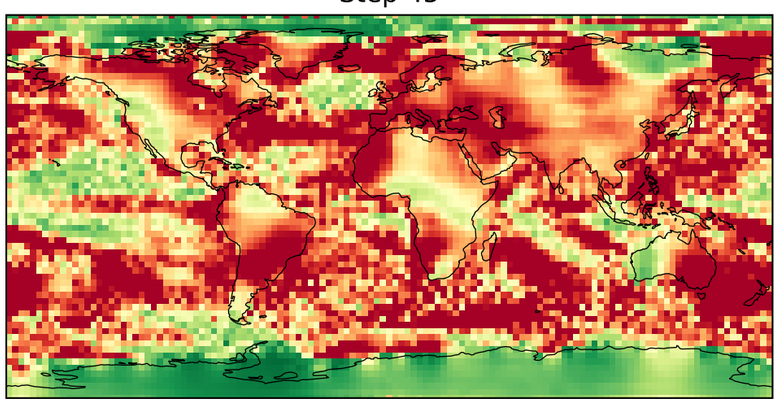
Step 21



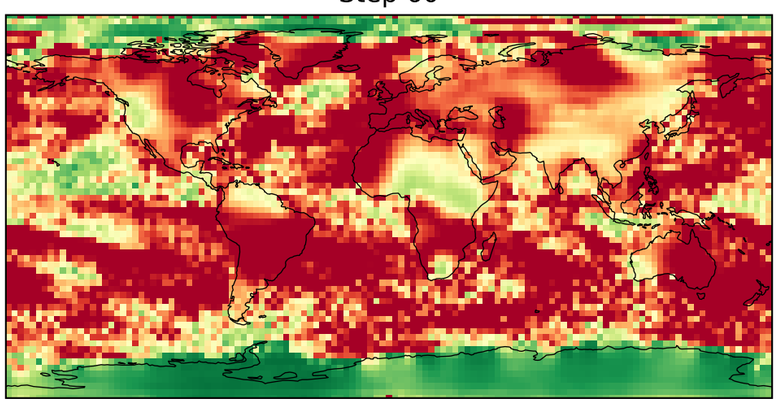
Step 30



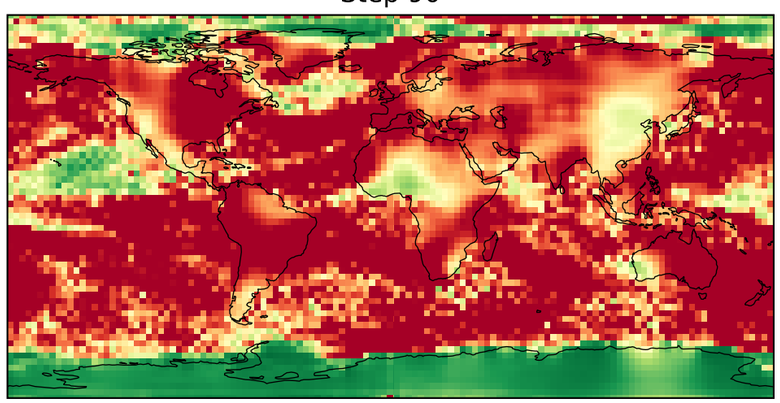
Step 45

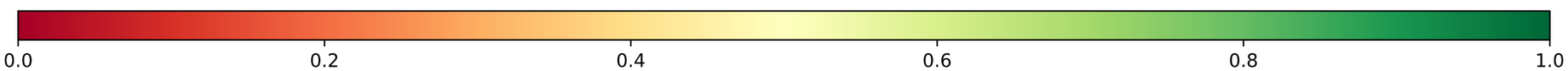


Step 60



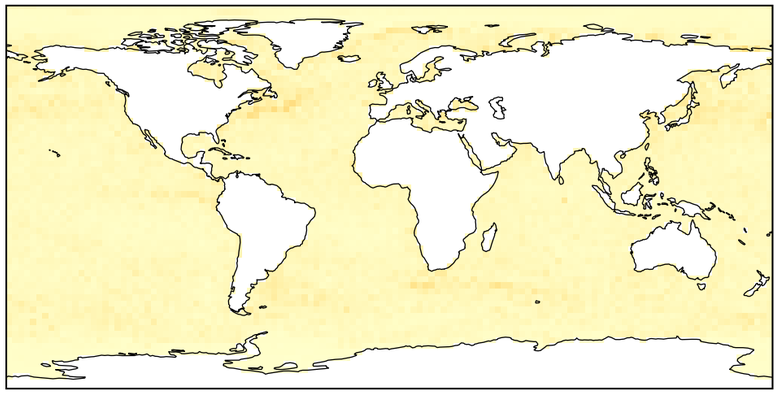
Step 90



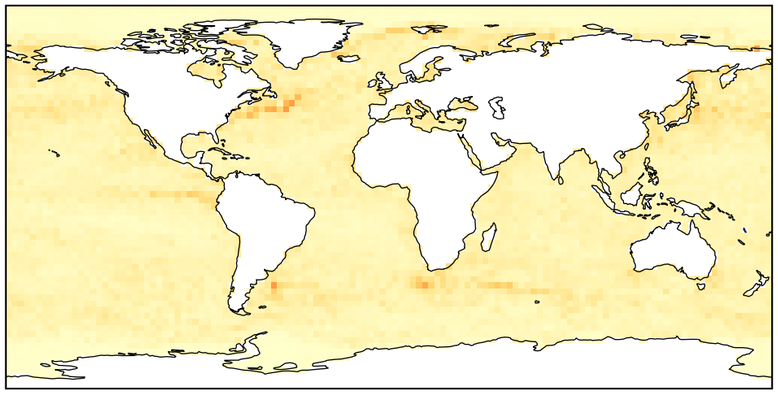


Spatial RMSE (ocean only) — Validation

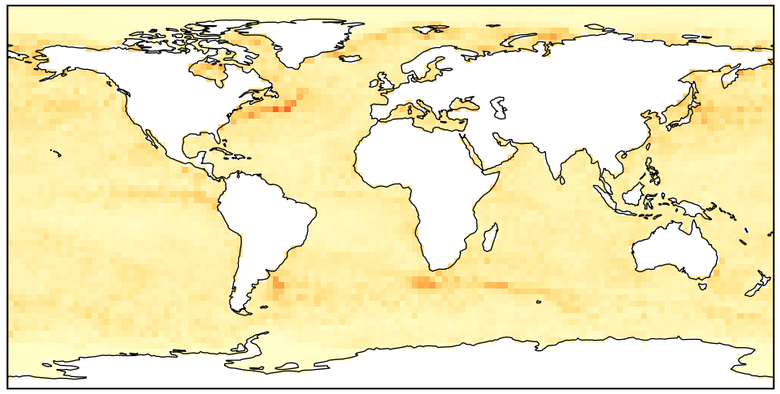
Step 1



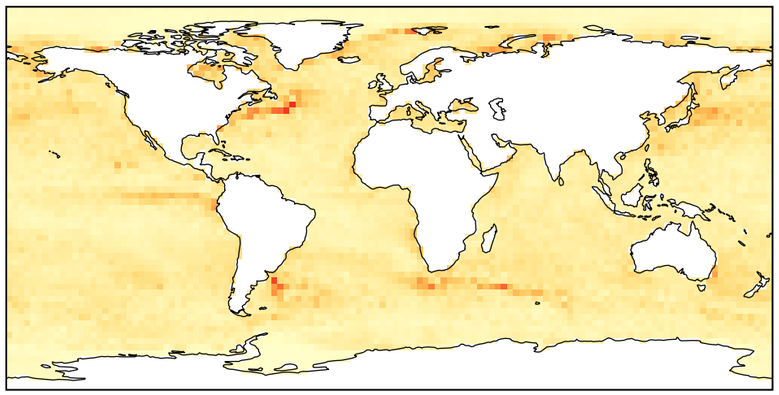
Step 3



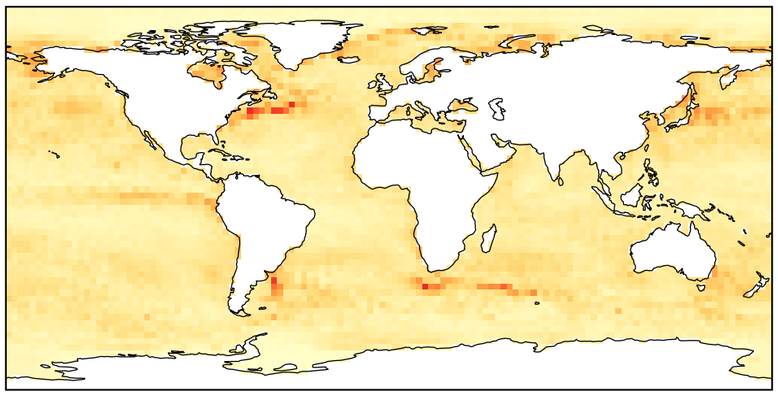
Step 7



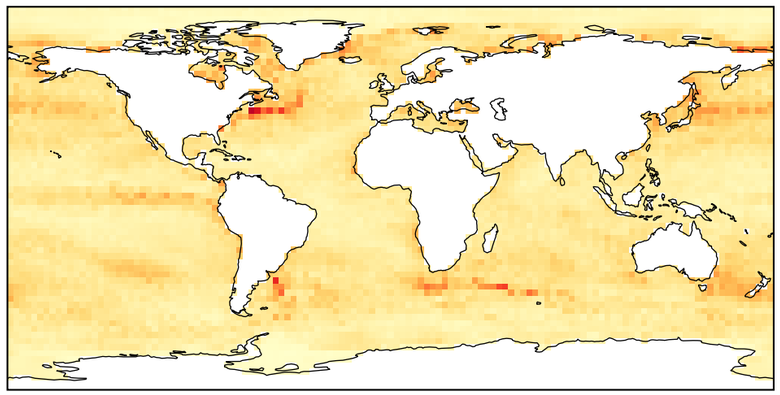
Step 14



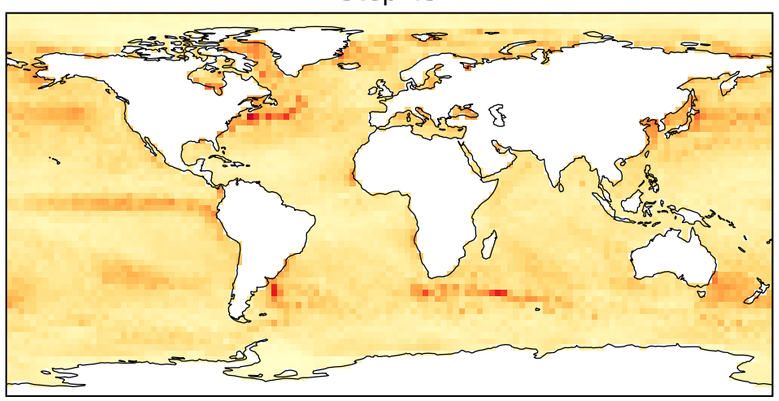
Step 21



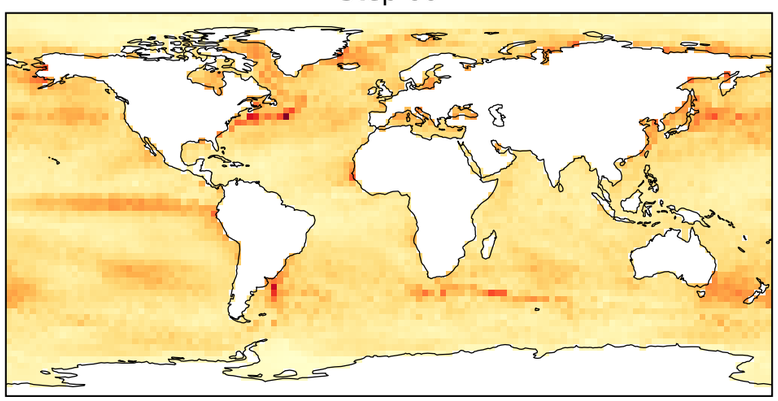
Step 30



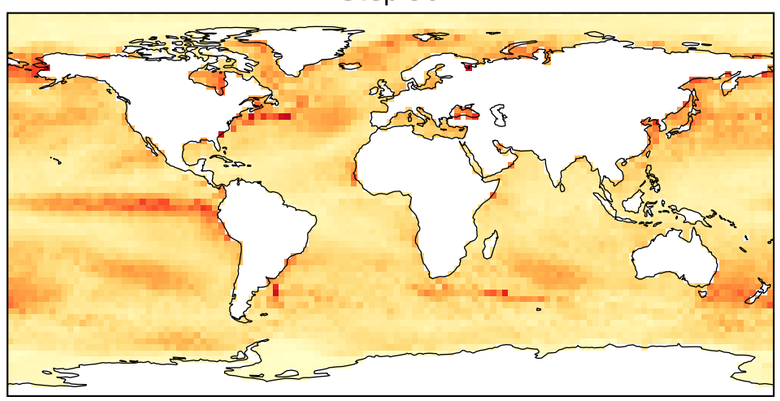
Step 45



Step 60

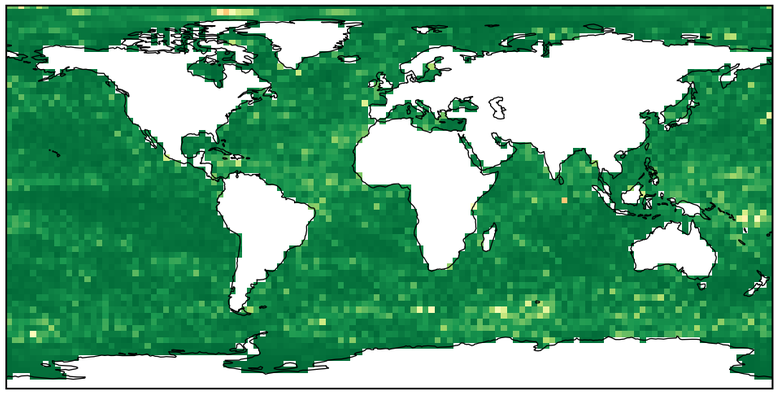


Step 90

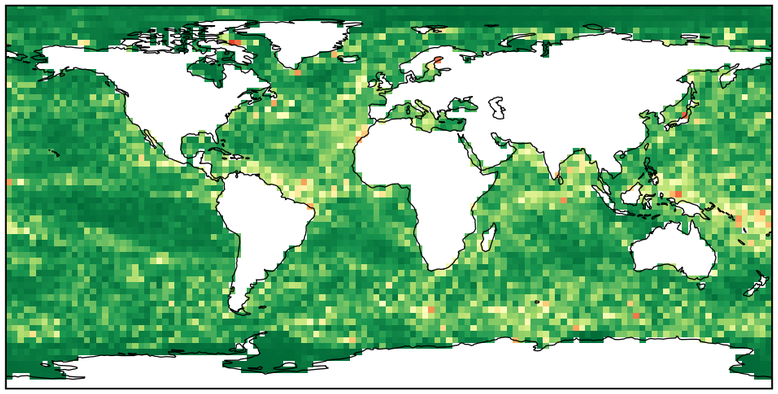


Spatial ACC (ocean only) — Validation

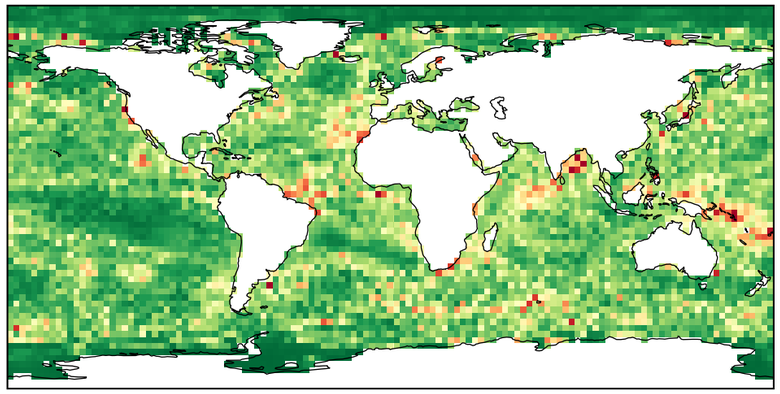
Step 1



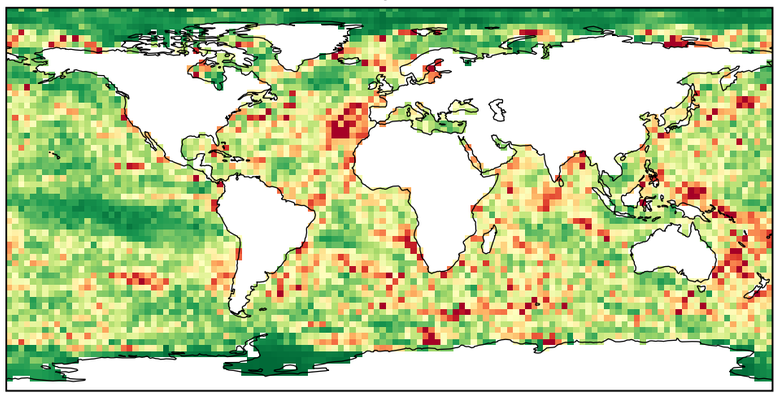
Step 3



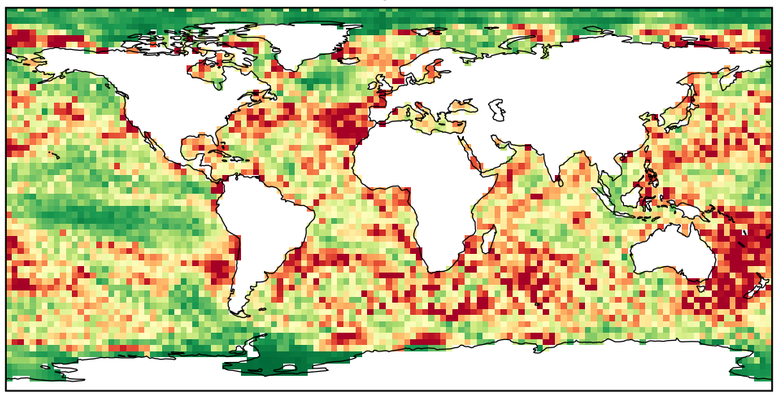
Step 7



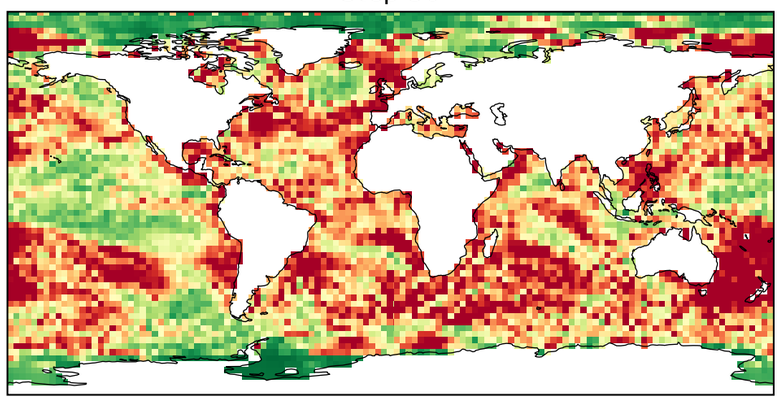
Step 14



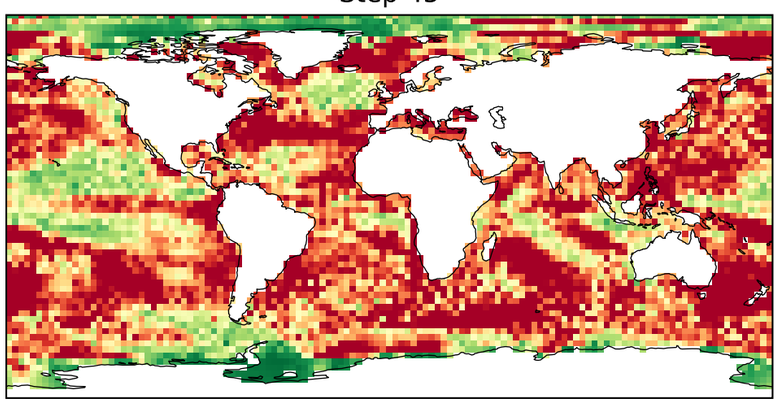
Step 21



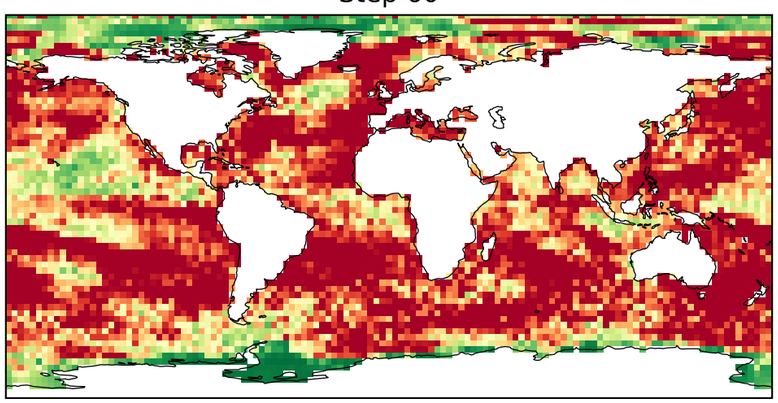
Step 30



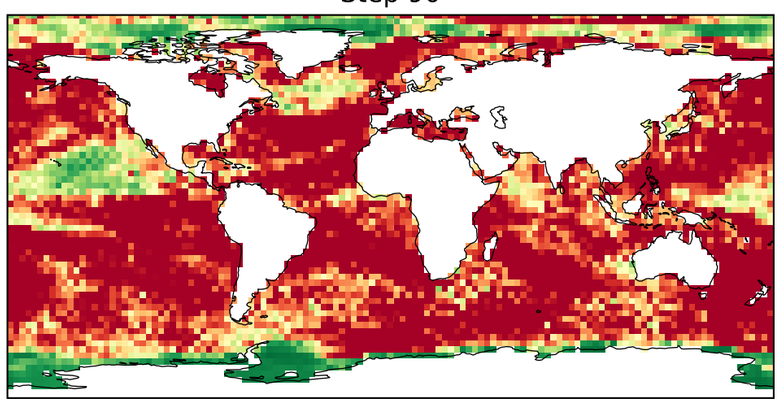
Step 45

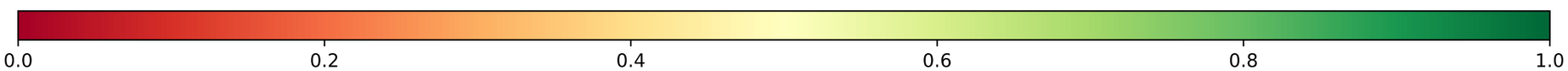


Step 60



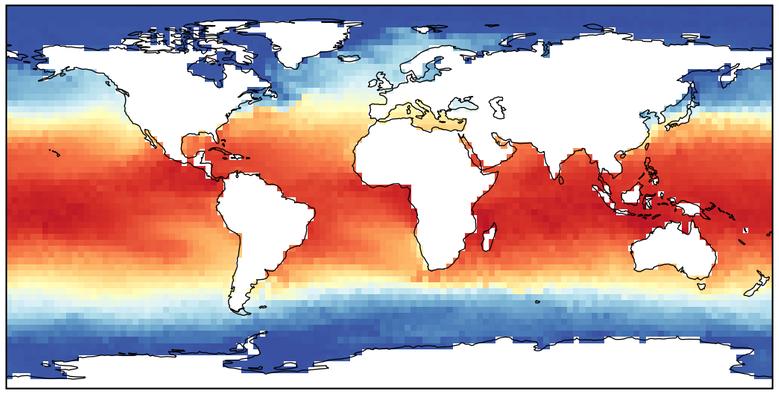
Step 90



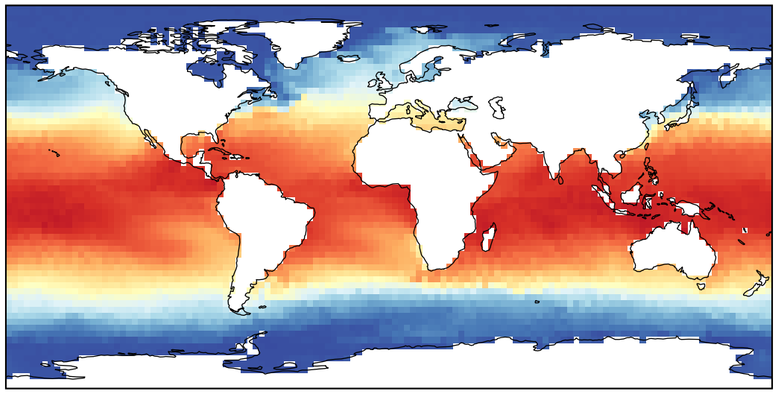


Chain init: 2016-01-11 — Validation (°C)

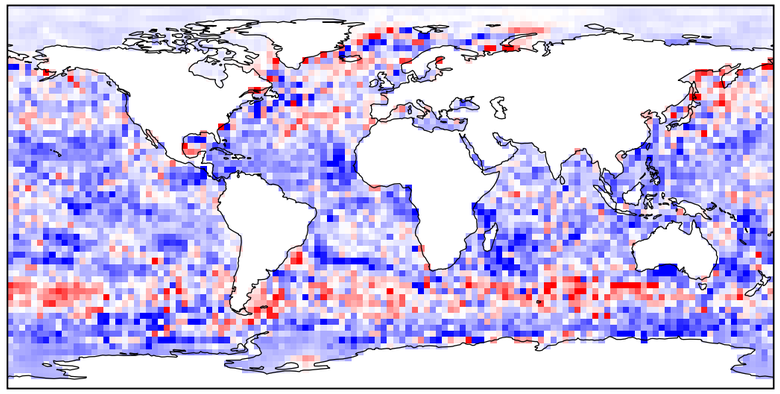
Actual — Step 1 (°C)



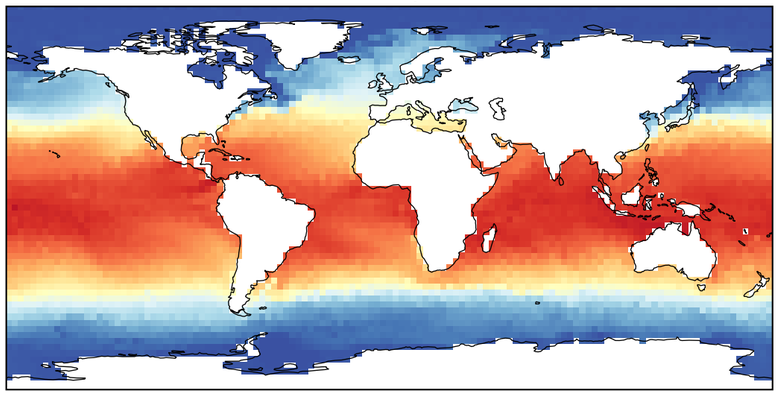
SFNO — Step 1 (°C)



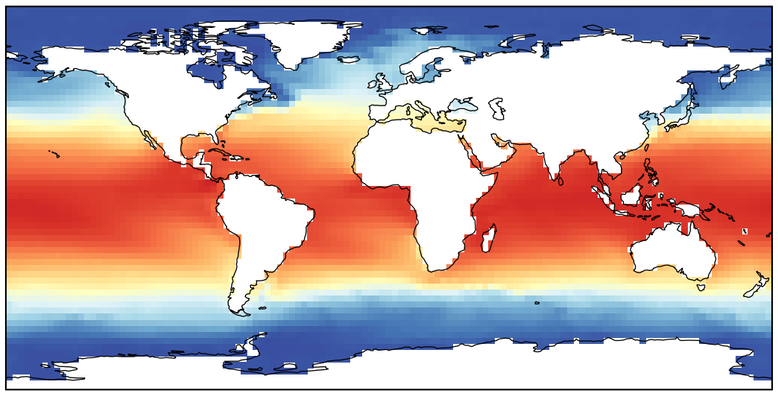
Error — Step 1 (°C)



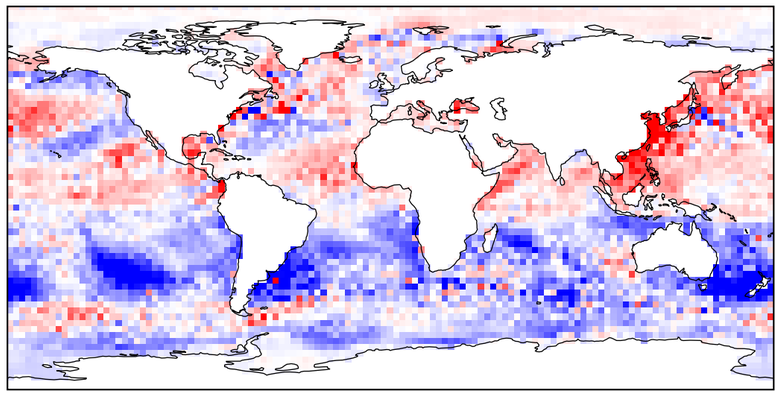
Actual — Step 46 (°C)



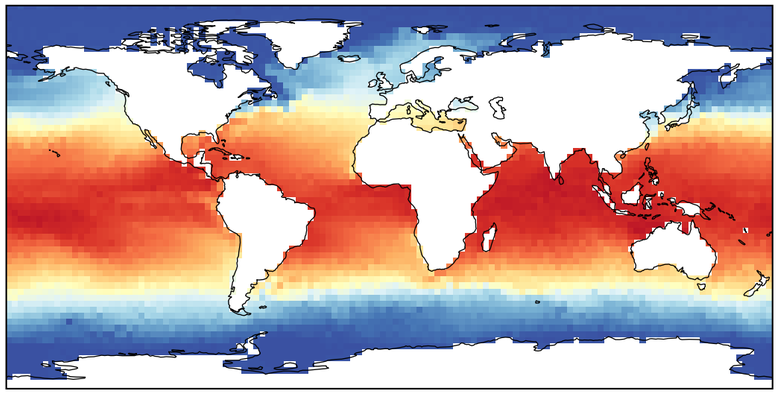
SFNO — Step 46 (°C)



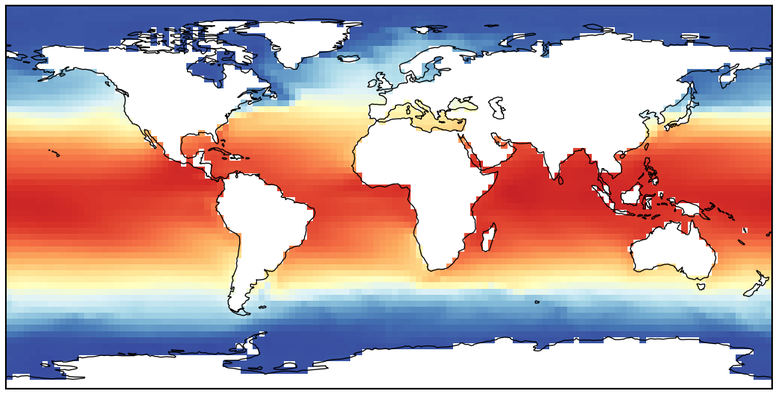
Error — Step 46 (°C)



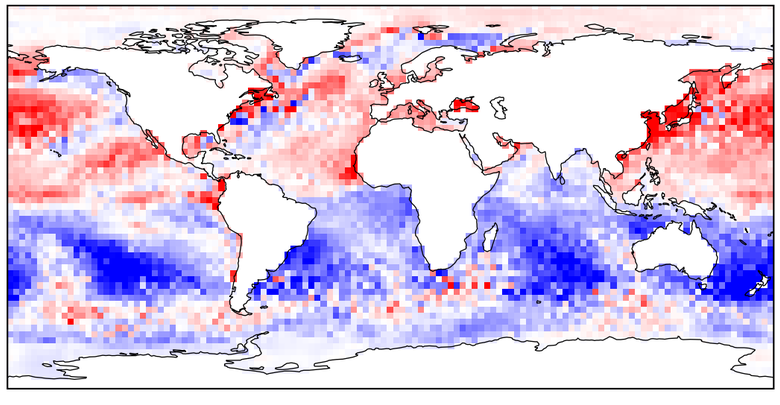
Actual — Step 90 (°C)

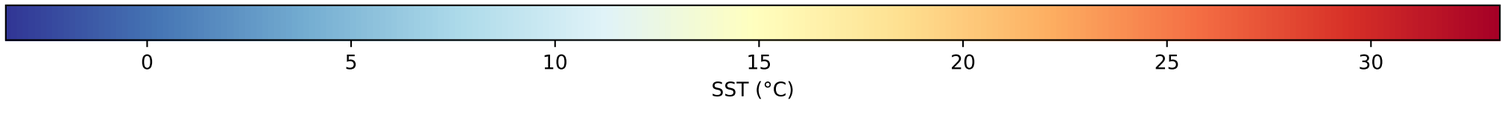


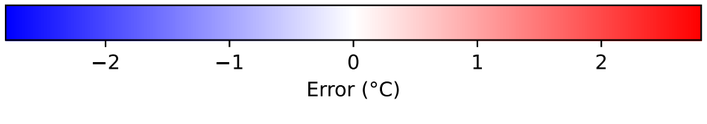
SFNO — Step 90 (°C)



Error — Step 90 (°C)

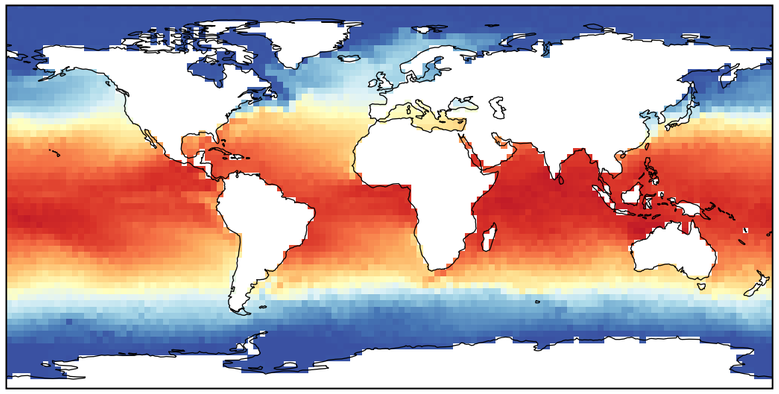




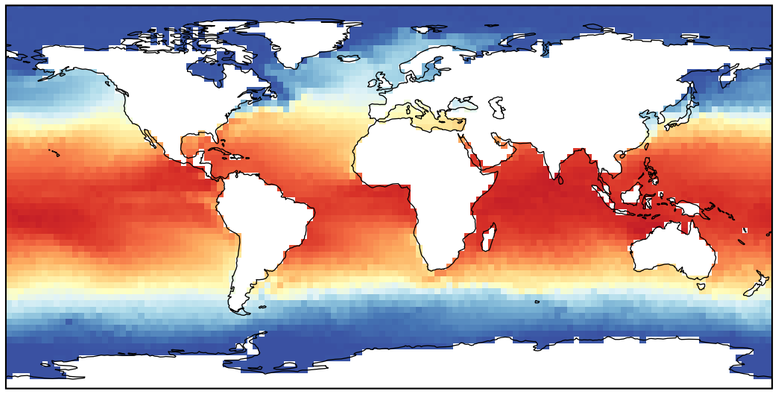


Chain init: 2016-04-10 — Validation (°C)

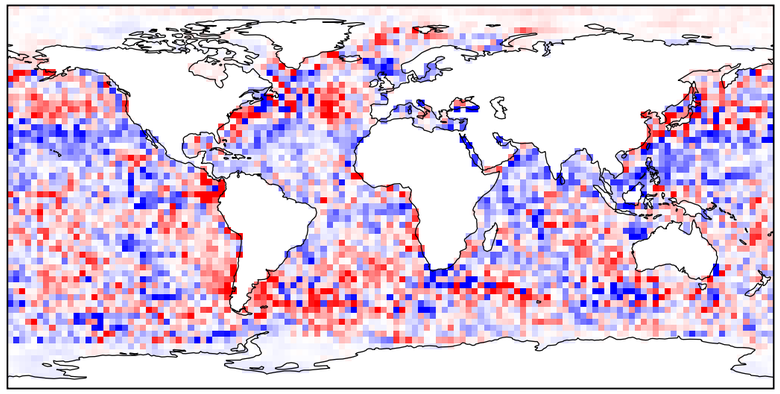
Actual — Step 1 (°C)



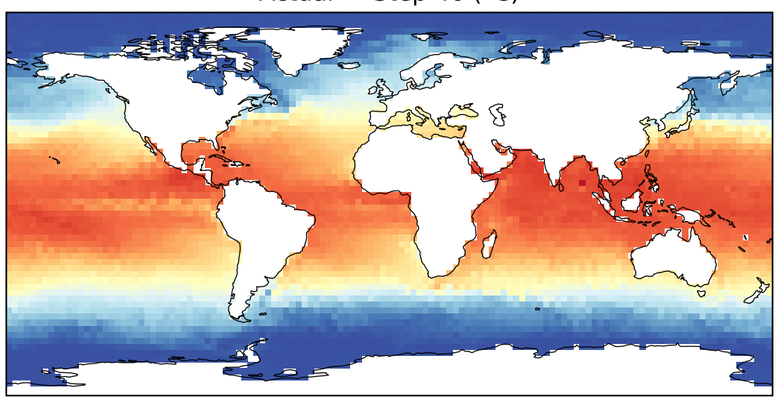
SFNO — Step 1 (°C)



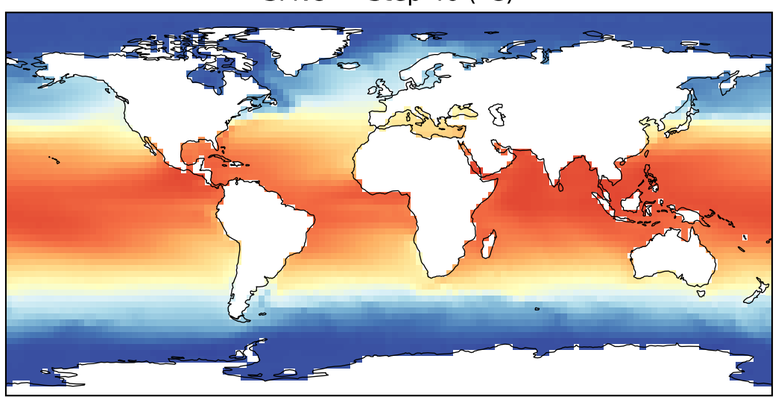
Error — Step 1 (°C)



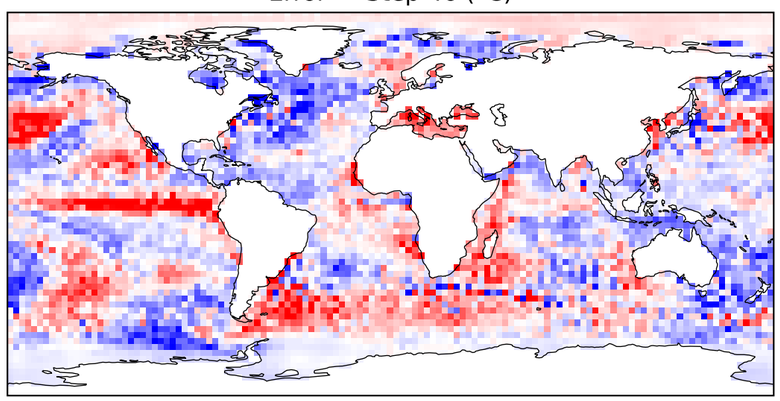
Actual — Step 46 (°C)



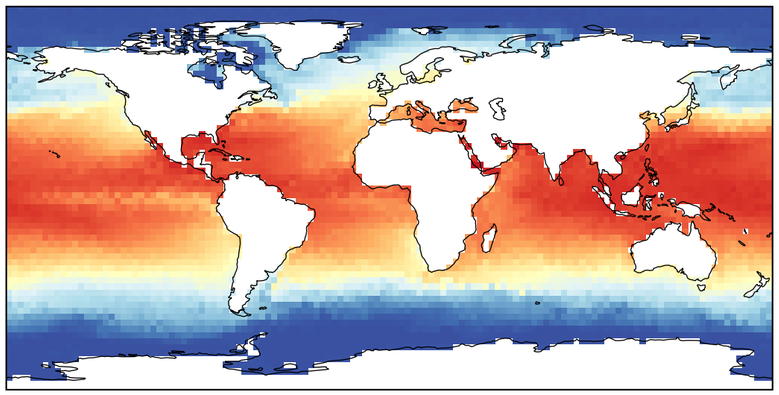
SFNO — Step 46 (°C)



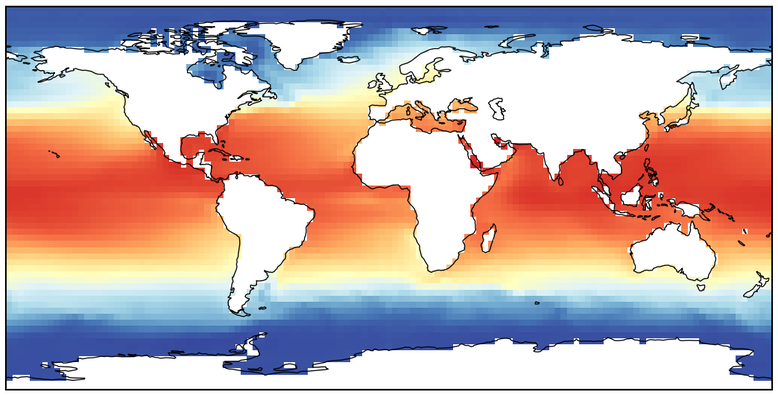
Error — Step 46 (°C)



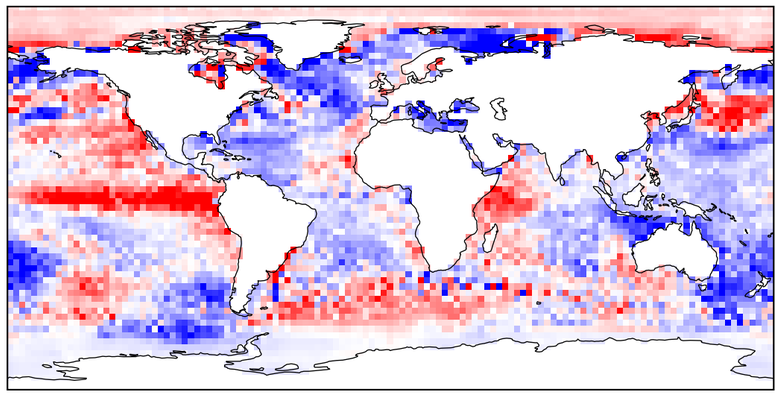
Actual — Step 90 (°C)

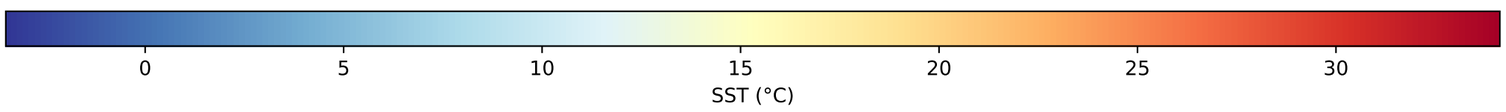


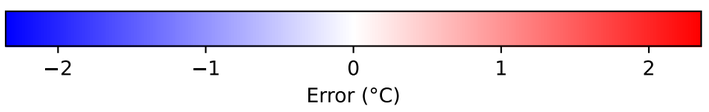
SFNO — Step 90 (°C)



Error — Step 90 (°C)

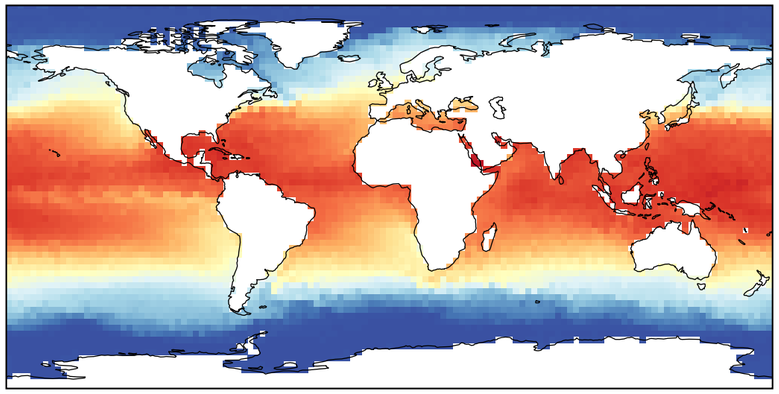




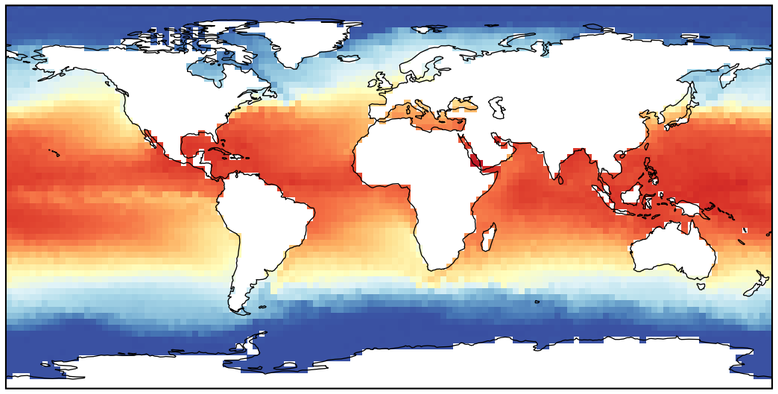


Chain init: 2016-10-07 — Validation (°C)

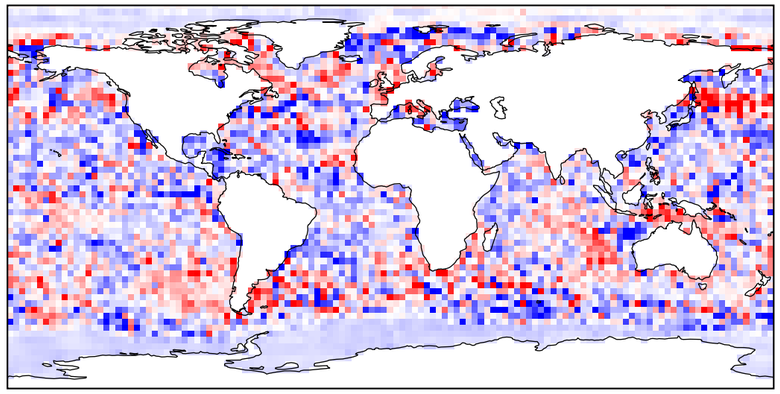
Actual — Step 1 (°C)



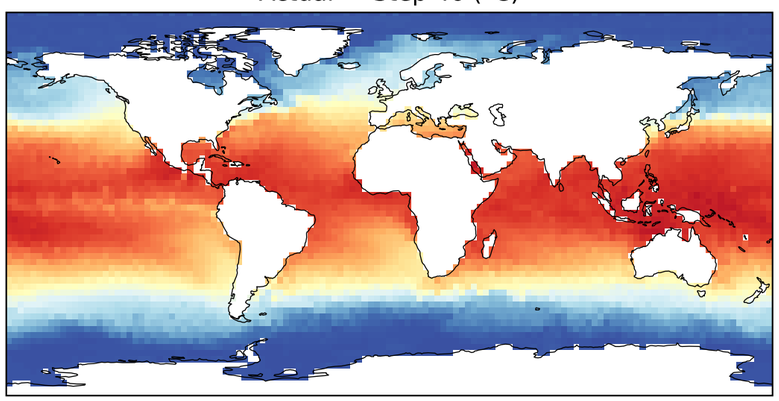
SFNO — Step 1 (°C)



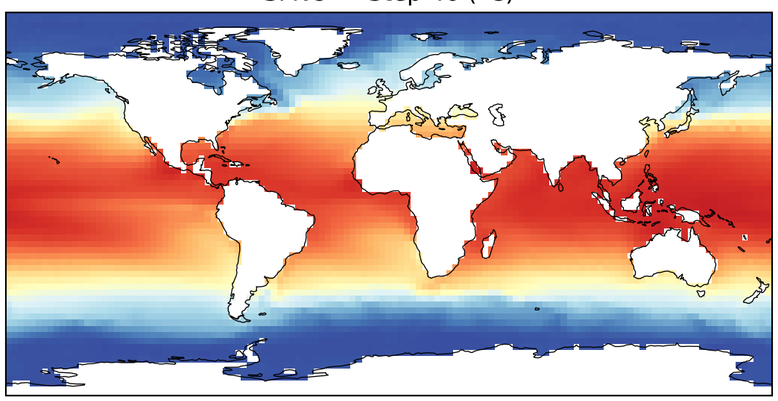
Error — Step 1 (°C)



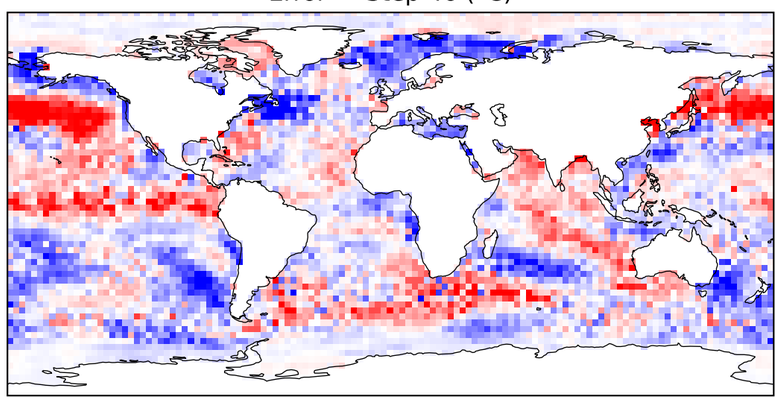
Actual — Step 46 (°C)



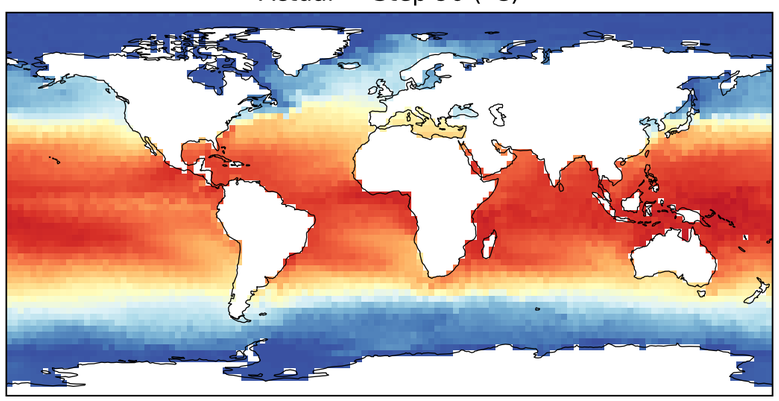
SFNO — Step 46 (°C)



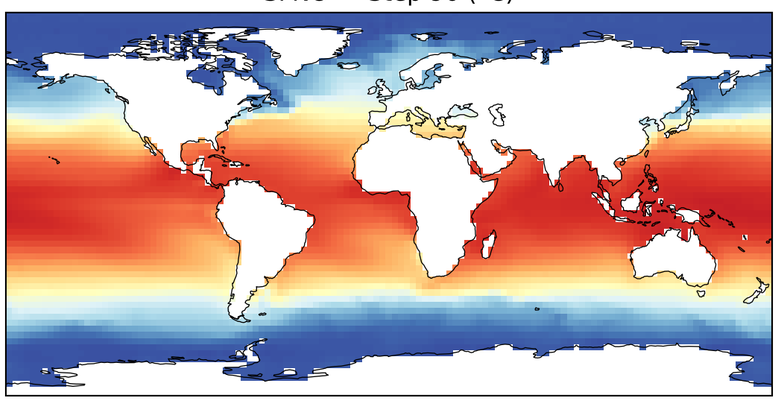
Error — Step 46 (°C)



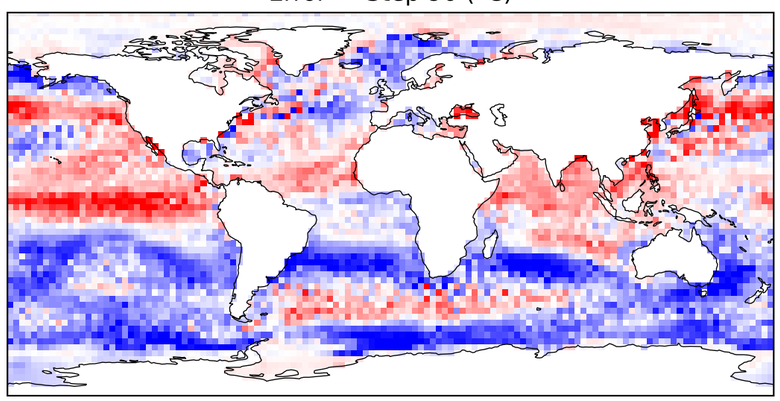
Actual — Step 90 (°C)

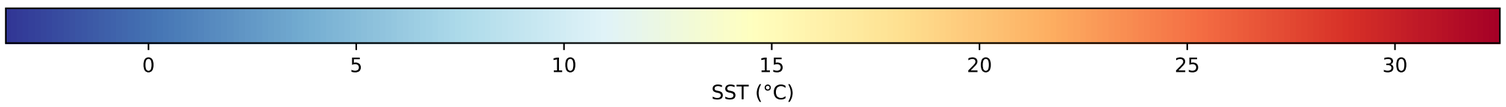


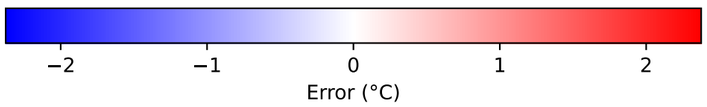
SFNO — Step 90 (°C)



Error — Step 90 (°C)

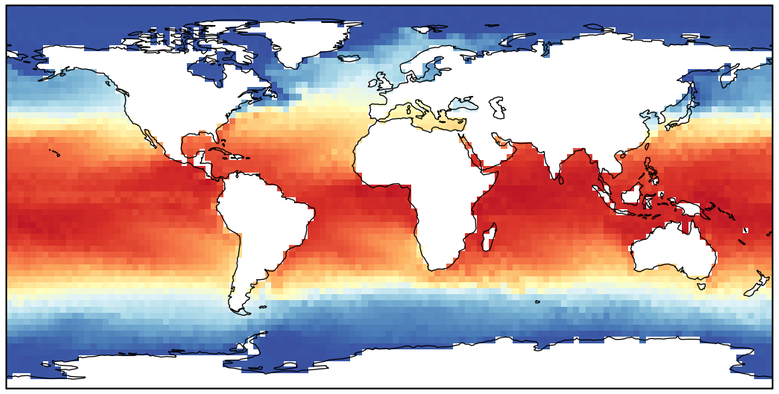




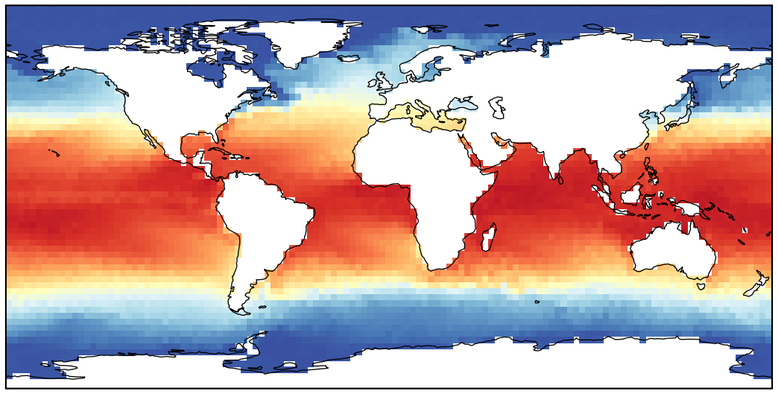


Chain init: 2017-04-05 — Validation (°C)

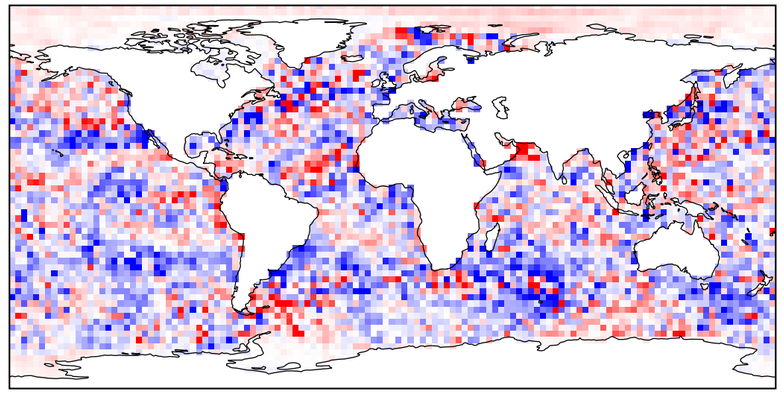
Actual — Step 1 (°C)



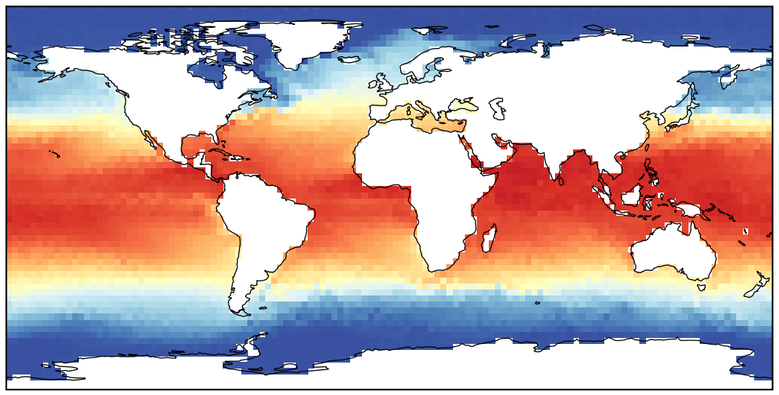
SFNO — Step 1 (°C)



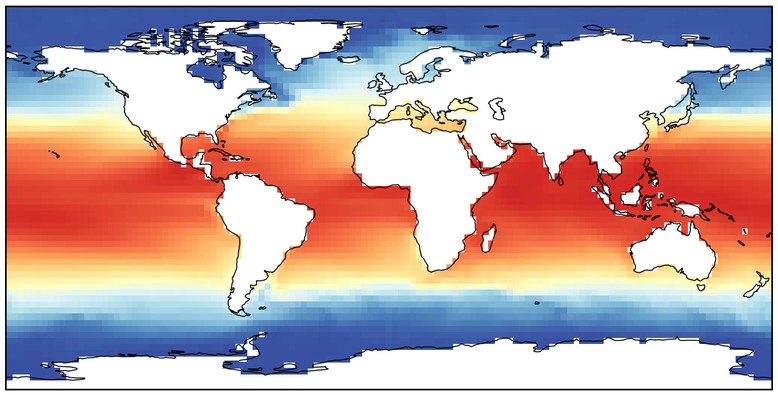
Error — Step 1 (°C)



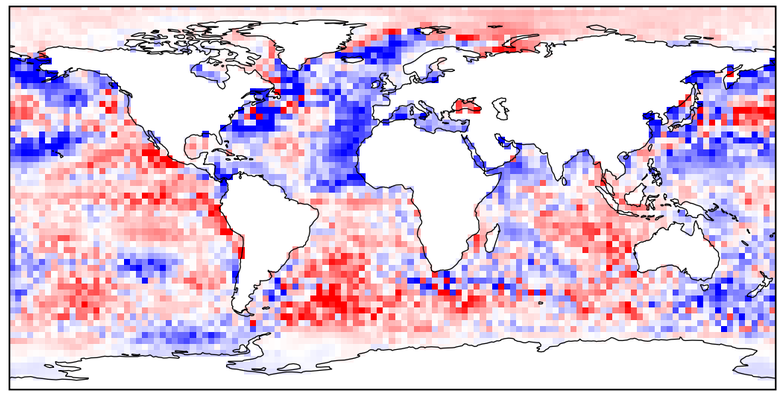
Actual — Step 46 (°C)



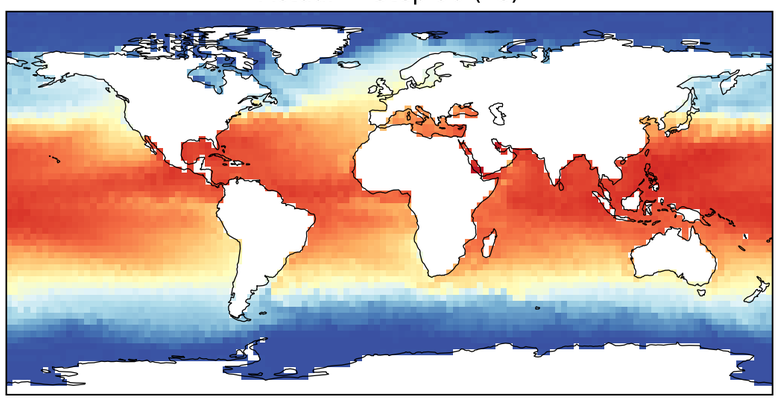
SFNO — Step 46 (°C)



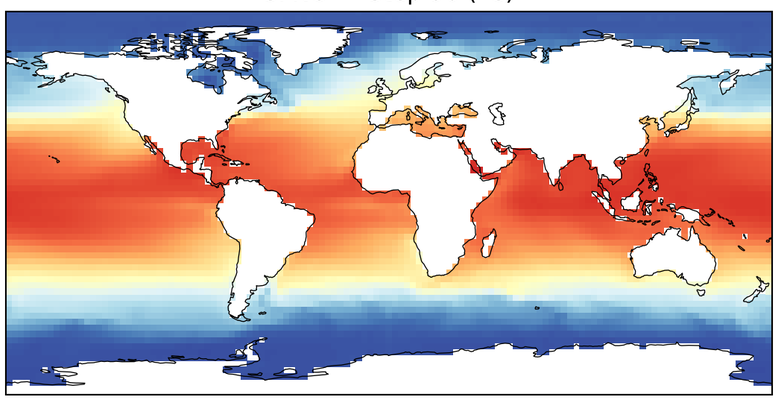
Error — Step 46 (°C)



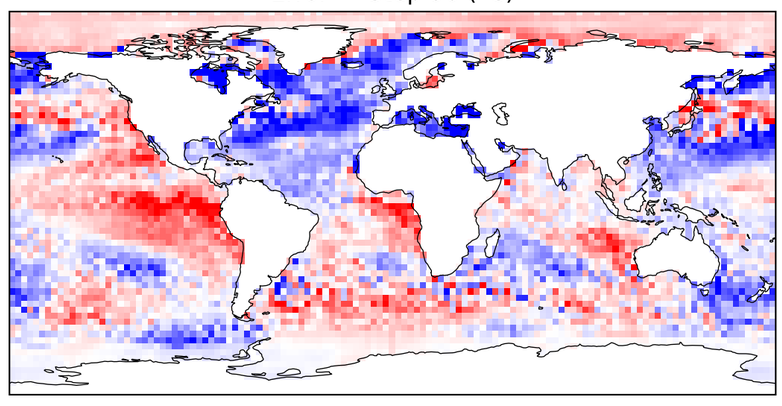
Actual — Step 90 (°C)

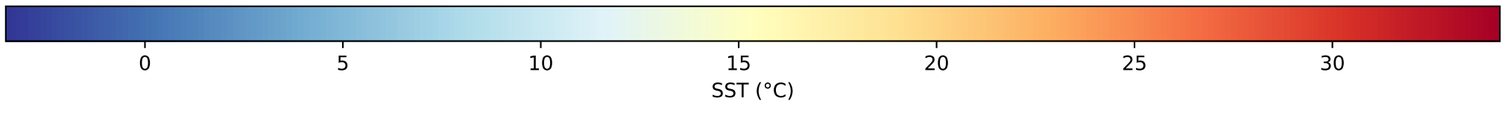


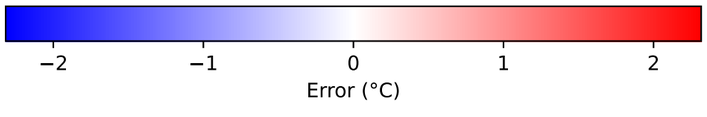
SFNO — Step 90 (°C)



Error — Step 90 (°C)

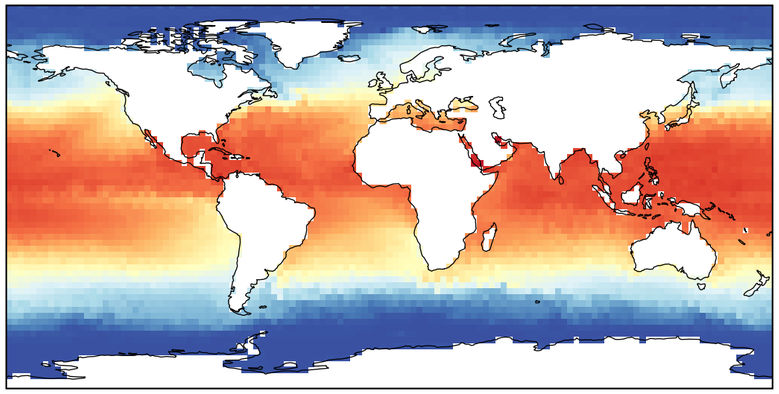




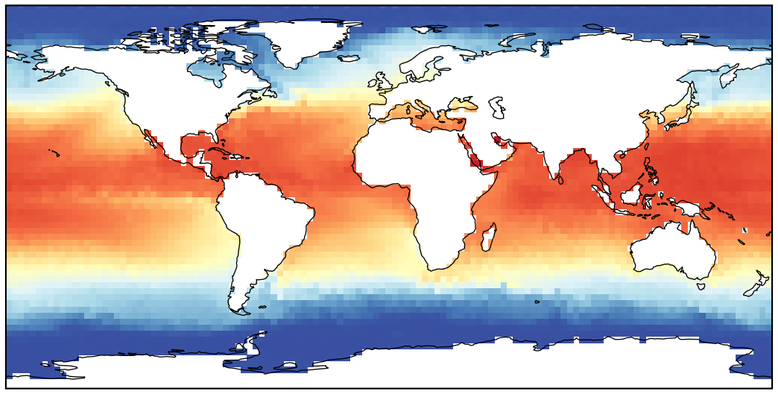


Chain init: 2017-10-02 — Validation (°C)

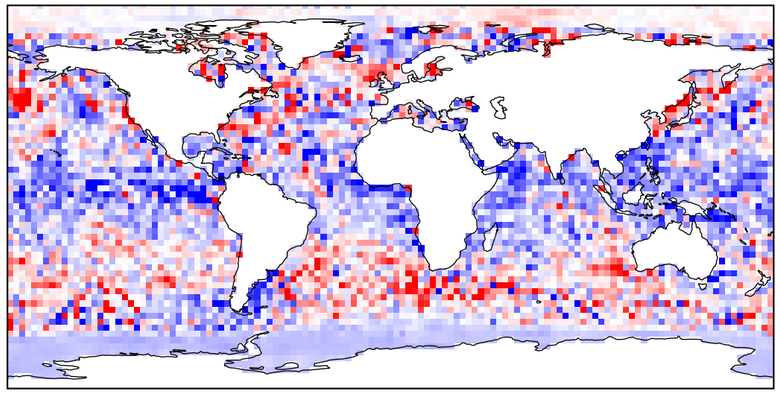
Actual — Step 1 (°C)



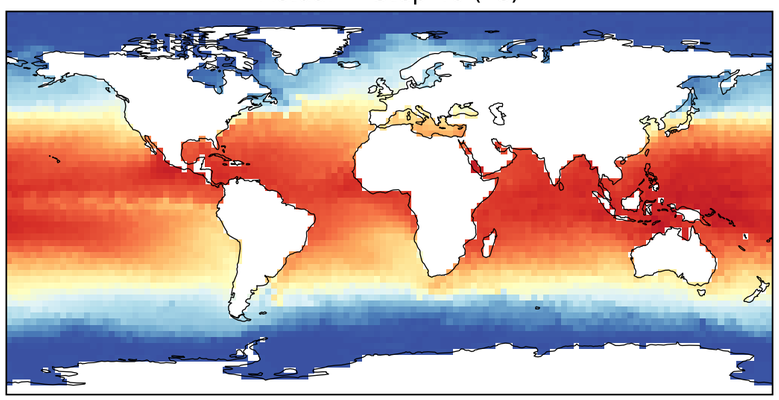
SFNO — Step 1 (°C)



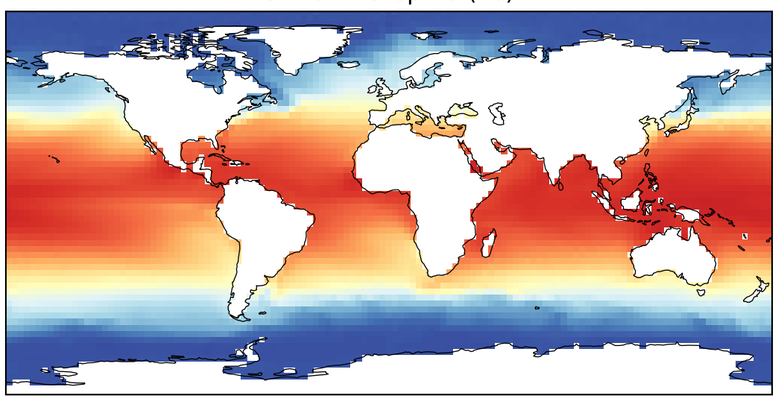
Error — Step 1 (°C)



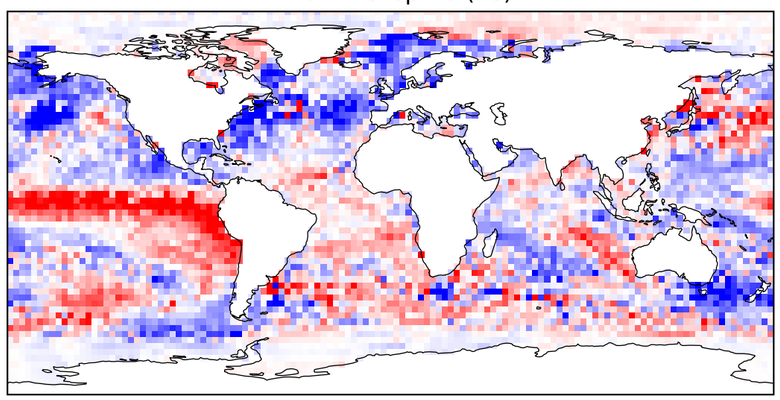
Actual — Step 46 (°C)



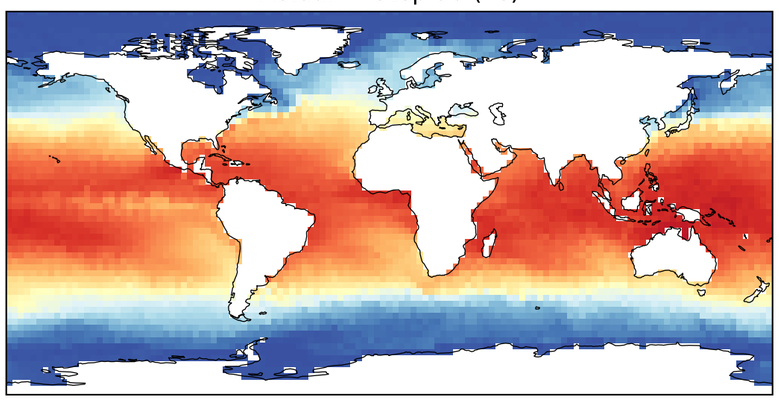
SFNO — Step 46 (°C)



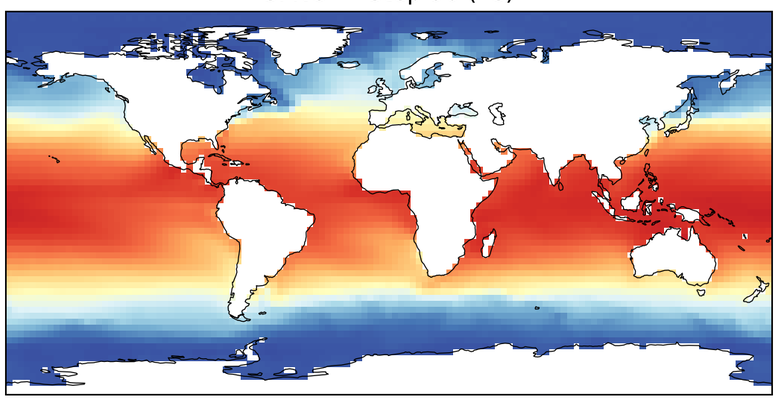
Error — Step 46 (°C)



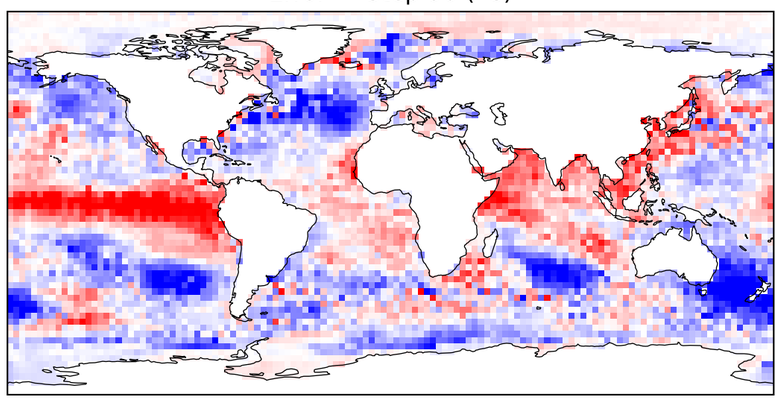
Actual — Step 90 (°C)

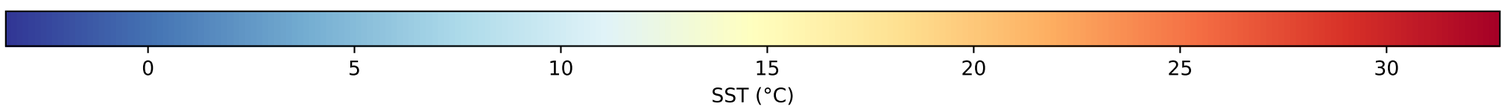


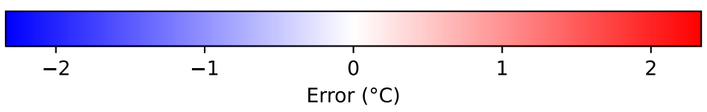
SFNO — Step 90 (°C)



Error — Step 90 (°C)

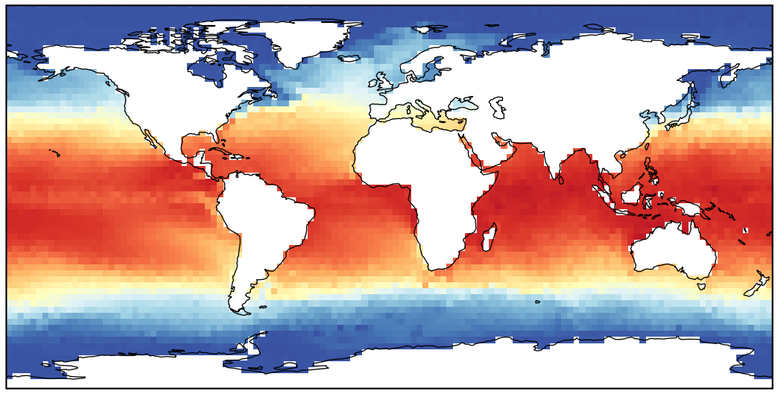




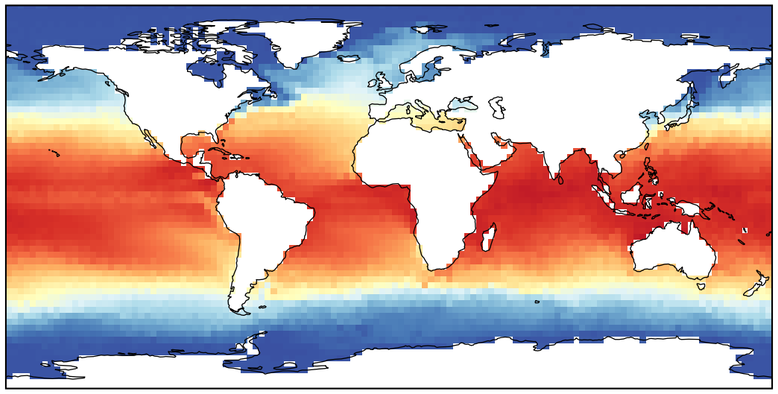


Chain init: 2018-03-31 — Validation (°C)

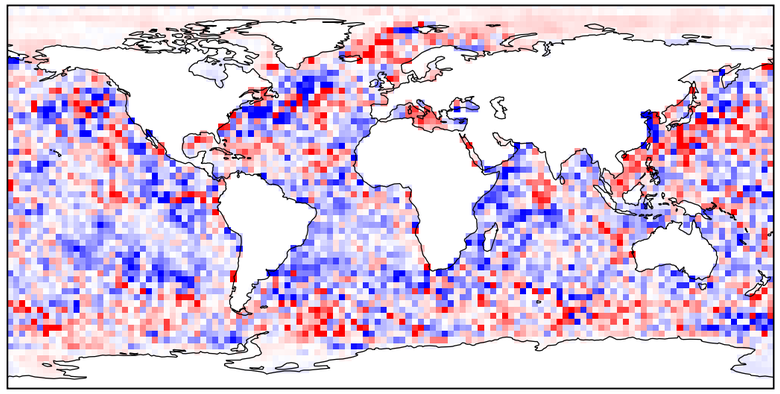
Actual — Step 1 (°C)



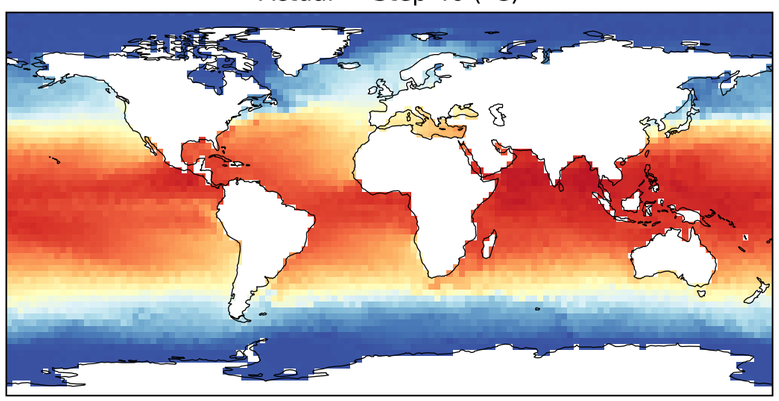
SFNO — Step 1 (°C)



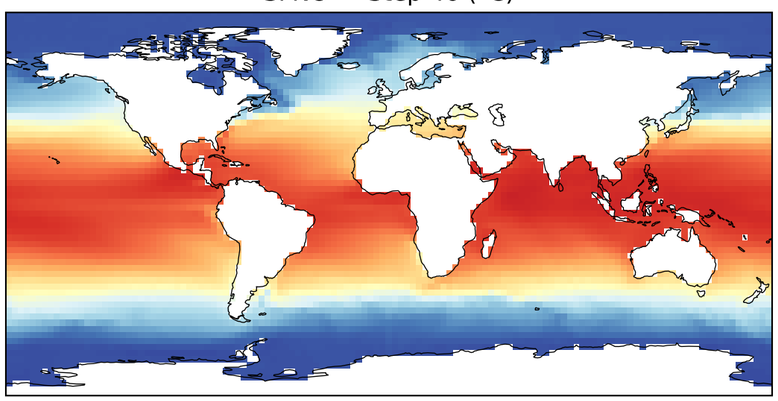
Error — Step 1 (°C)



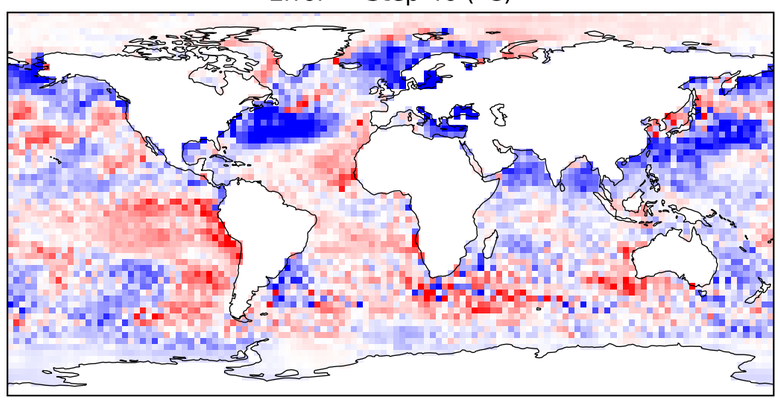
Actual — Step 46 (°C)



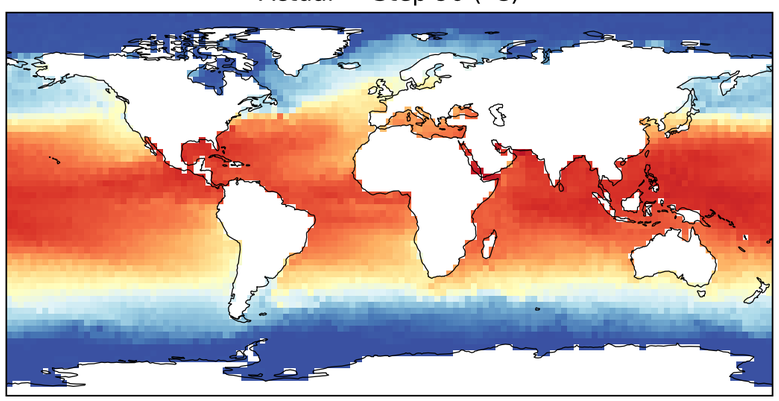
SFNO — Step 46 (°C)



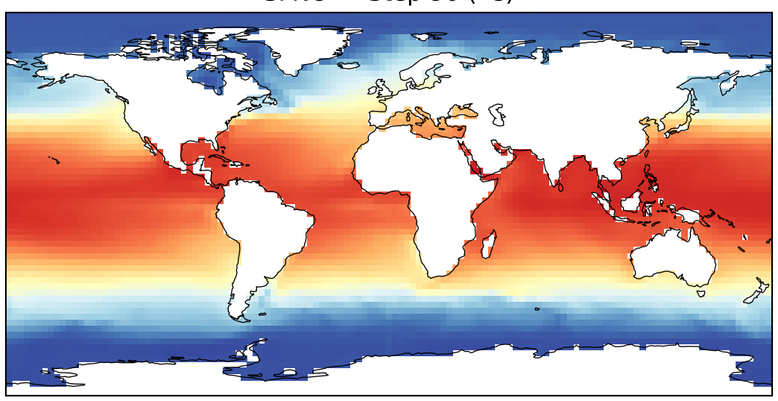
Error — Step 46 (°C)



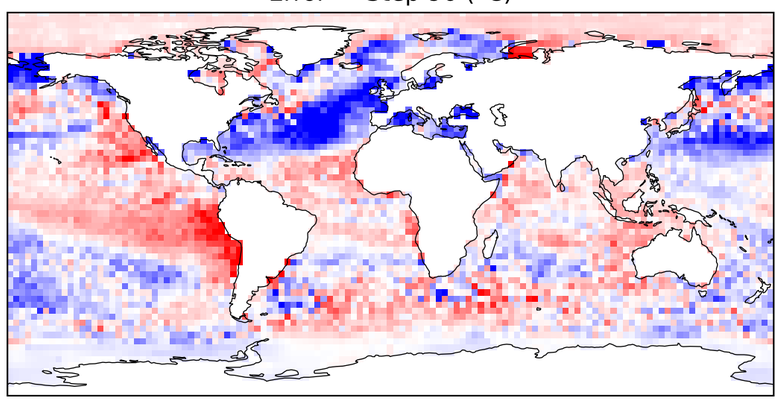
Actual — Step 90 (°C)

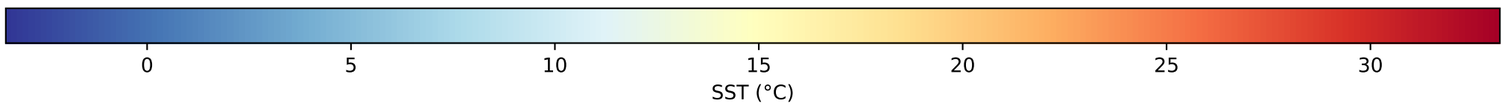


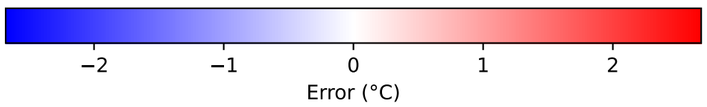
SFNO — Step 90 (°C)



Error — Step 90 (°C)

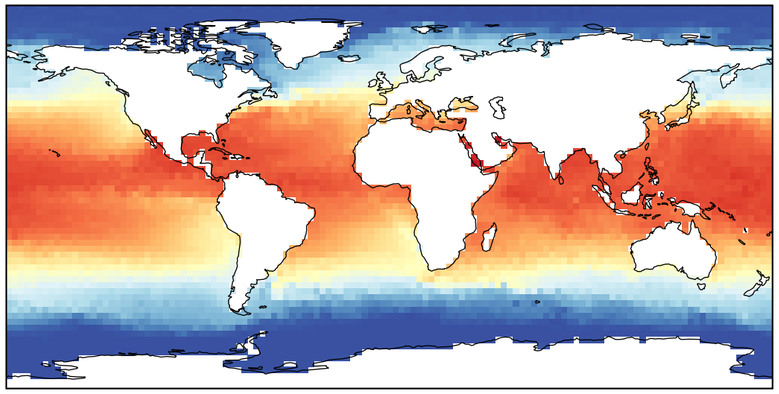




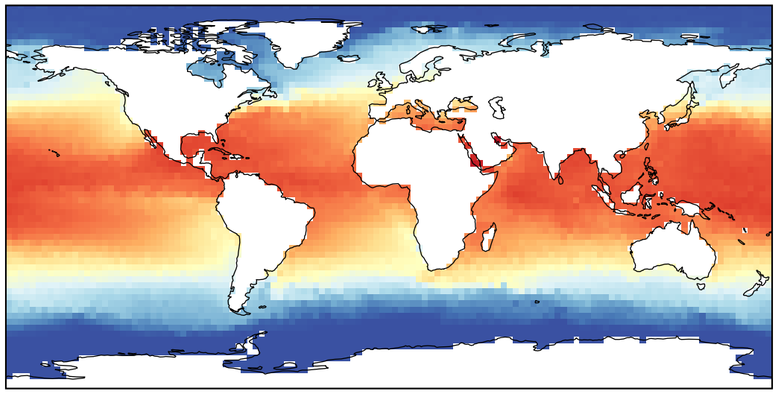


Chain init: 2018-09-27 — Validation (°C)

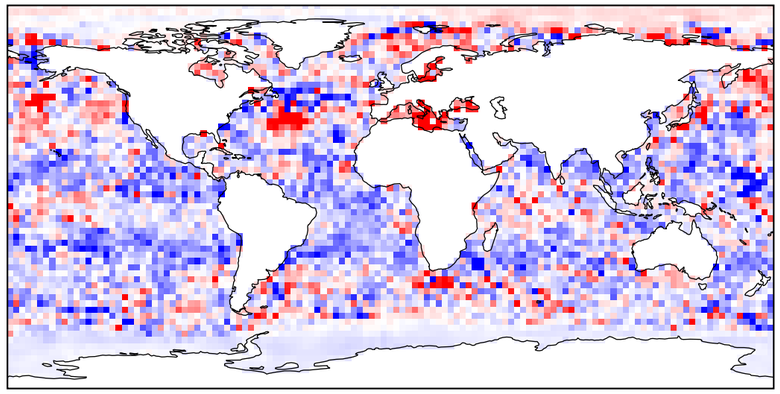
Actual — Step 1 (°C)



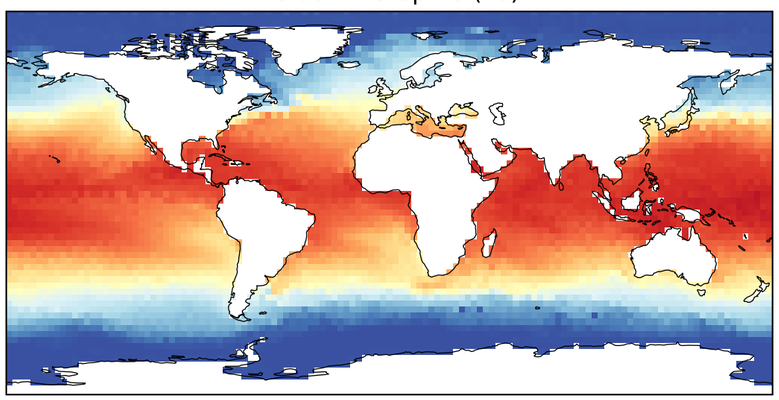
SFNO — Step 1 (°C)



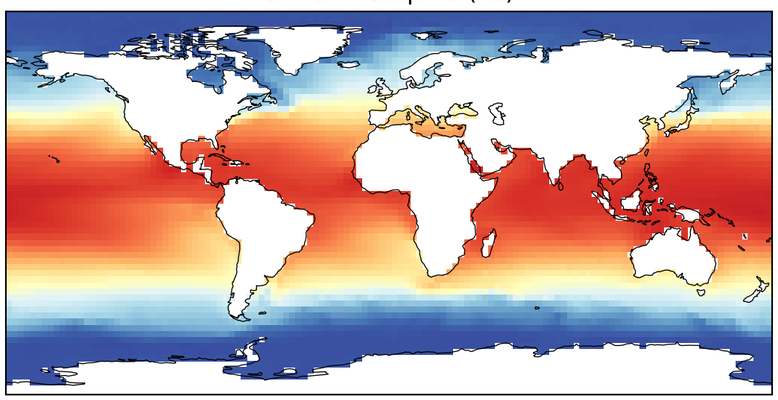
Error — Step 1 (°C)



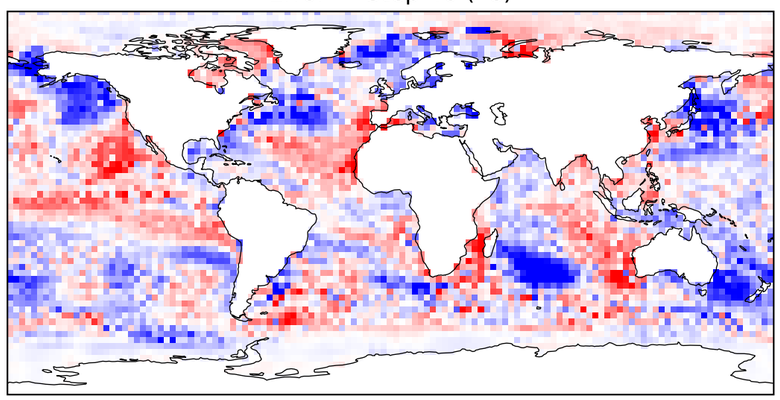
Actual — Step 46 (°C)



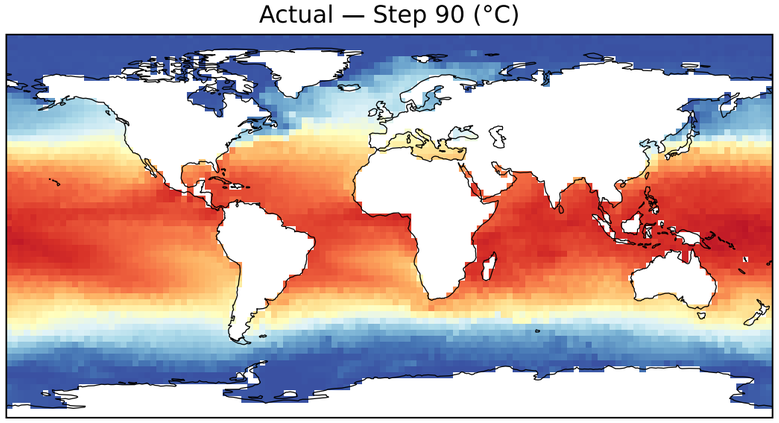
SFNO — Step 46 (°C)



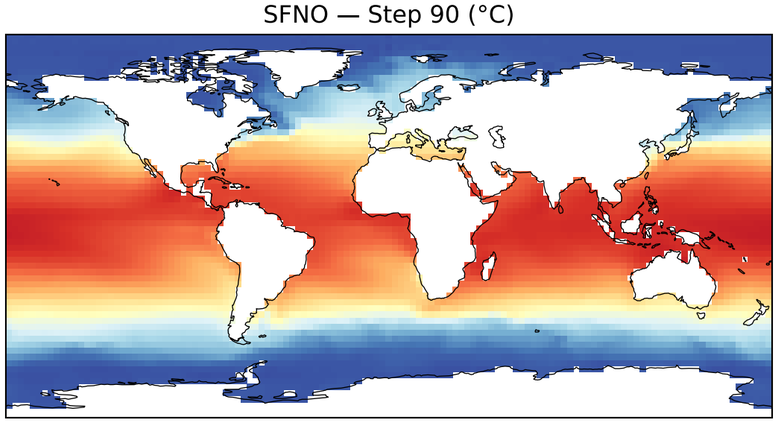
Error — Step 46 (°C)



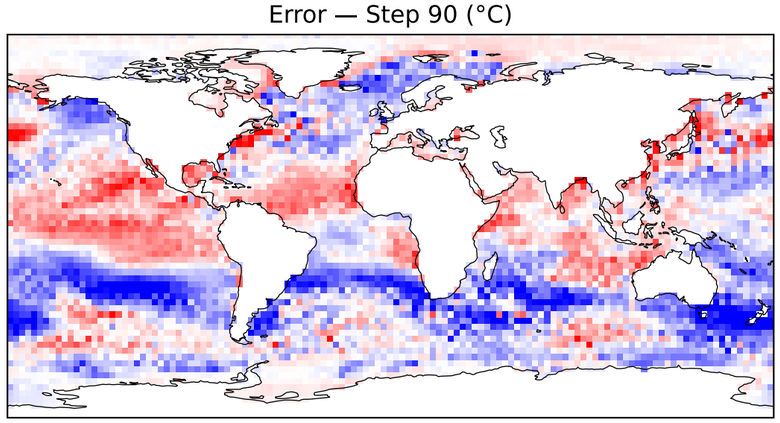
Actual — Step 90 (°C)

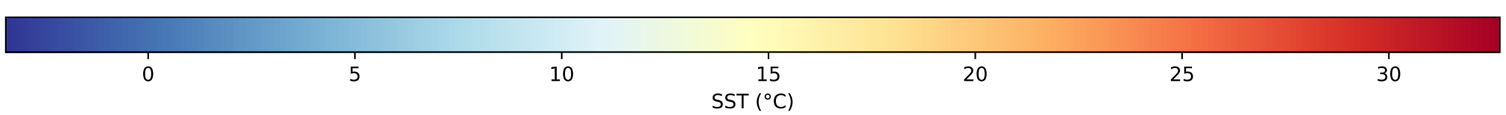


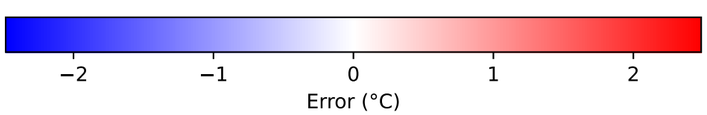
SFNO — Step 90 (°C)



Error — Step 90 (°C)

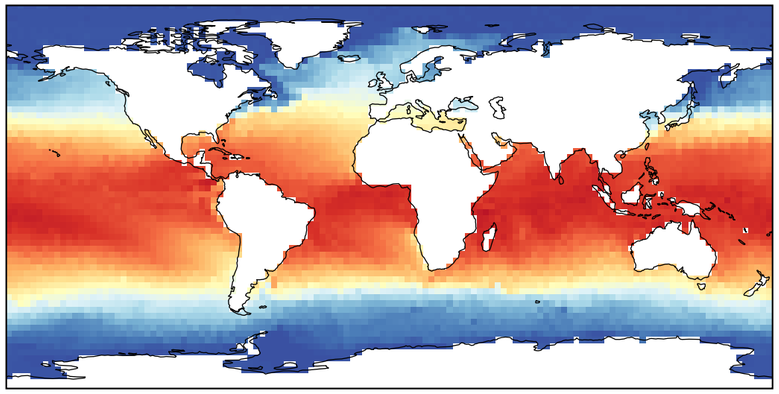




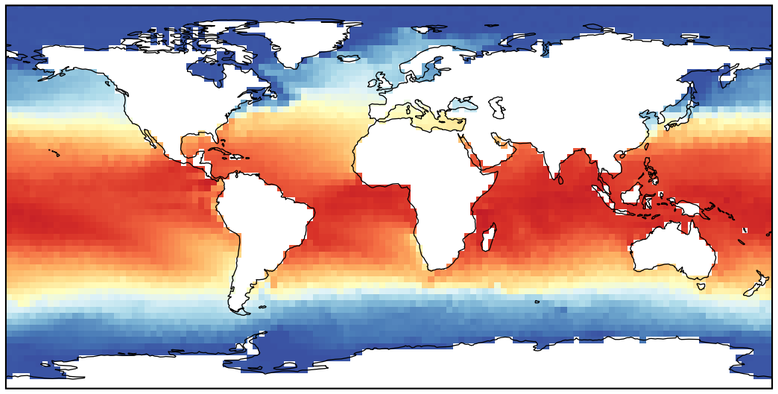


Chain init: 2019-03-26 — Validation (°C)

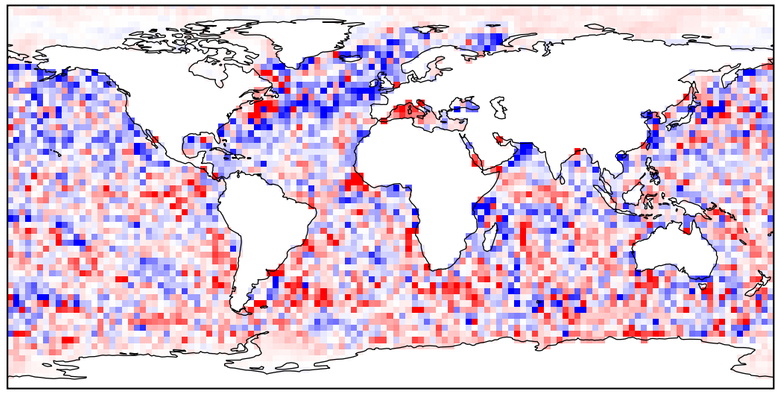
Actual — Step 1 (°C)



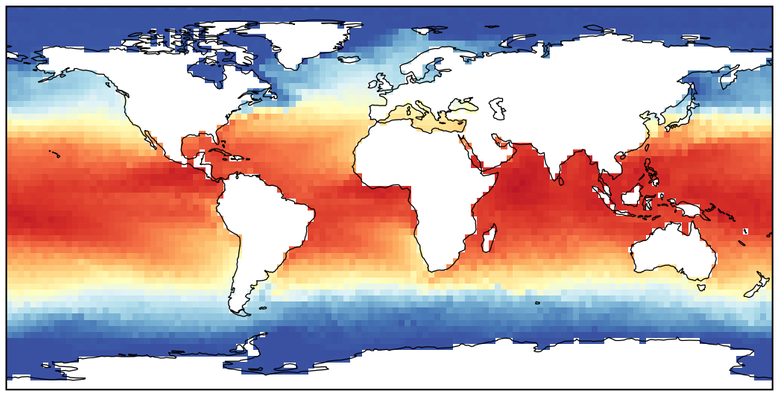
SFNO — Step 1 (°C)



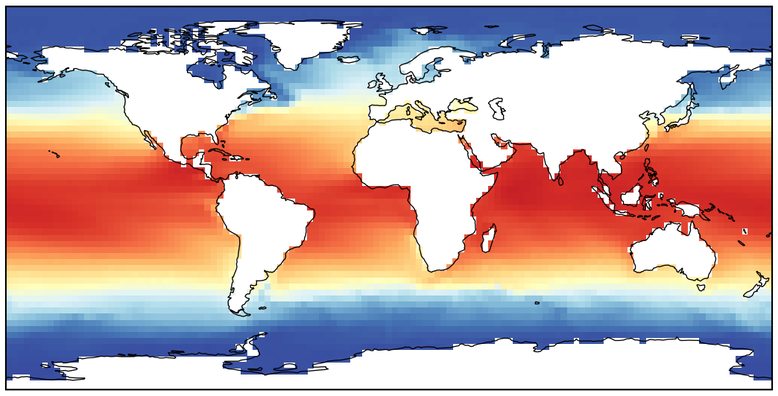
Error — Step 1 (°C)



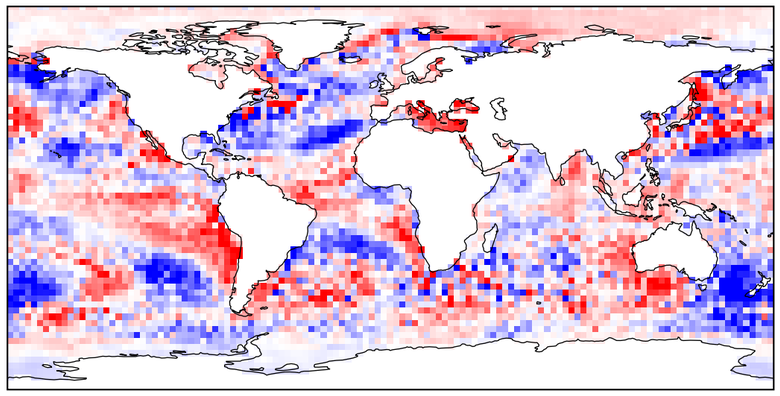
Actual — Step 46 (°C)



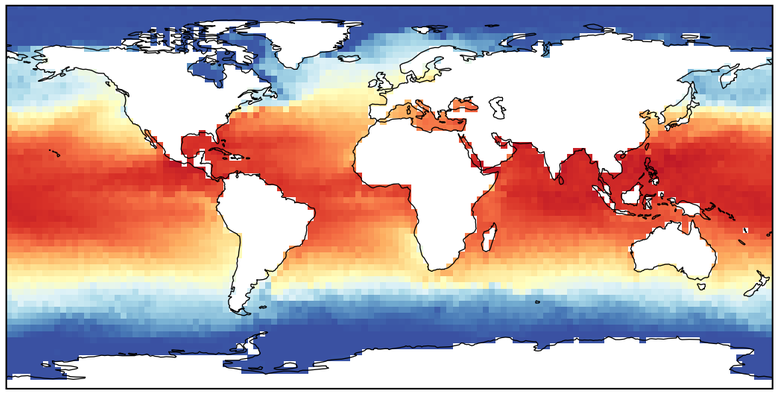
SFNO — Step 46 (°C)



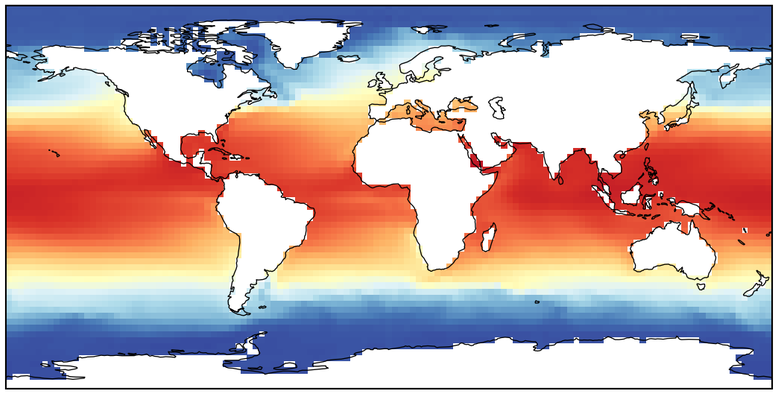
Error — Step 46 (°C)



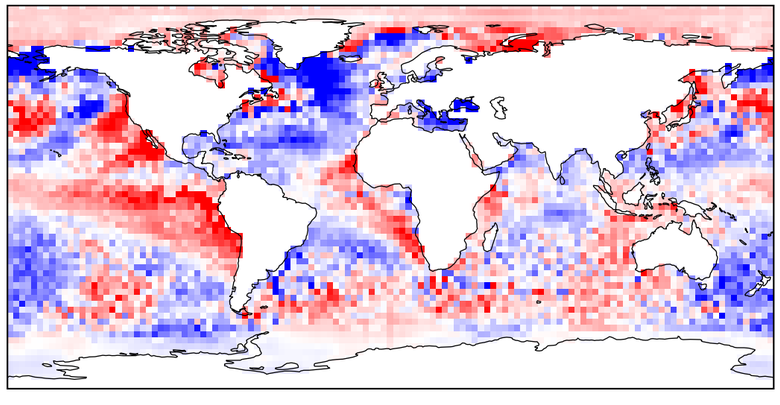
Actual — Step 90 (°C)

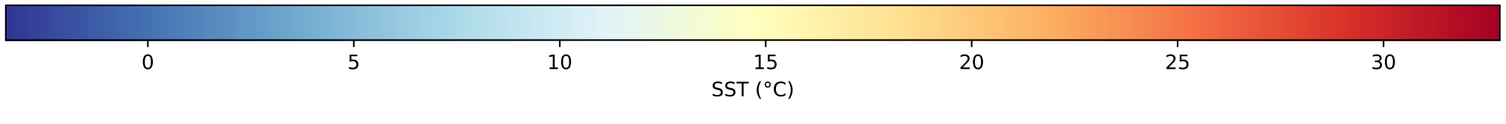


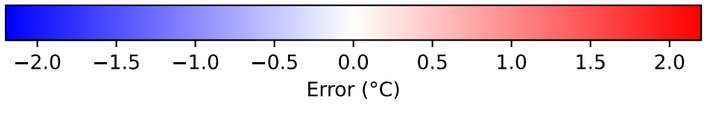
SFNO — Step 90 (°C)



Error — Step 90 (°C)

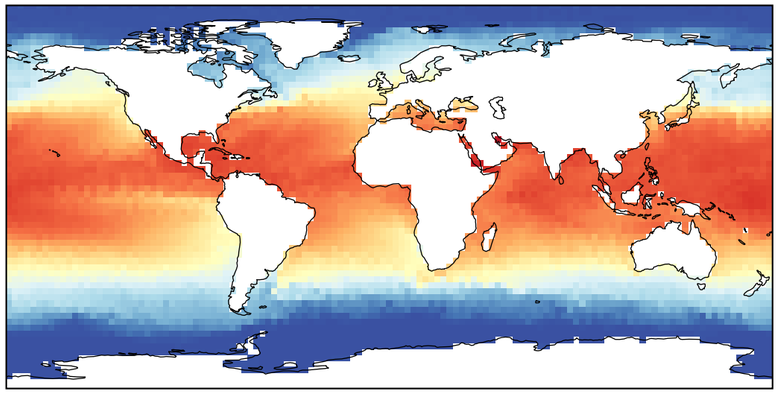




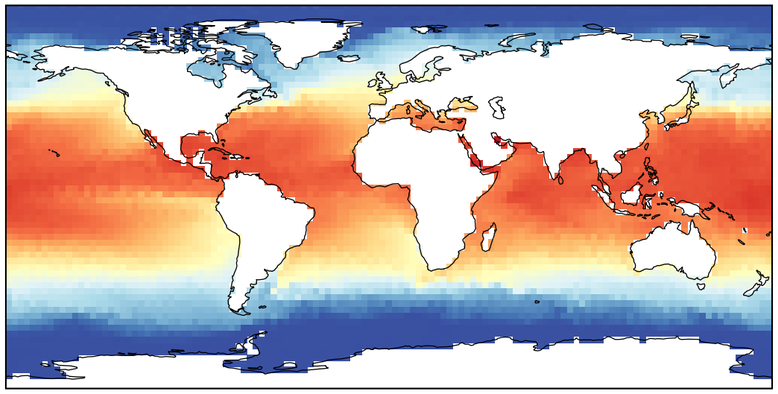


Chain init: 2019-09-22 — Validation (°C)

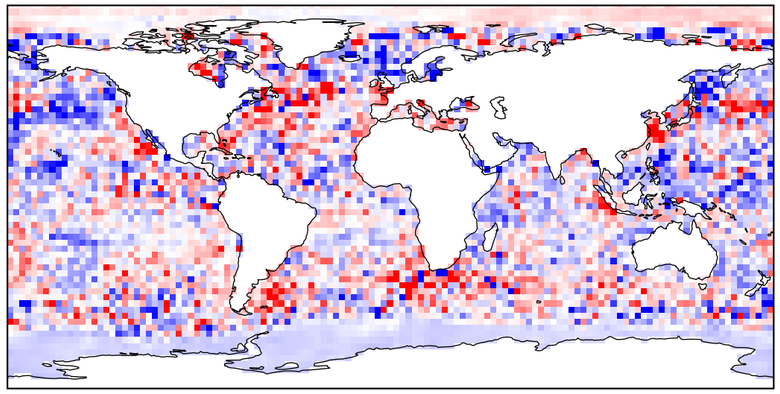
Actual — Step 1 (°C)



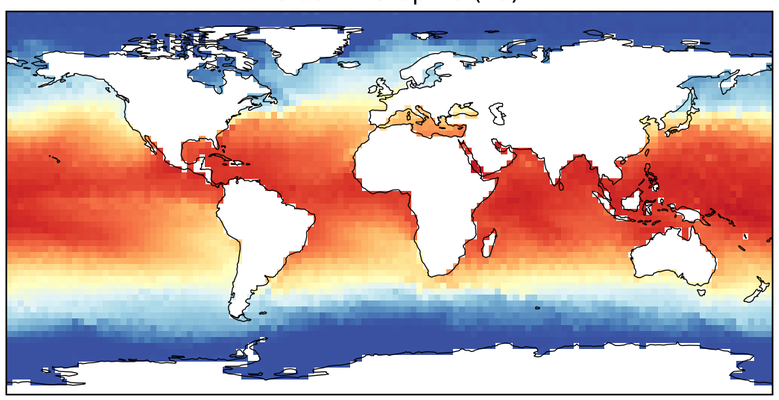
SFNO — Step 1 (°C)



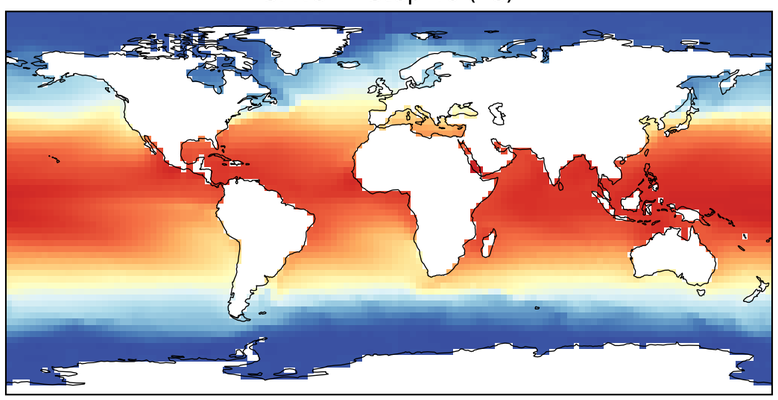
Error — Step 1 (°C)



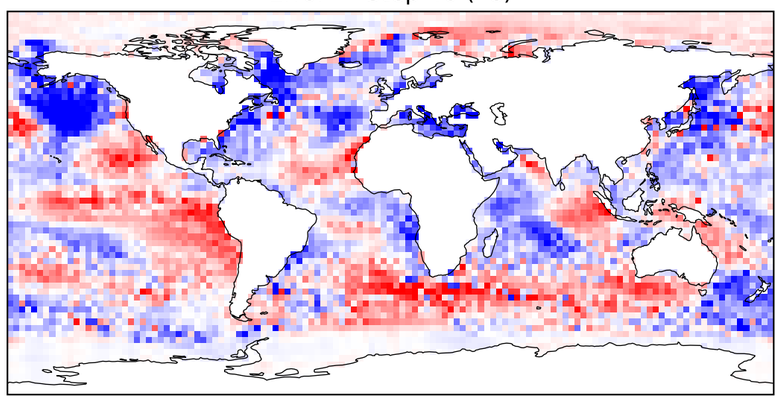
Actual — Step 46 (°C)



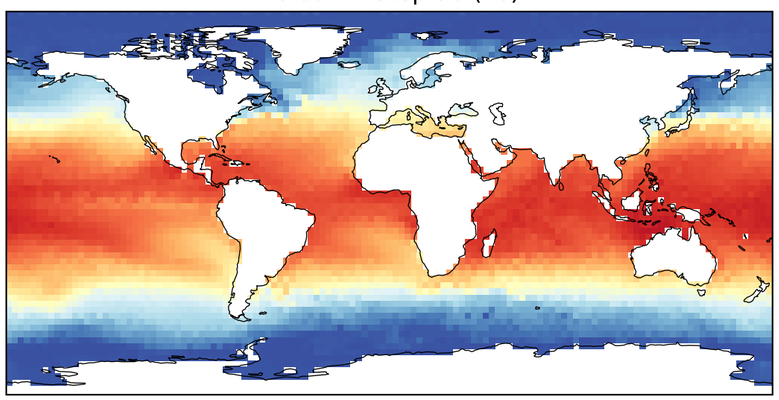
SFNO — Step 46 (°C)



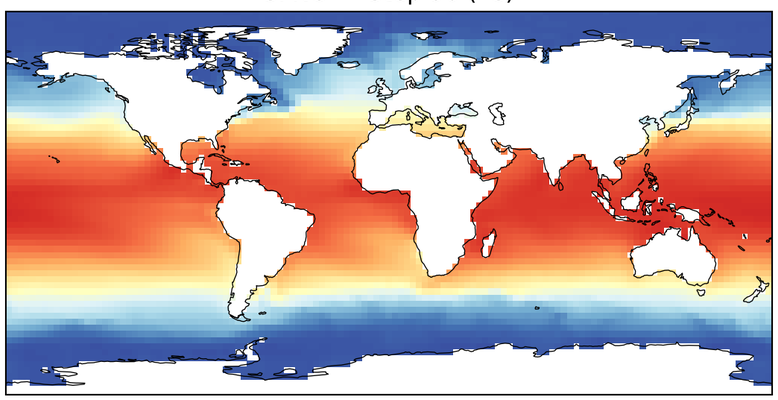
Error — Step 46 (°C)



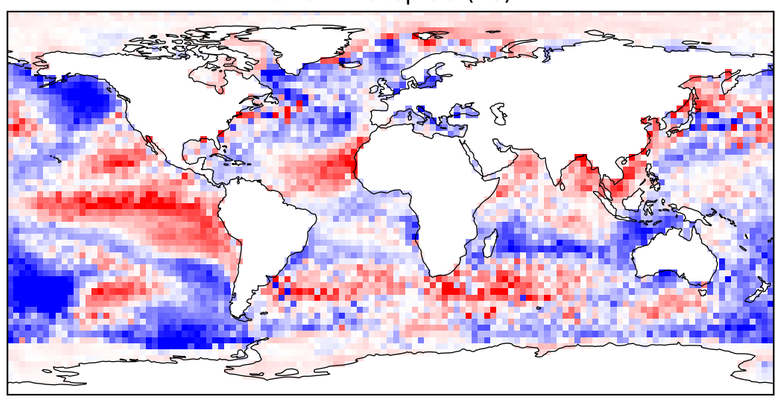
Actual — Step 90 (°C)

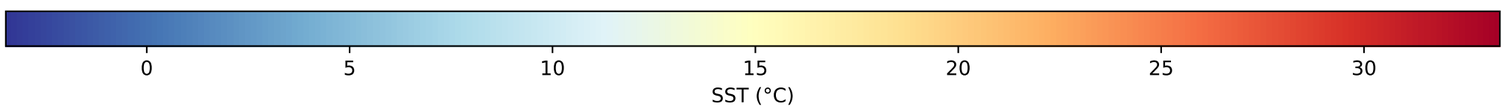


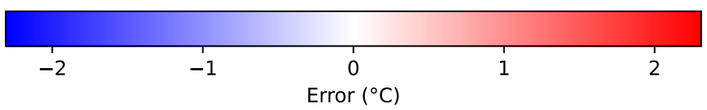
SFNO — Step 90 (°C)



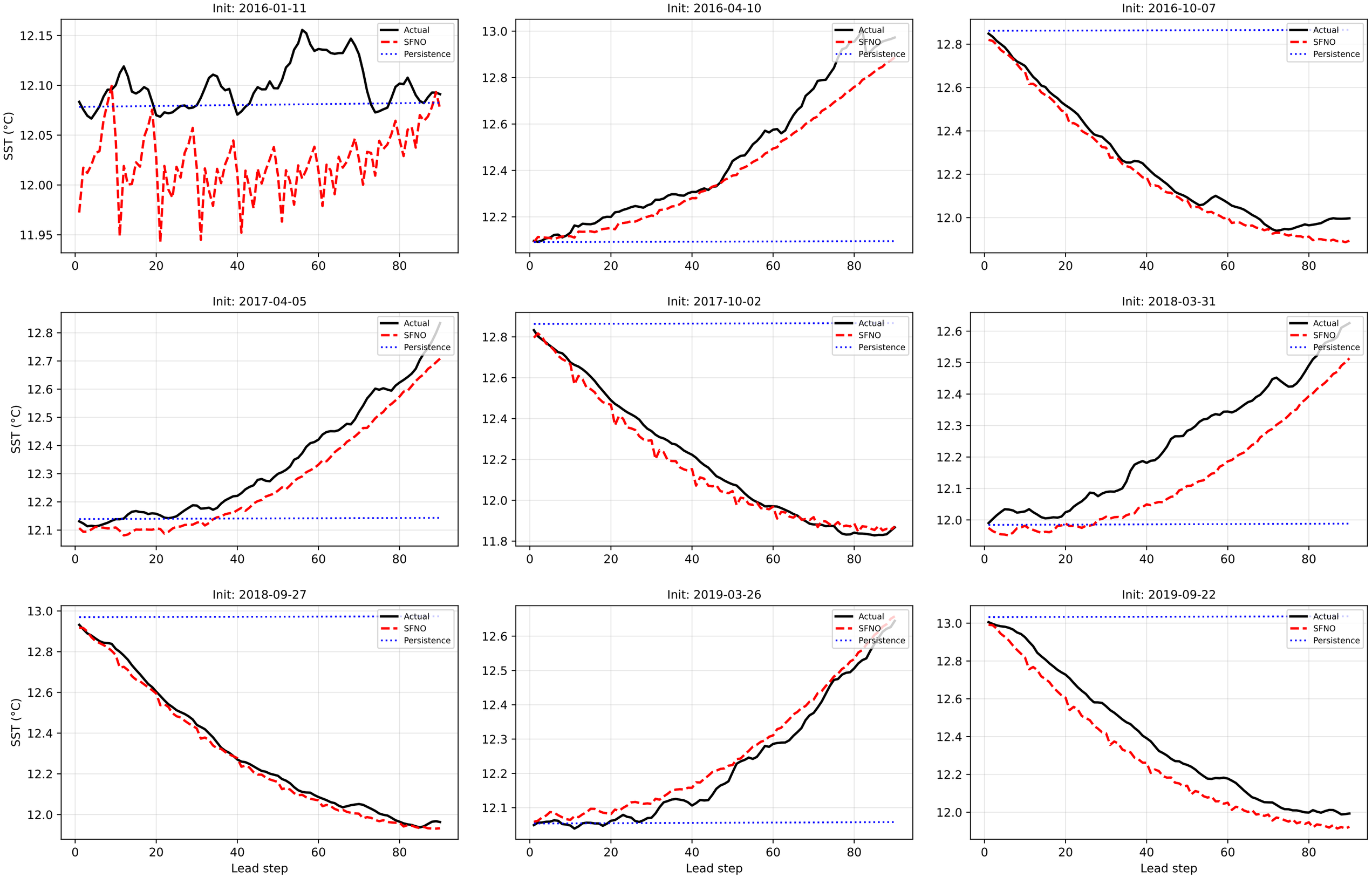
Error — Step 90 (°C)







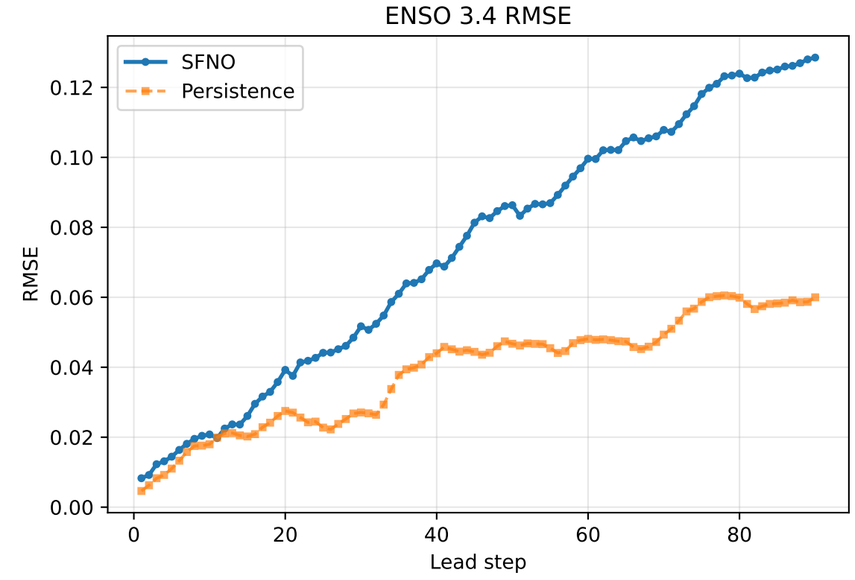
Global-Mean SST — Validation (°C)

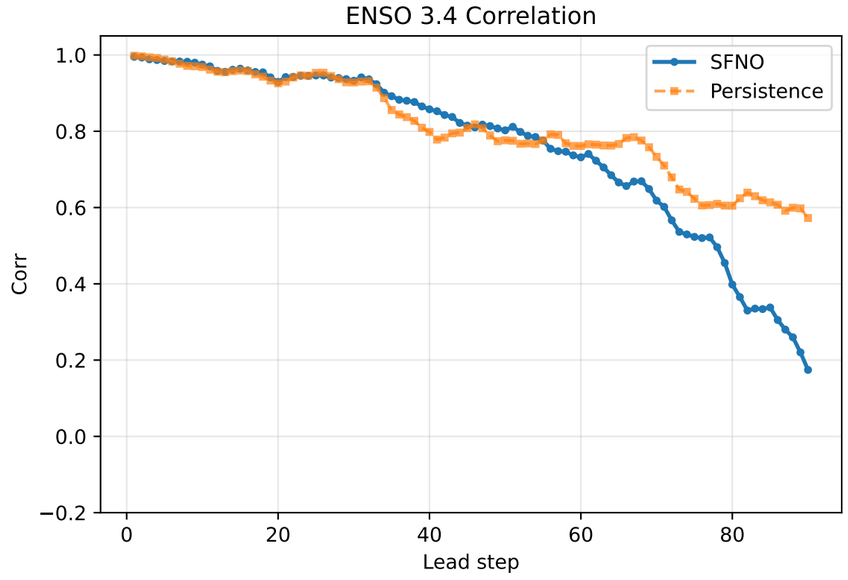


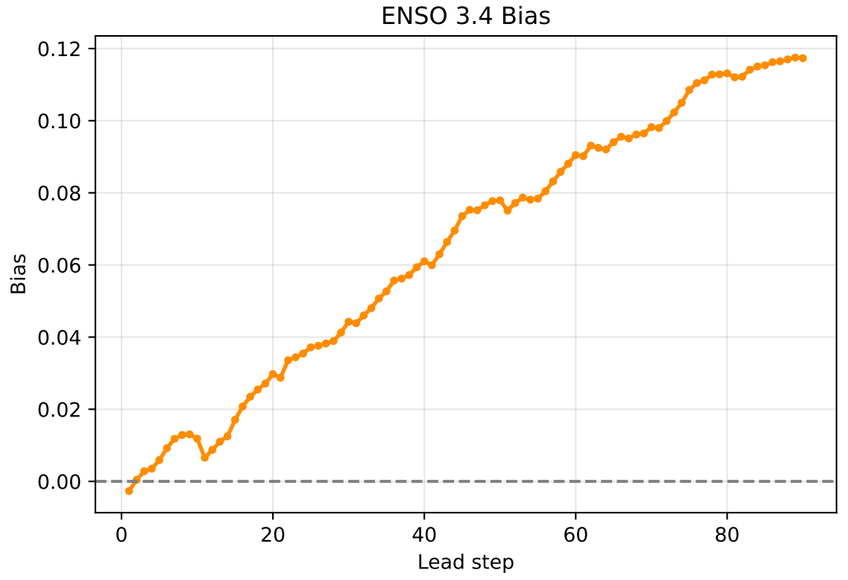
ENSO 3.4 Metrics — Validation

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0083	0.0046	0.9952	0.9981	-0.0027
2	0.0092	0.0063	0.9933	0.9964	0.0004
3	0.0123	0.0083	0.9886	0.9938	0.0028
4	0.0131	0.0092	0.9868	0.9920	0.0035
5	0.0144	0.0110	0.9844	0.9881	0.0059
6	0.0164	0.0133	0.9830	0.9833	0.0092
7	0.0181	0.0158	0.9829	0.9771	0.0118
8	0.0195	0.0175	0.9823	0.9714	0.0129
9	0.0204	0.0176	0.9799	0.9705	0.0131
10	0.0208	0.0180	0.9748	0.9684	0.0118
11	0.0198	0.0199	0.9702	0.9615	0.0066
12	0.0225	0.0211	0.9583	0.9562	0.0087
13	0.0237	0.0212	0.9559	0.9548	0.0110
14	0.0236	0.0205	0.9613	0.9576	0.0125
15	0.0261	0.0202	0.9641	0.9592	0.0171
16	0.0295	0.0209	0.9595	0.9576	0.0208
17	0.0316	0.0228	0.9555	0.9494	0.0235
18	0.0330	0.0241	0.9545	0.9431	0.0255
19	0.0358	0.0261	0.9411	0.9328	0.0271
20	0.0392	0.0275	0.9294	0.9257	0.0298
21	0.0375	0.0270	0.9420	0.9303	0.0287
22	0.0414	0.0256	0.9426	0.9410	0.0336
23	0.0418	0.0243	0.9455	0.9466	0.0344
24	0.0427	0.0245	0.9451	0.9458	0.0355
25	0.0441	0.0227	0.9463	0.9531	0.0372
26	0.0442	0.0222	0.9462	0.9533	0.0376
27	0.0452	0.0238	0.9408	0.9448	0.0382
28	0.0461	0.0252	0.9397	0.9366	0.0389
29	0.0485	0.0268	0.9368	0.9284	0.0413
30	0.0517	0.0271	0.9324	0.9280	0.0443
31	0.0507	0.0268	0.9413	0.9302	0.0438
32	0.0524	0.0264	0.9362	0.9312	0.0460
33	0.0548	0.0293	0.9235	0.9144	0.0480
34	0.0586	0.0338	0.9012	0.8869	0.0507
35	0.0610	0.0378	0.8920	0.8557	0.0527
36	0.0640	0.0394	0.8826	0.8439	0.0557
37	0.0641	0.0399	0.8800	0.8370	0.0562
38	0.0651	0.0408	0.8768	0.8272	0.0572
39	0.0678	0.0429	0.8651	0.8091	0.0594
40	0.0697	0.0440	0.8576	0.7981	0.0610
41	0.0688	0.0458	0.8526	0.7782	0.0599
42	0.0712	0.0451	0.8426	0.7835	0.0630
43	0.0744	0.0445	0.8375	0.7945	0.0664
44	0.0776	0.0449	0.8219	0.7965	0.0695
45	0.0813	0.0444	0.8148	0.8085	0.0735
46	0.0831	0.0436	0.8100	0.8178	0.0753
47	0.0826	0.0441	0.8174	0.8079	0.0752
48	0.0846	0.0460	0.8136	0.7888	0.0766
49	0.0861	0.0474	0.8074	0.7738	0.0777
50	0.0863	0.0467	0.8025	0.7763	0.0779
51	0.0833	0.0462	0.8117	0.7751	0.0751
52	0.0853	0.0468	0.7983	0.7672	0.0772
53	0.0867	0.0467	0.7880	0.7680	0.0787
54	0.0866	0.0466	0.7849	0.7665	0.0781
55	0.0869	0.0454	0.7756	0.7763	0.0784
56	0.0893	0.0441	0.7541	0.7919	0.0804
57	0.0919	0.0446	0.7478	0.7902	0.0832
58	0.0945	0.0469	0.7464	0.7685	0.0858
59	0.0969	0.0477	0.7368	0.7616	0.0881
60	0.0996	0.0481	0.7317	0.7614	0.0905
61	0.0995	0.0478	0.7406	0.7659	0.0902
62	0.1020	0.0480	0.7228	0.7649	0.0931
63	0.1021	0.0477	0.7051	0.7628	0.0925
64	0.1021	0.0474	0.6848	0.7625	0.0920
65	0.1046	0.0473	0.6658	0.7666	0.0941
66	0.1057	0.0458	0.6564	0.7823	0.0955
67	0.1047	0.0452	0.6682	0.7845	0.0951
68	0.1055	0.0459	0.6690	0.7758	0.0962
69	0.1060	0.0472	0.6486	0.7581	0.0965
70	0.1078	0.0493	0.6182	0.7330	0.0982
71	0.1073	0.0510	0.6018	0.7100	0.0980
72	0.1095	0.0533	0.5665	0.6791	0.0999
73	0.1123	0.0559	0.5361	0.6473	0.1023
74	0.1146	0.0567	0.5296	0.6412	0.1050
75	0.1181	0.0587	0.5231	0.6229	0.1085
76	0.1199	0.0600	0.5200	0.6055	0.1104
77	0.1210	0.0603	0.5217	0.6065	0.1112
78	0.1232	0.0605	0.4962	0.6097	0.1128
79	0.1234	0.0603	0.4547	0.6050	0.1128
80	0.1239	0.0599	0.3981	0.6048	0.1131
81	0.1227	0.0581	0.3653	0.6243	0.1120
82	0.1228	0.0566	0.3299	0.6391	0.1122
83	0.1242	0.0574	0.3351	0.6298	0.1141
84	0.1248	0.0581	0.3338	0.6190	0.1150
85	0.1251	0.0583	0.3380	0.6136	0.1154
86	0.1260	0.0585	0.3052	0.6072	0.1162
87	0.1262	0.0591	0.2800	0.5917	0.1165
88	0.1269	0.0586	0.2598	0.5994	0.1170
89	0.1280	0.0587	0.2203	0.5977	0.1175
90	0.1285	0.0600	0.1743	0.5725	0.1173

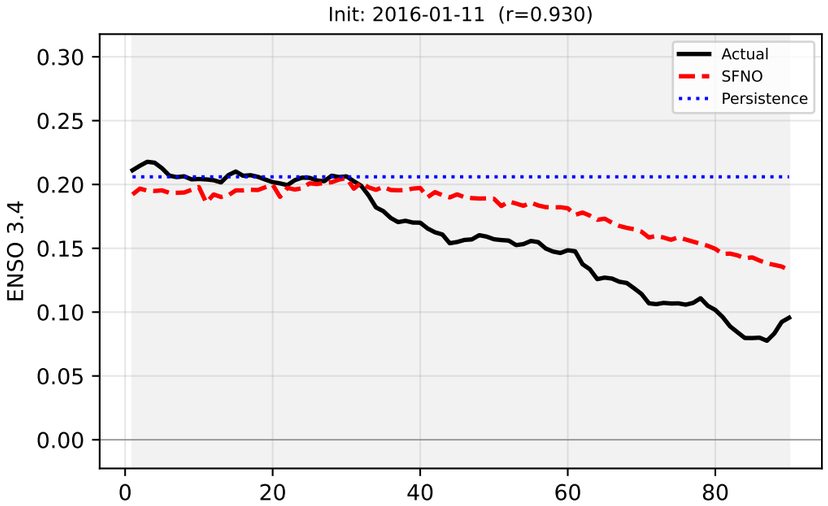
ENSO 3.4 Forecast Skill — Validation

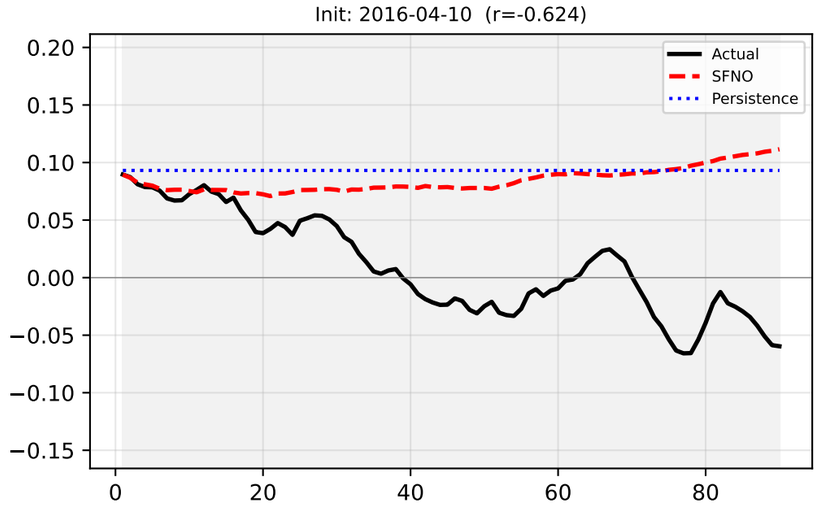


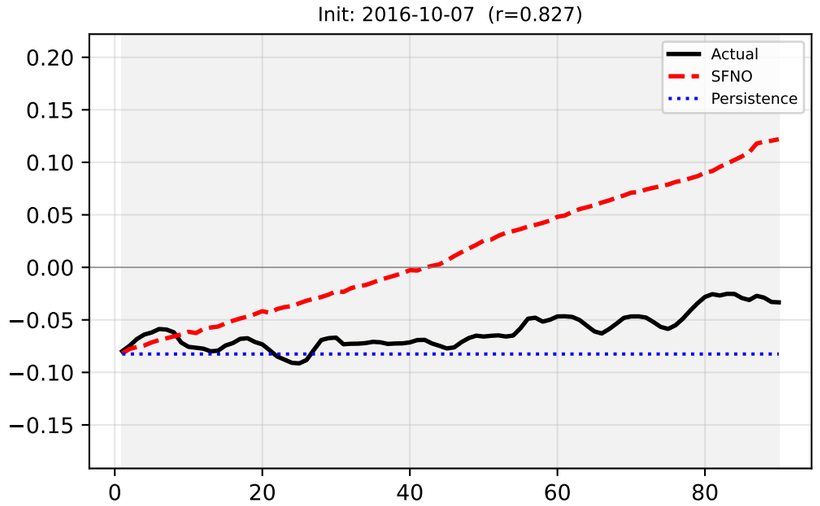


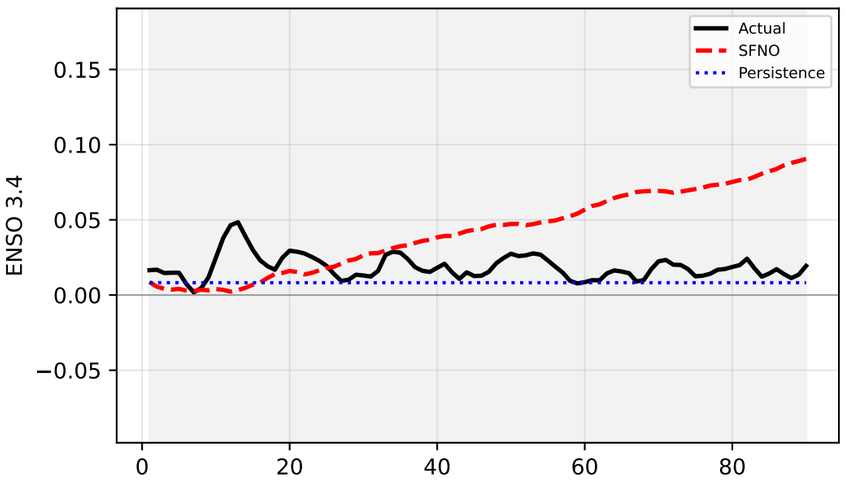


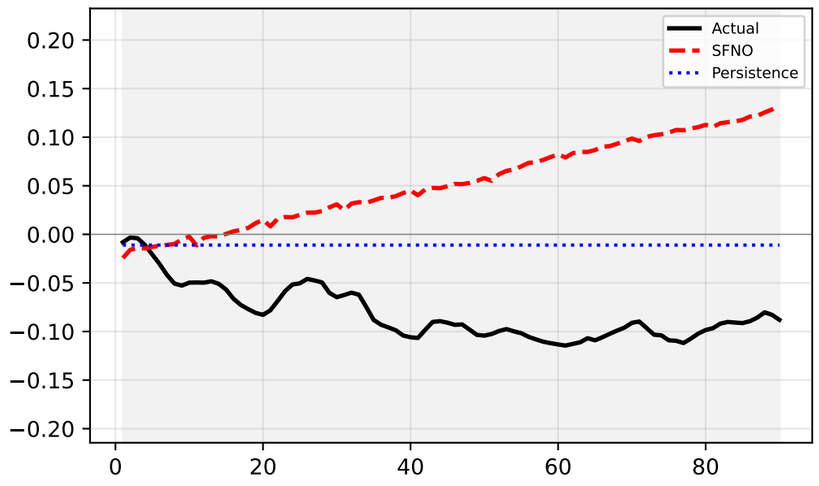
ENSO 3.4 Index — Validation

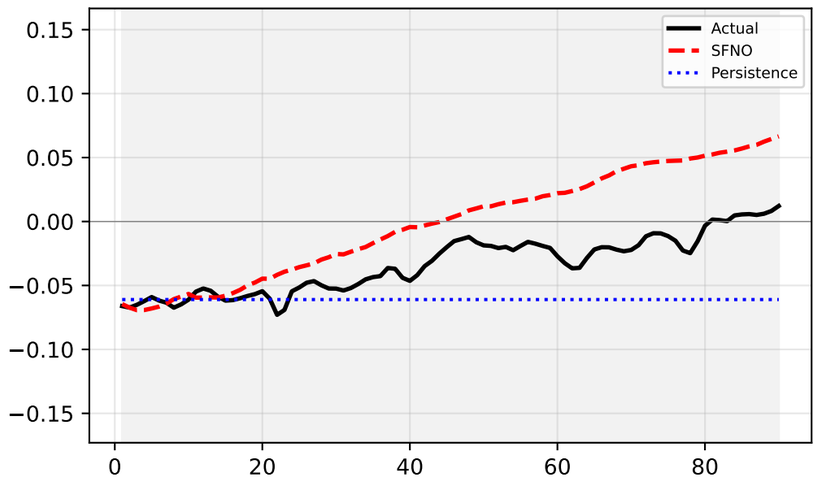


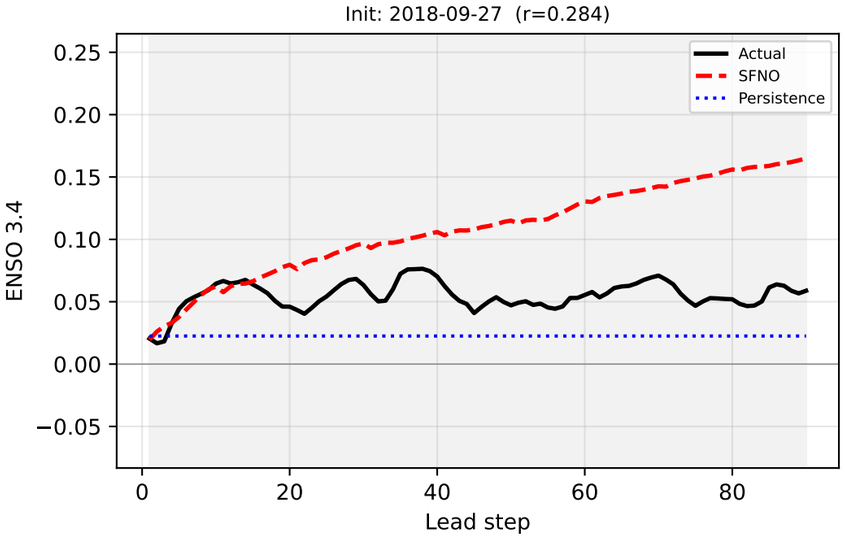


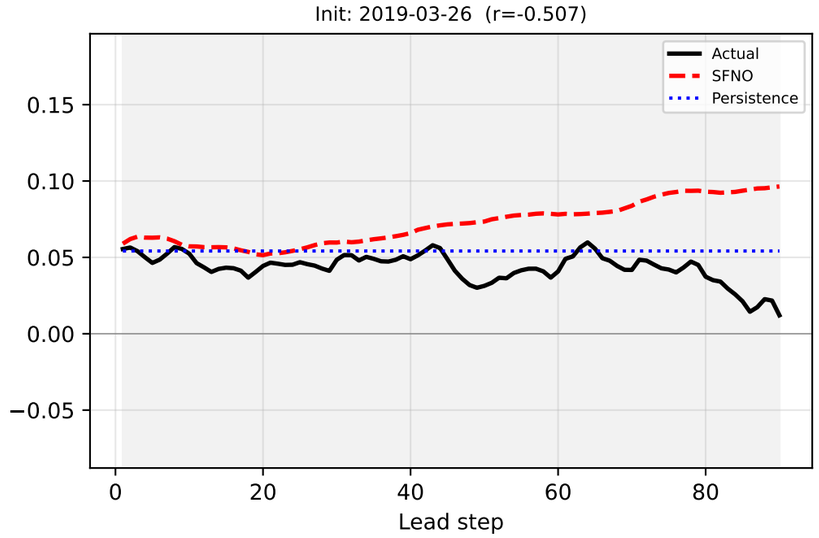


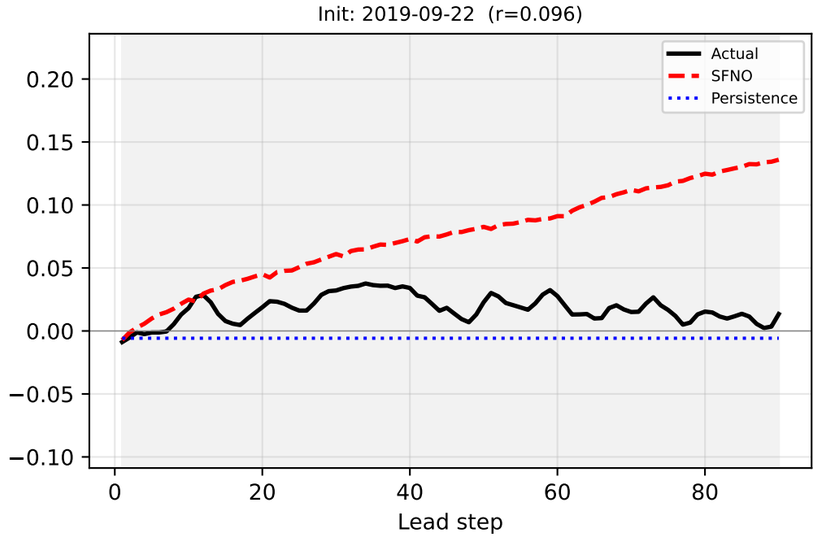




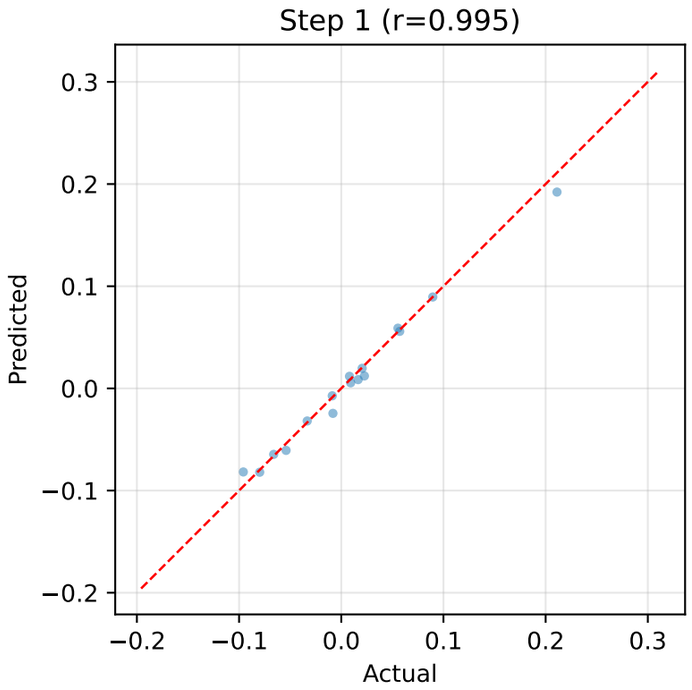


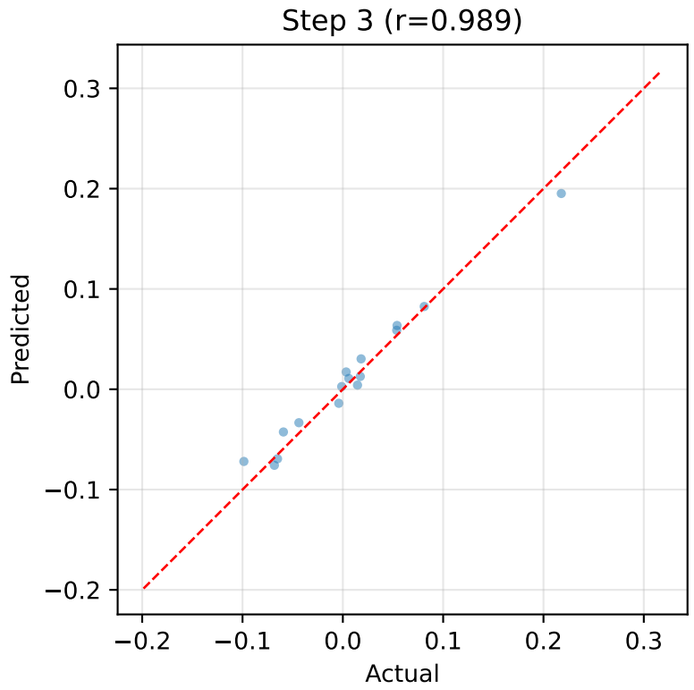


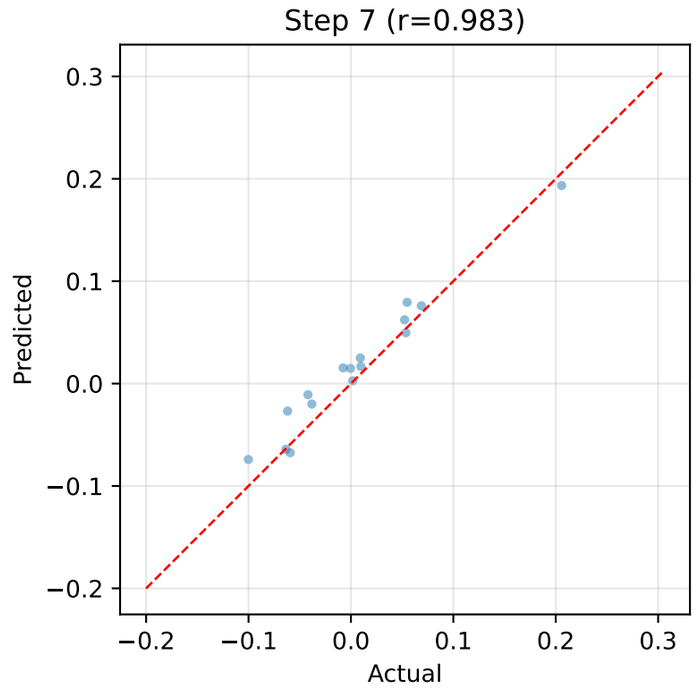


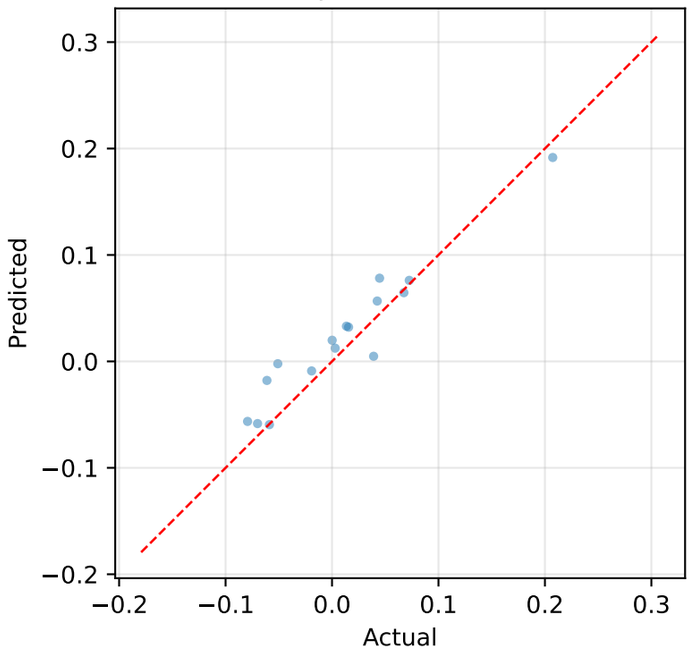


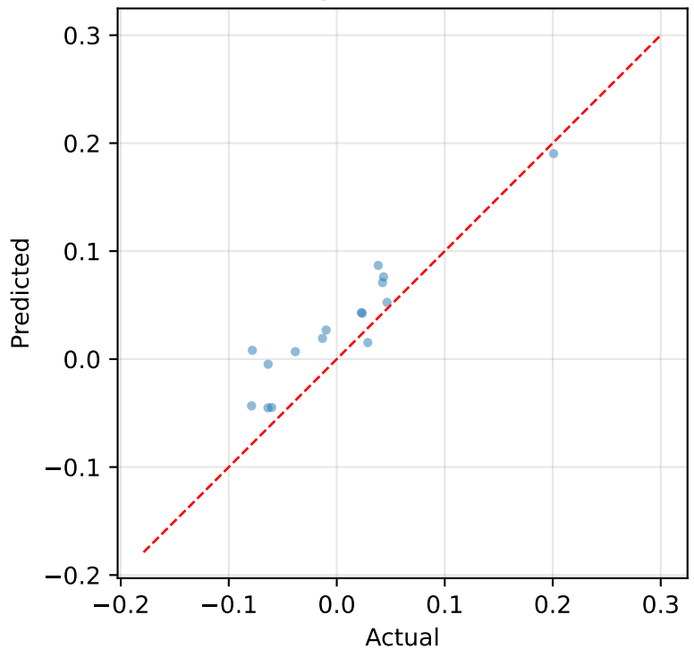
ENSO 3.4 Scatter — Validation

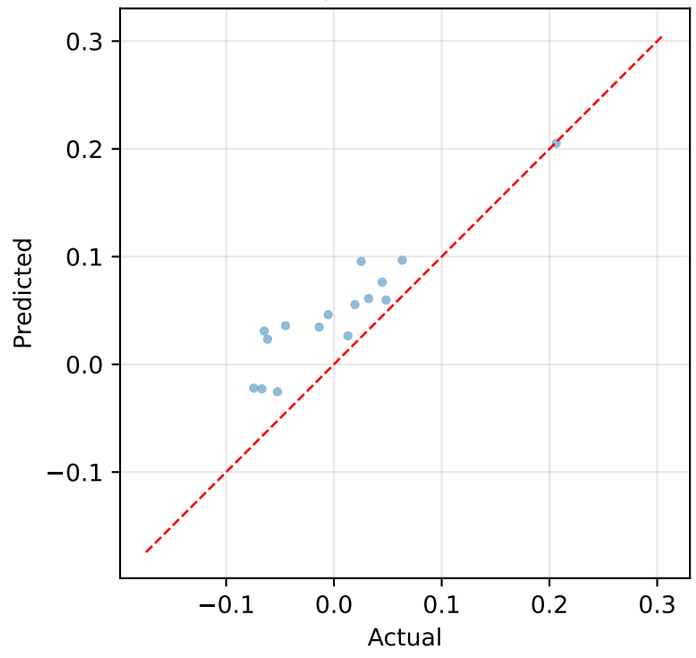


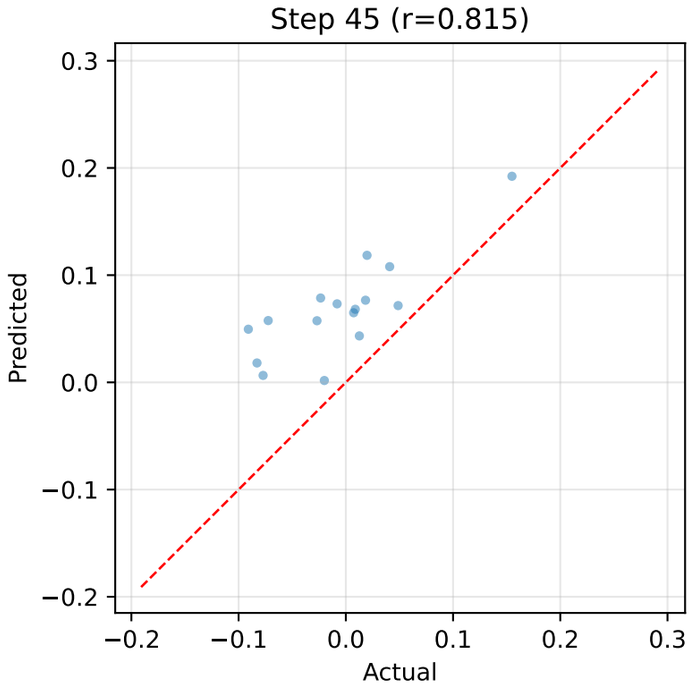


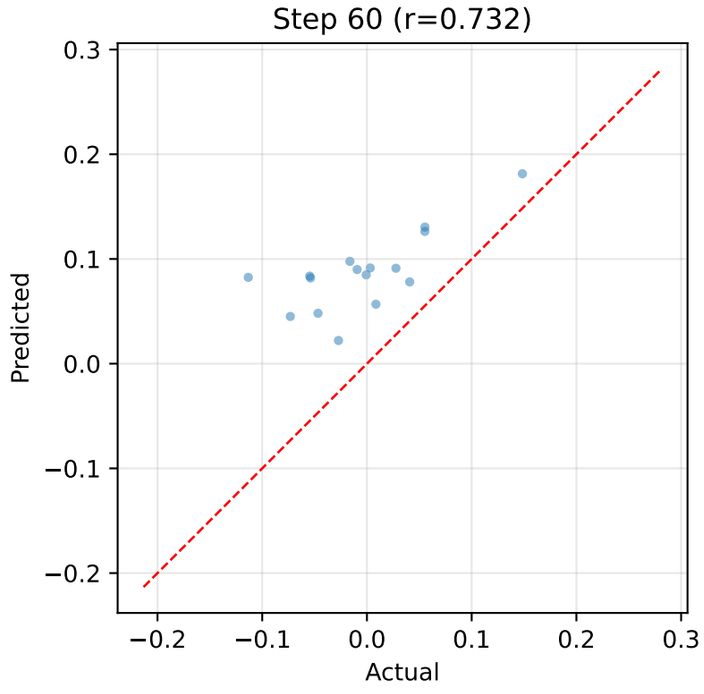


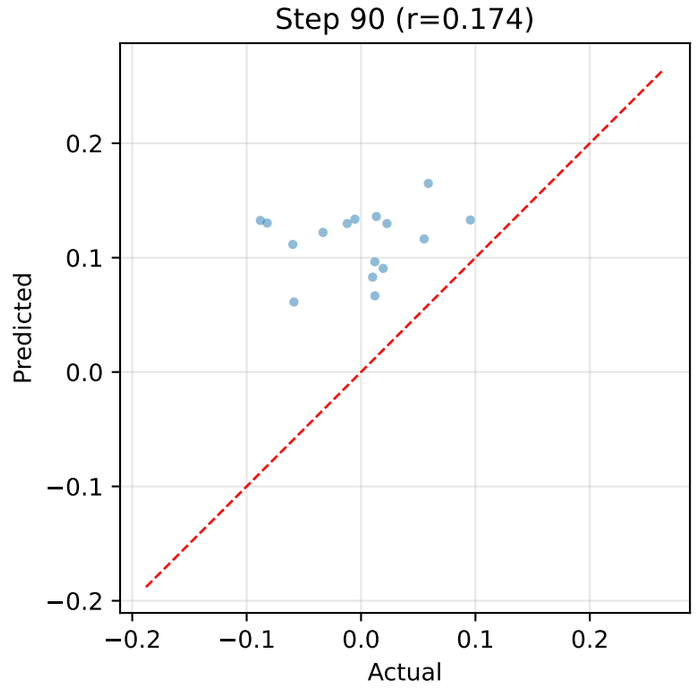








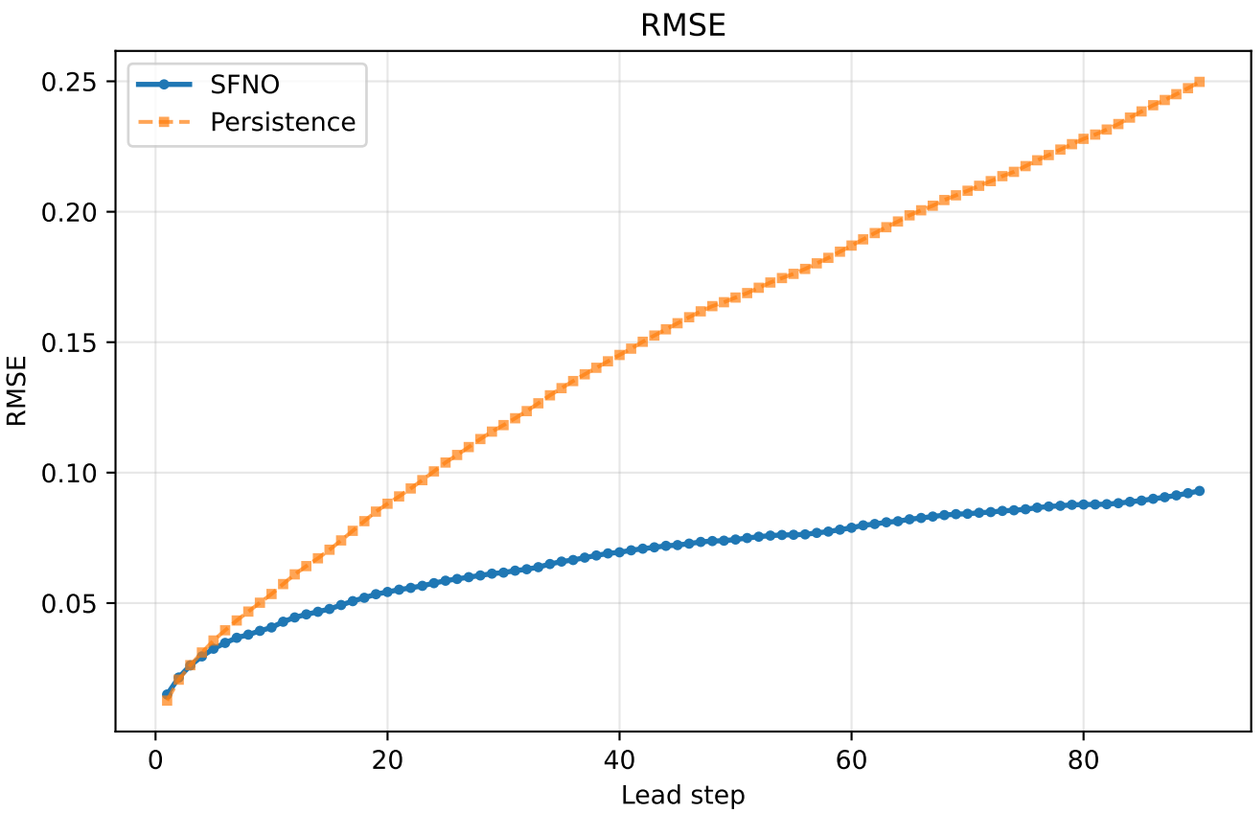


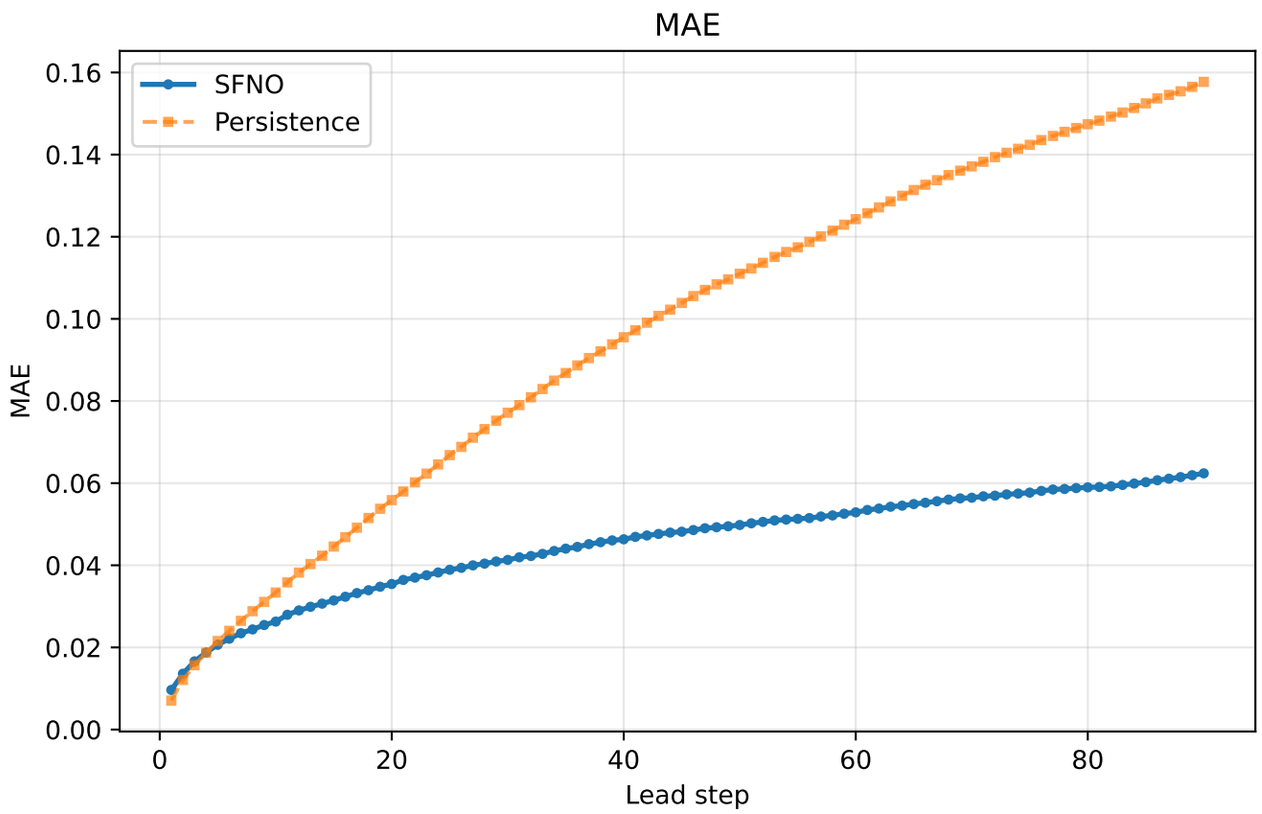


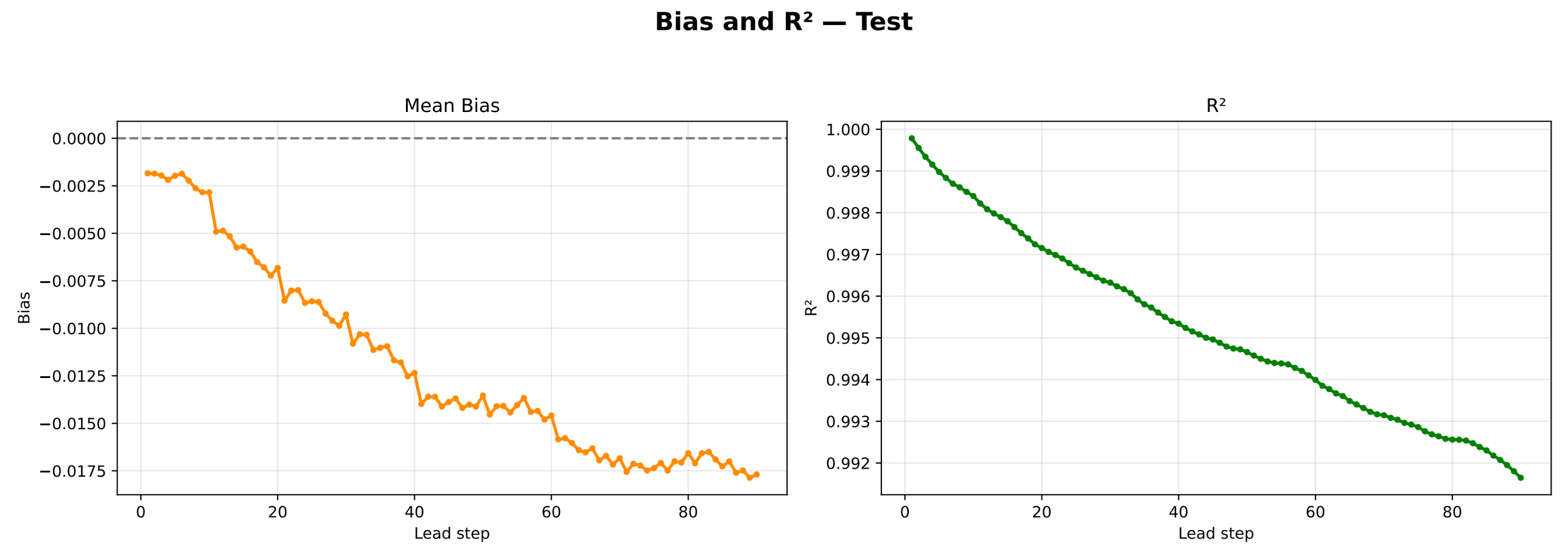
Metrics by Lead Step — Test

Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0150	0.0126	-18.5%	0.0096	-0.0018	0.9740	0.9796	0.9998
2	0.0215	0.0206	-3.9%	0.0136	-0.0019	0.9440	0.9462	0.9996
3	0.0261	0.0262	0.3%	0.0166	-0.0020	0.9141	0.9132	0.9993
4	0.0296	0.0311	4.8%	0.0188	-0.0022	0.8866	0.8780	0.9992
5	0.0325	0.0357	8.8%	0.0207	-0.0020	0.8603	0.8411	0.9990
6	0.0347	0.0396	12.3%	0.0221	-0.0019	0.8371	0.8072	0.9988
7	0.0367	0.0433	15.3%	0.0235	-0.0022	0.8153	0.7728	0.9987
8	0.0379	0.0467	18.9%	0.0244	-0.0026	0.7975	0.7389	0.9986
9	0.0394	0.0501	21.5%	0.0254	-0.0028	0.7802	0.7063	0.9985
10	0.0407	0.0535	24.0%	0.0263	-0.0028	0.7645	0.6743	0.9984
11	0.0429	0.0573	25.2%	0.0279	-0.0049	0.7413	0.6410	0.9982
12	0.0445	0.0610	27.1%	0.0290	-0.0049	0.7206	0.6068	0.9981
13	0.0457	0.0642	28.8%	0.0299	-0.0052	0.7032	0.5769	0.9980
14	0.0467	0.0672	30.5%	0.0307	-0.0057	0.6871	0.5500	0.9979
15	0.0477	0.0705	32.2%	0.0314	-0.0057	0.6715	0.5219	0.9978
16	0.0493	0.0740	33.5%	0.0323	-0.0060	0.6514	0.4938	0.9977
17	0.0507	0.0777	34.7%	0.0332	-0.0065	0.6313	0.4629	0.9975
18	0.0520	0.0814	36.1%	0.0339	-0.0068	0.6128	0.4320	0.9974
19	0.0534	0.0851	37.2%	0.0348	-0.0072	0.5933	0.4030	0.9972
20	0.0543	0.0881	38.4%	0.0354	-0.0068	0.5769	0.3801	0.9972
21	0.0551	0.0909	39.3%	0.0364	-0.0085	0.5549	0.3592	0.9971
22	0.0558	0.0939	40.6%	0.0370	-0.0080	0.5405	0.3356	0.9970
23	0.0566	0.0971	41.7%	0.0376	-0.0080	0.5265	0.3131	0.9969
24	0.0576	0.1005	42.6%	0.0383	-0.0087	0.5107	0.2889	0.9968
25	0.0586	0.1039	43.6%	0.0389	-0.0086	0.4960	0.2670	0.9967
26	0.0592	0.1068	44.5%	0.0394	-0.0086	0.4802	0.2487	0.9966
27	0.0599	0.1098	45.4%	0.0400	-0.0092	0.4638	0.2287	0.9965
28	0.0606	0.1129	46.3%	0.0404	-0.0096	0.4507	0.2138	0.9965
29	0.0613	0.1157	47.0%	0.0409	-0.0099	0.4339	0.2032	0.9964
30	0.0617	0.1182	47.8%	0.0413	-0.0093	0.4171	0.1947	0.9963
31	0.0624	0.1208	48.3%	0.0419	-0.0108	0.3942	0.1854	0.9962
32	0.0630	0.1236	49.0%	0.0423	-0.0103	0.3797	0.1758	0.9962
33	0.0638	0.1266	49.6%	0.0428	-0.0103	0.3612	0.1635	0.9961
34	0.0650	0.1296	49.9%	0.0435	-0.0111	0.3412	0.1532	0.9959
35	0.0659	0.1324	50.2%	0.0441	-0.0110	0.3234	0.1445	0.9958
36	0.0665	0.1351	50.8%	0.0445	-0.0109	0.3052	0.1347	0.9957
37	0.0674	0.1377	51.0%	0.0452	-0.0117	0.2859	0.1282	0.9956
38	0.0682	0.1402	51.3%	0.0456	-0.0118	0.2699	0.1243	0.9955
39	0.0690	0.1427	51.6%	0.0461	-0.0125	0.2576	0.1209	0.9954
40	0.0695	0.1451	52.1%	0.0463	-0.0123	0.2484	0.1145	0.9953
41	0.0702	0.1475	52.4%	0.0469	-0.0140	0.2306	0.1073	0.9952
42	0.0709	0.1502	52.8%	0.0473	-0.0136	0.2225	0.1002	0.9952
43	0.0714	0.1526	53.2%	0.0476	-0.0136	0.2142	0.0960	0.9951
44	0.0719	0.1549	53.6%	0.0480	-0.0141	0.2019	0.0912	0.9950
45	0.0722	0.1573	54.1%	0.0482	-0.0139	0.1926	0.0831	0.9950
46	0.0728	0.1596	54.4%	0.0486	-0.0137	0.1803	0.0757	0.9949
47	0.0734	0.1618	54.6%	0.0490	-0.0142	0.1710	0.0700	0.9948
48	0.0738	0.1638	55.0%	0.0493	-0.0140	0.1625	0.0685	0.9947
49	0.0739	0.1653	55.3%	0.0495	-0.0141	0.1537	0.0718	0.9947
50	0.0744	0.1671	55.5%	0.0498	-0.0135	0.1423	0.0747	0.9947
51	0.0749	0.1688	55.6%	0.0502	-0.0145	0.1258	0.0768	0.9946
52	0.0754	0.1709	55.8%	0.0506	-0.0141	0.1135	0.0753	0.9945
53	0.0759	0.1729	56.1%	0.0509	-0.0141	0.1034	0.0730	0.9944
54	0.0761	0.1746	56.4%	0.0511	-0.0144	0.0970	0.0736	0.9944
55	0.0762	0.1762	56.8%	0.0513	-0.0140	0.0944	0.0758	0.9944
56	0.0763	0.1781	57.1%	0.0515	-0.0137	0.0895	0.0736	0.9944
57	0.0769	0.1802	57.3%	0.0519	-0.0144	0.0834	0.0694	0.9943
58	0.0774	0.1823	57.5%	0.0521	-0.0143	0.0783	0.0654	0.9942
59	0.0781	0.1847	57.7%	0.0526	-0.0148	0.0728	0.0584	0.9941
60	0.0789	0.1870	57.8%	0.0529	-0.0146	0.0697	0.0535	0.9940
61	0.0798	0.1894	57.9%	0.0535	-0.0158	0.0648	0.0470	0.9939
62	0.0803	0.1918	58.1%	0.0538	-0.0158	0.0627	0.0385	0.9938
63	0.0810	0.1941	58.3%	0.0542	-0.0160	0.0562	0.0316	0.9937
64	0.0814	0.1963	58.5%	0.0545	-0.0164	0.0512	0.0248	0.9936
65	0.0821	0.1986	58.7%	0.0549	-0.0165	0.0468	0.0192	0.9935
66	0.0826	0.2006	58.8%	0.0552	-0.0163	0.0406	0.0171	0.9934
67	0.0832	0.2023	58.9%	0.0556	-0.0169	0.0381	0.0183	0.9933
68	0.0837	0.2045	59.1%	0.0560	-0.0167	0.0330	0.0135	0.9932
69	0.0841	0.2063	59.2%	0.0563	-0.0172	0.0319	0.0125	0.9932
70	0.0842	0.2080	59.5%	0.0564	-0.0168	0.0307	0.0122	0.9931
71	0.0846	0.2099	59.7%	0.0568	-0.0176	0.0266	0.0110	0.9931
72	0.0849	0.2117	59.9%	0.0570	-0.0171	0.0259	0.0120	0.9930
73	0.0853	0.2136	60.1%	0.0573	-0.0172	0.0236	0.0109	0.9930
74	0.0856	0.2153	60.3%	0.0575	-0.0175	0.0230	0.0133	0.9929
75	0.0860	0.2175	60.5%	0.0577	-0.0174	0.0233	0.0086	0.9929
76	0.0866	0.2197	60.6%	0.0581	-0.0171	0.0184	0.0050	0.9928
77	0.0870	0.2217	60.8%	0.0585	-0.0175	0.0164	0.0049	0.9927
78	0.0873	0.2238	61.0%	0.0586	-0.0170	0.0144	0.0037	0.9926
79	0.0877	0.2259	61.2%	0.0588	-0.0171	0.0118	0.0019	0.9926
80	0.0878	0.2279	61.5%	0.0590	-0.0166	0.0103	-0.0015	0.9926
81	0.0878	0.2296	61.8%	0.0591	-0.0171	0.0112	-0.0020	0.9926
82	0.0879	0.2315	62.0%	0.0592	-0.0166	0.0102	-0.0056	0.9925
83	0.0883	0.2336	62.2%	0.0596	-0.0165	0.0091	-0.0076	0.9925
84	0.0888	0.2361	62.4%	0.0599	-0.0169	0.0112	-0.0090	0.9924
85	0.0893	0.2385	62.6%	0.0602	-0.0173	0.0119	-0.0119	0.9923
86	0.0900	0.2408	62.6%	0.0607	-0.0170	0.0088	-0.0147	0.9922
87	0.0906	0.2428	62.7%	0.0611	-0.0176	0.0063	-0.0142	0.9921
88	0.0913	0.2451	62.7%	0.0615	-0.0175	0.0028	-0.0133	0.9920
89	0.0921	0.2474	62.8%	0.0619	-0.0179	-0.0010	-0.0151	0.9918
90	0.0930	0.2498	62.8%	0.0624	-0.0177	-0.0068	-0.0180	0.9916

Forecast Skill vs Lead Step — Test

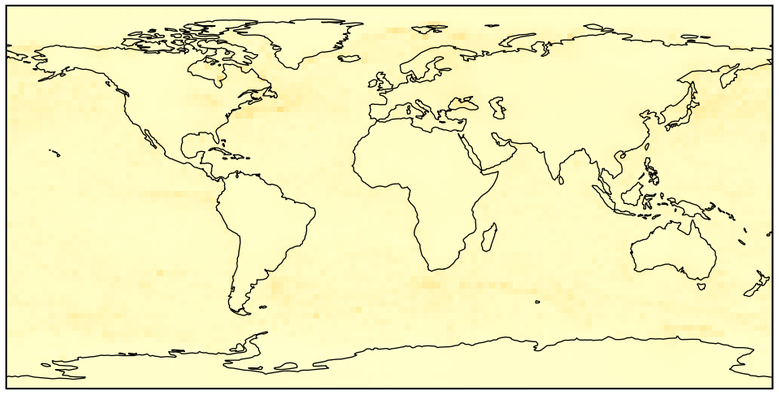




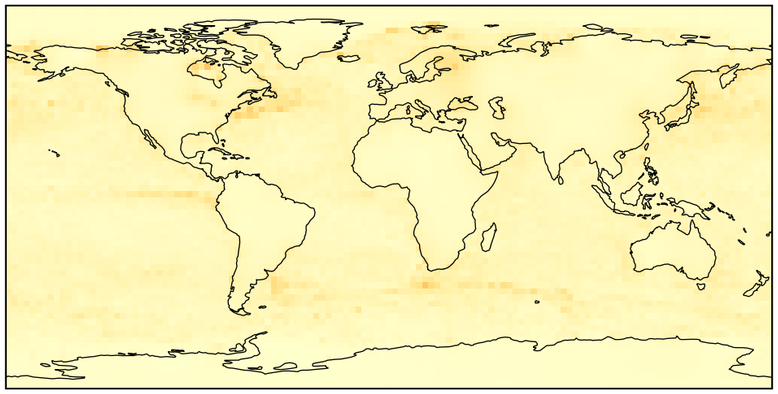

Bias and R² — Test
Mean Bias
Bias
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
−0.0125
−0.0150
−0.0175
0
20
40
60
80
Lead step
R²
R²
1.000
0.999
0.998
0.997
0.996
0.995
0.994
0.993
0.992
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Test

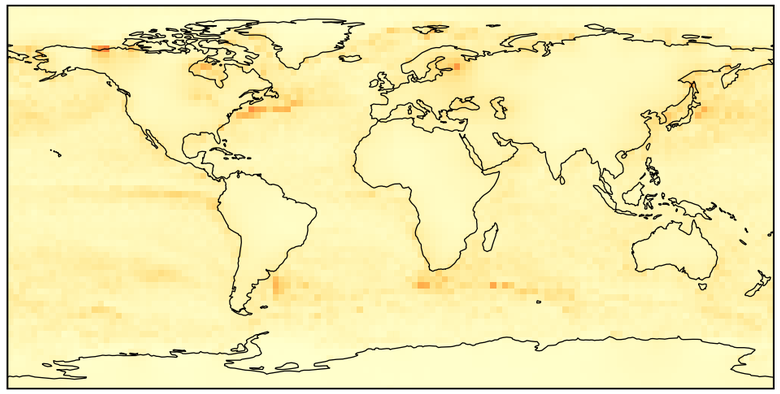
Step 1



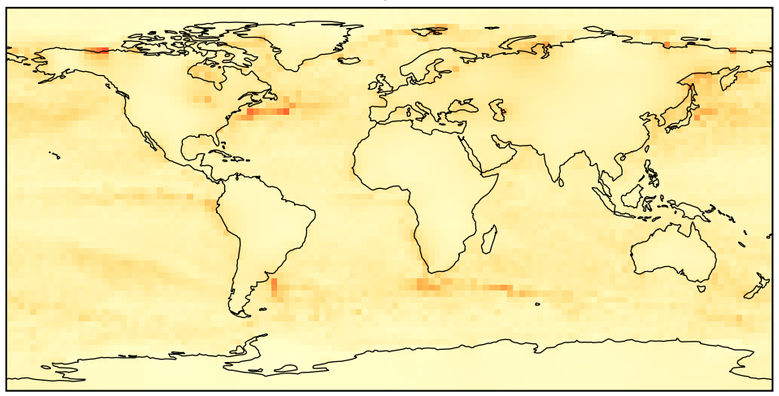
Step 3



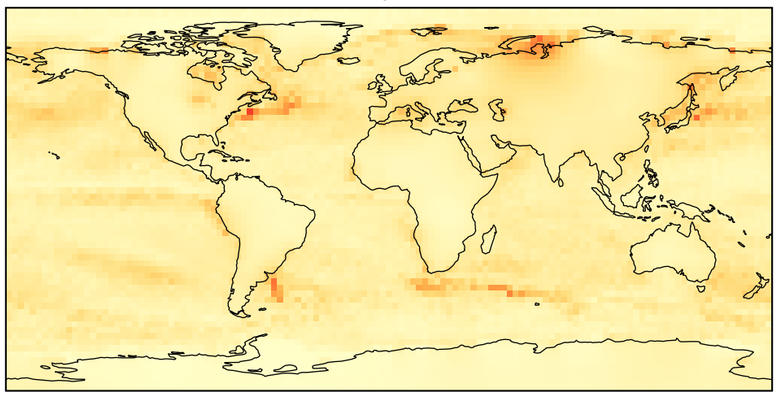
Step 7



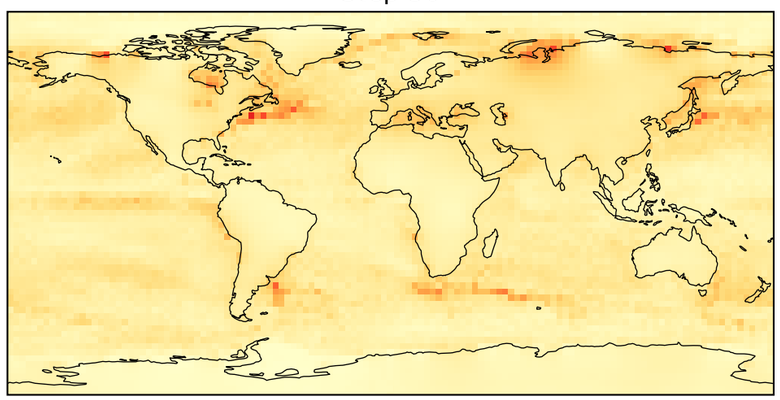
Step 14



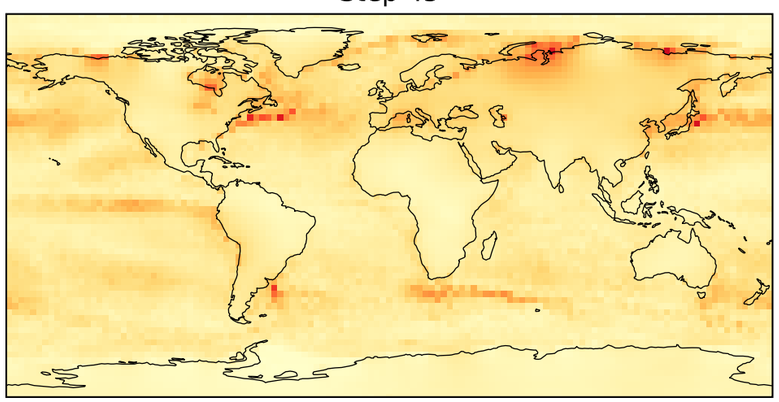
Step 21



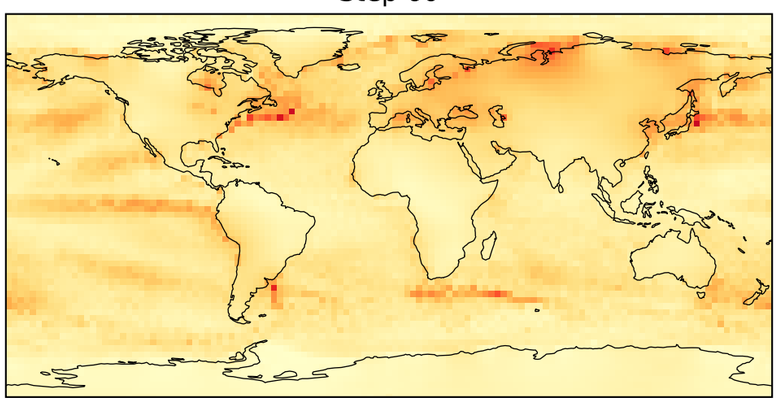
Step 30



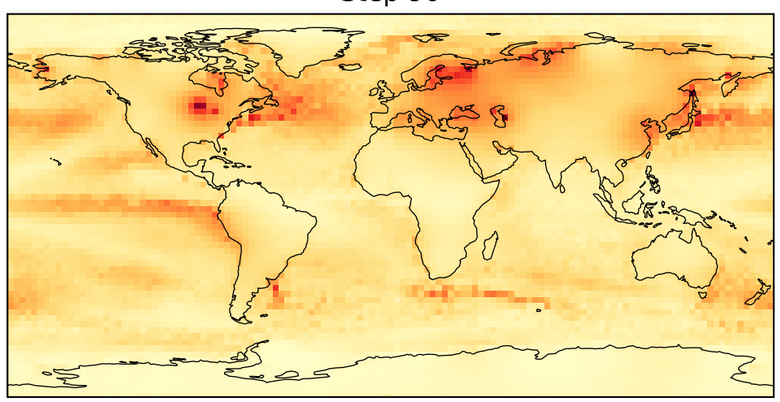
Step 45

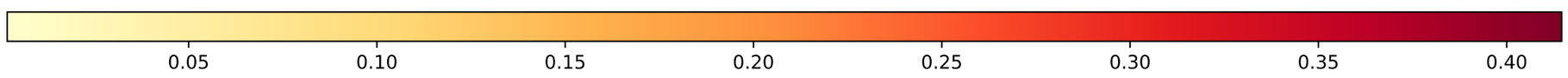


Step 60



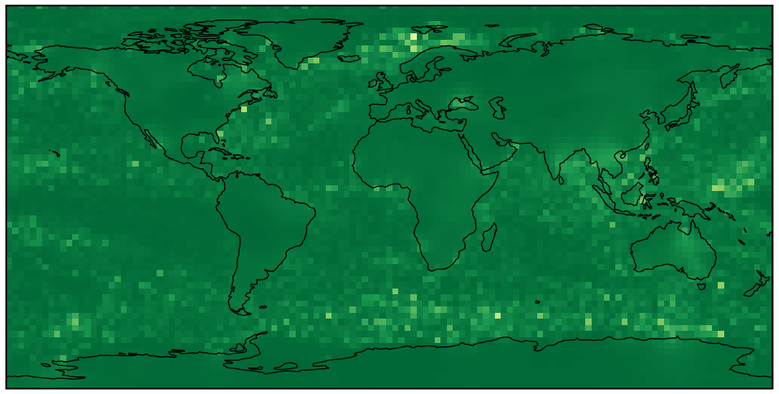
Step 90



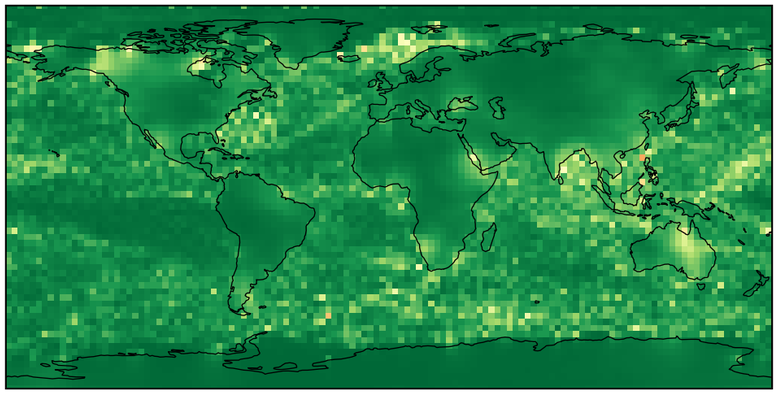


Spatial ACC (all grid) — Test

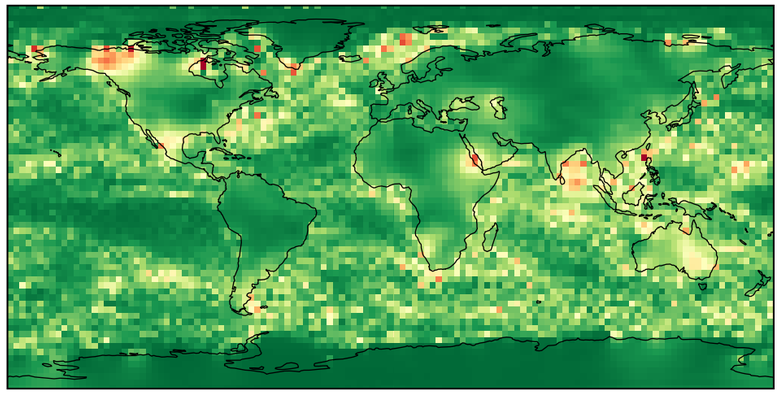
Step 1



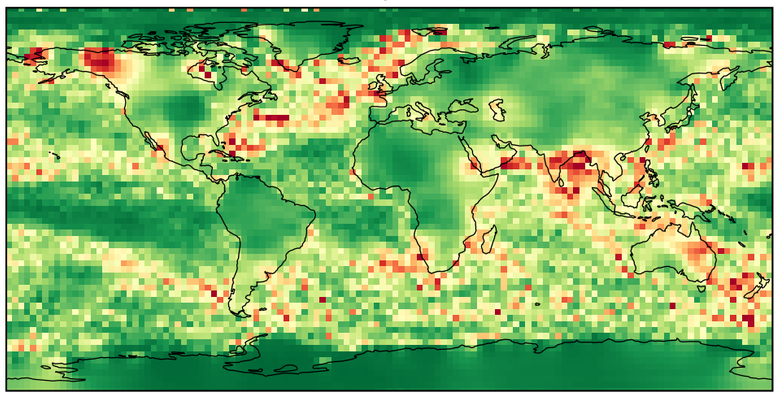
Step 3



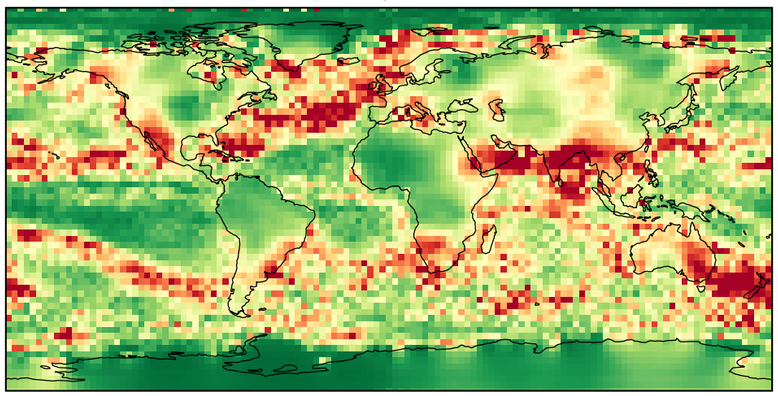
Step 7



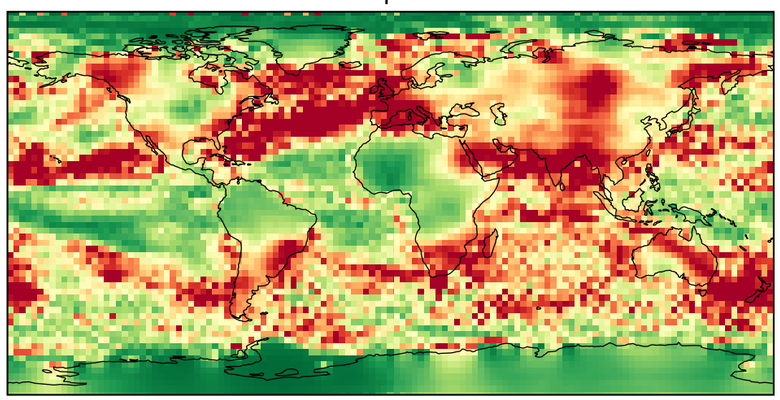
Step 14



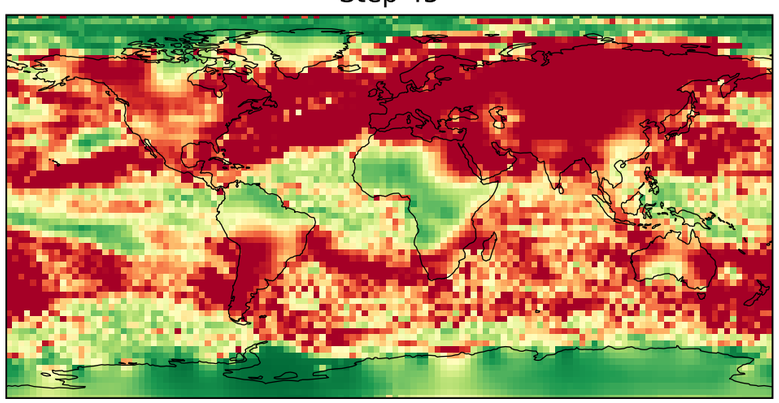
Step 21



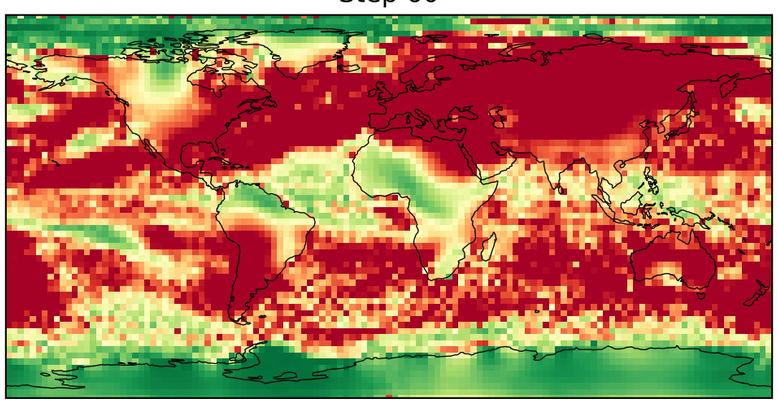
Step 30



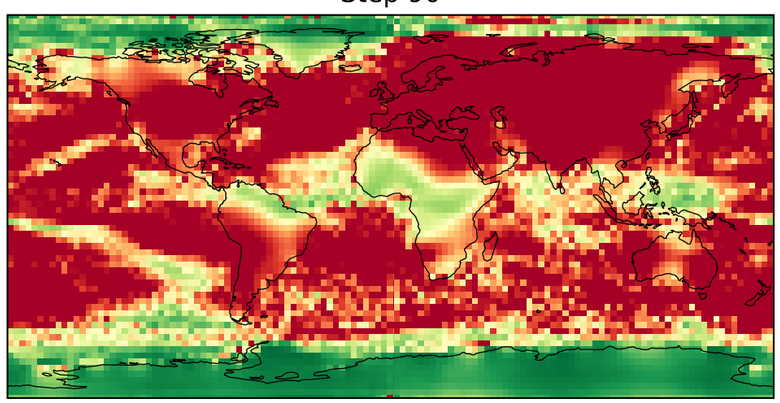
Step 45



Step 60

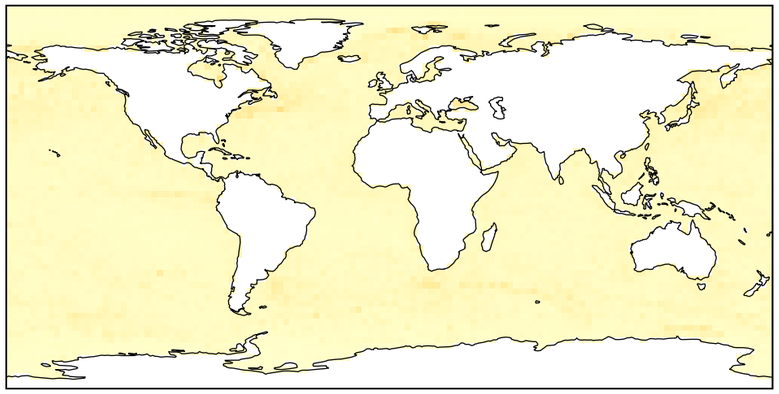


Step 90

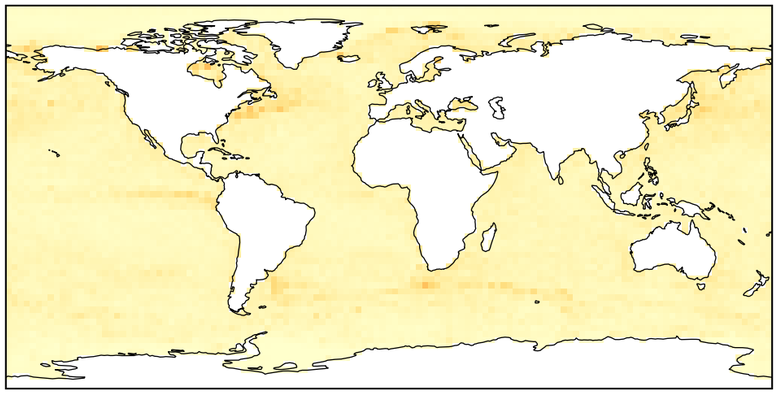


Spatial RMSE (ocean only) — Test

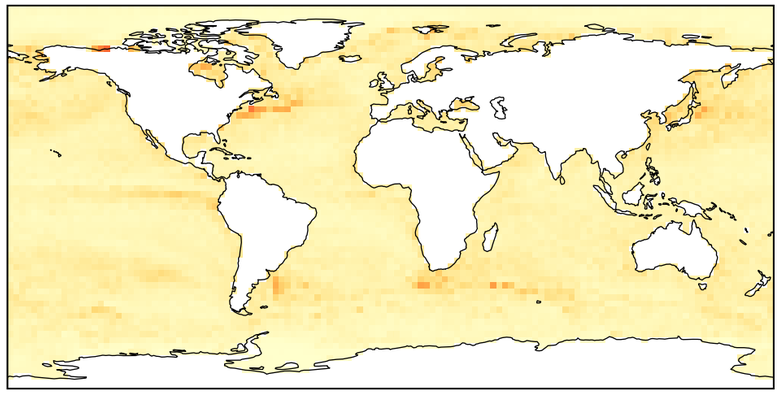
Step 1



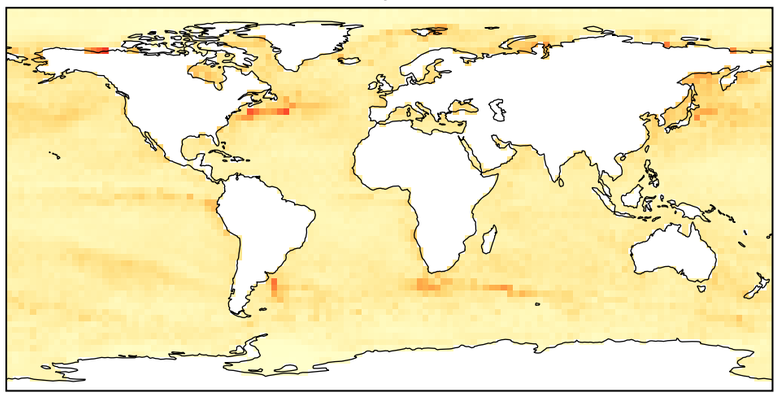
Step 3



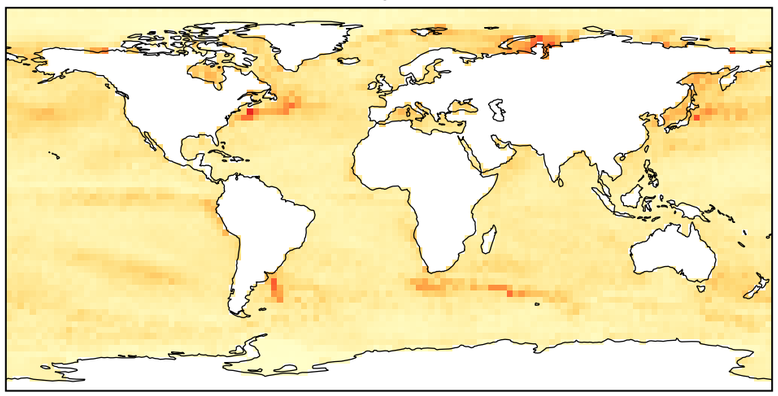
Step 7



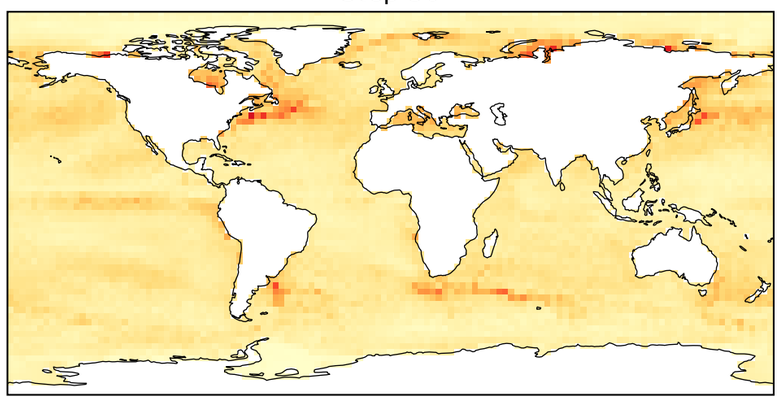
Step 14



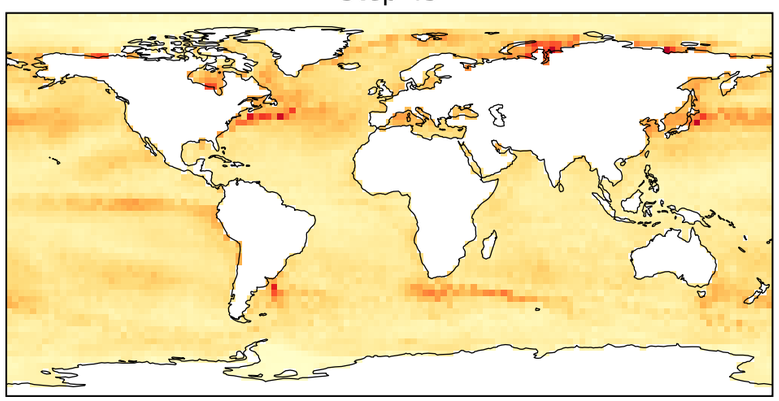
Step 21



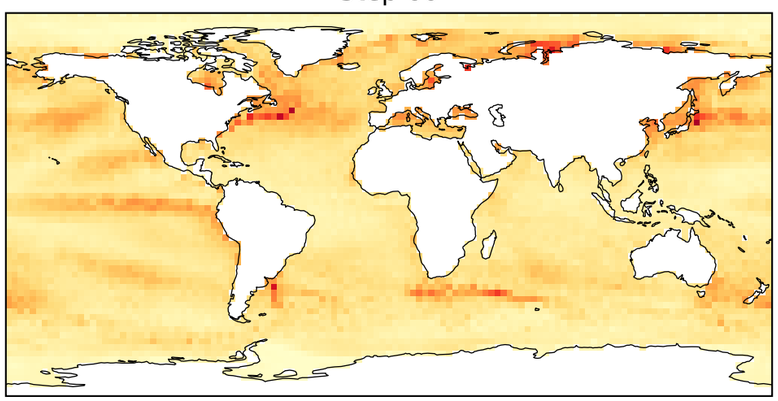
Step 30



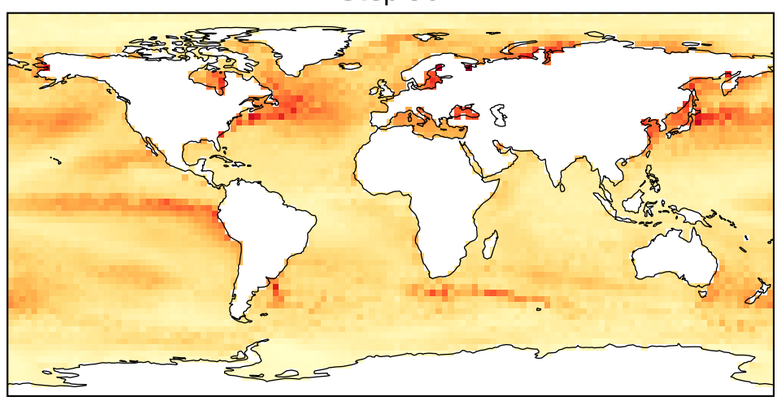
Step 45

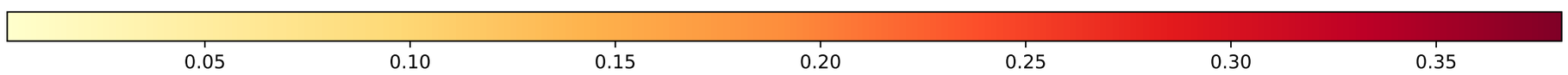


Step 60



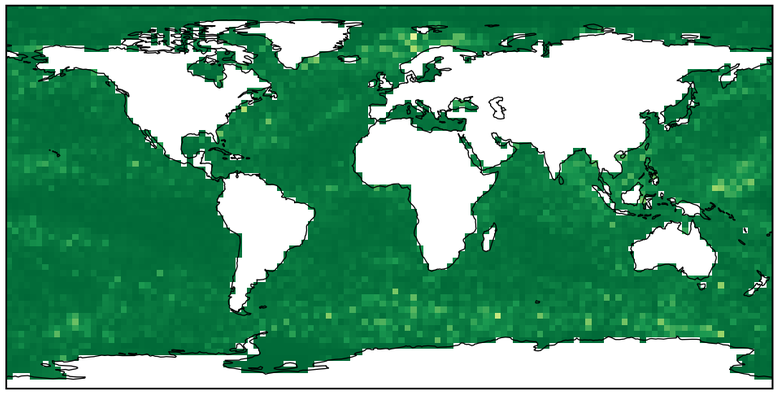
Step 90



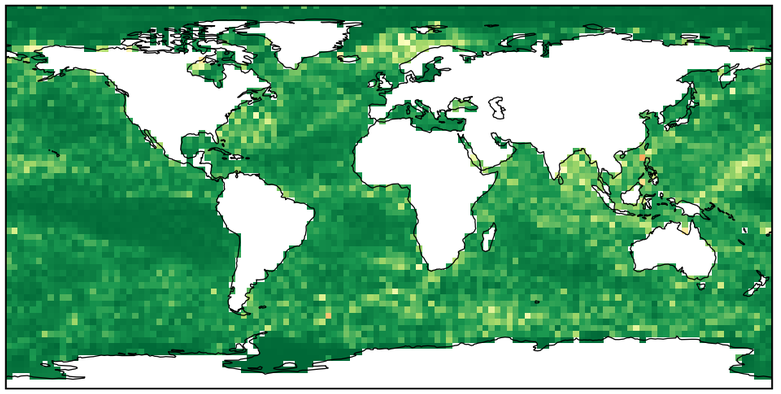


Spatial ACC (ocean only) — Test

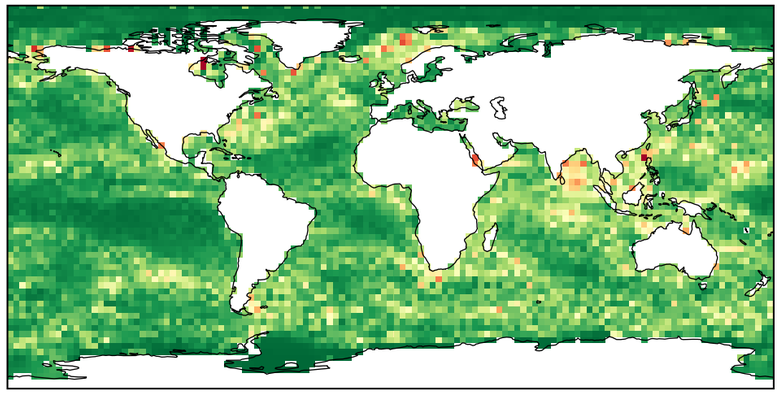
Step 1



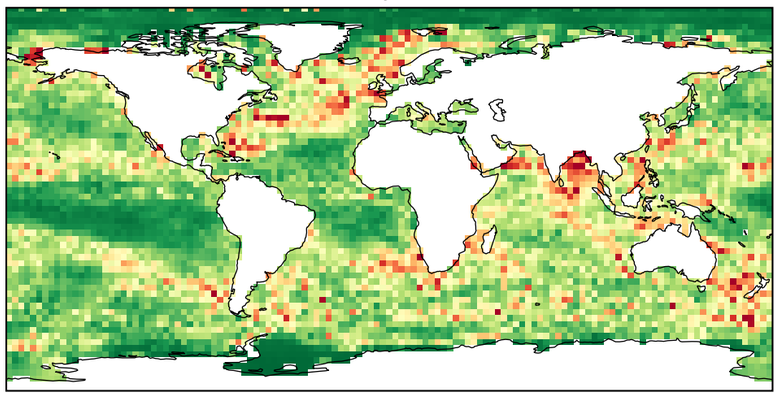
Step 3



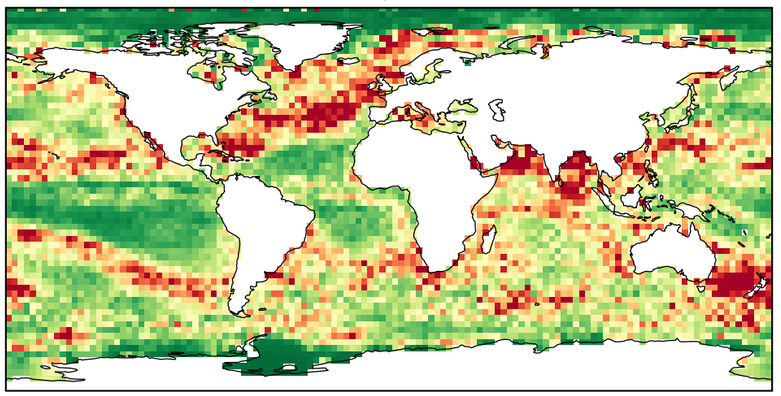
Step 7



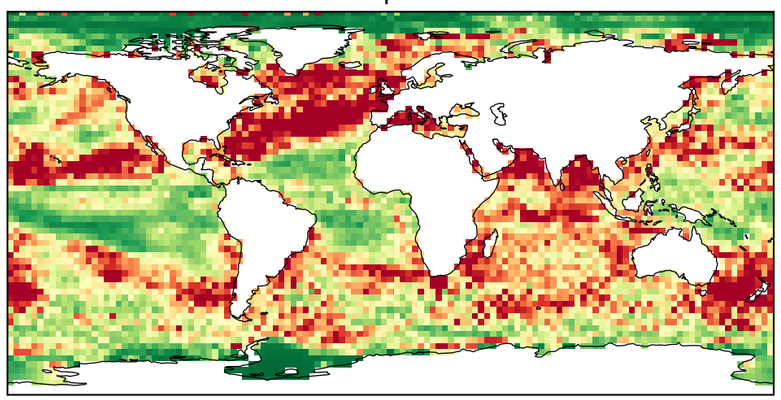
Step 14



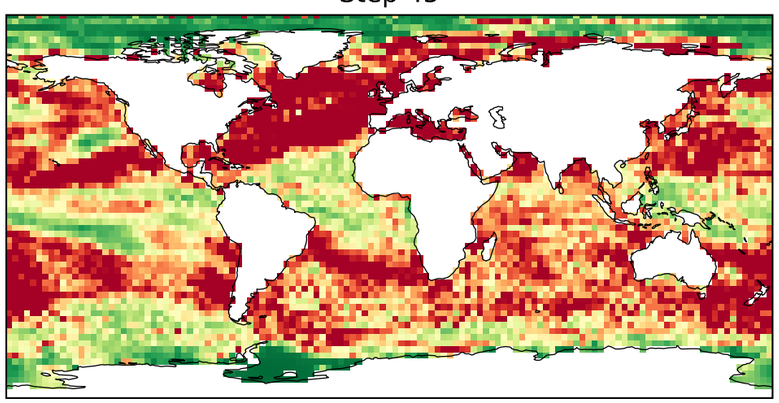
Step 21



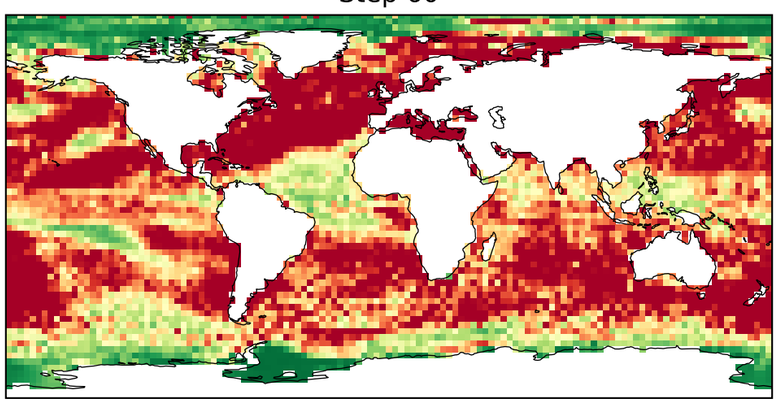
Step 30



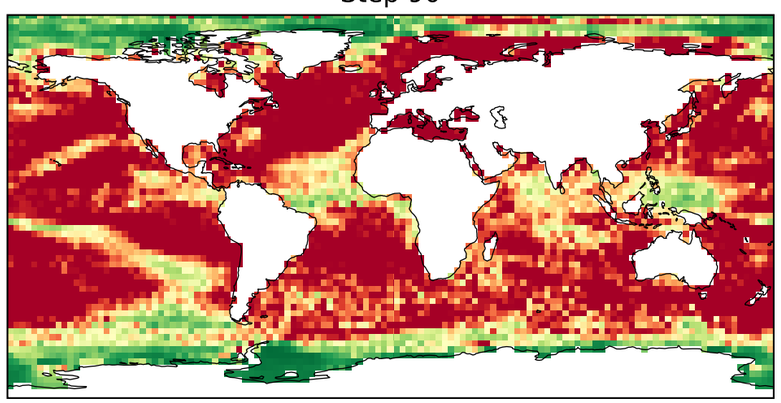
Step 45



Step 60

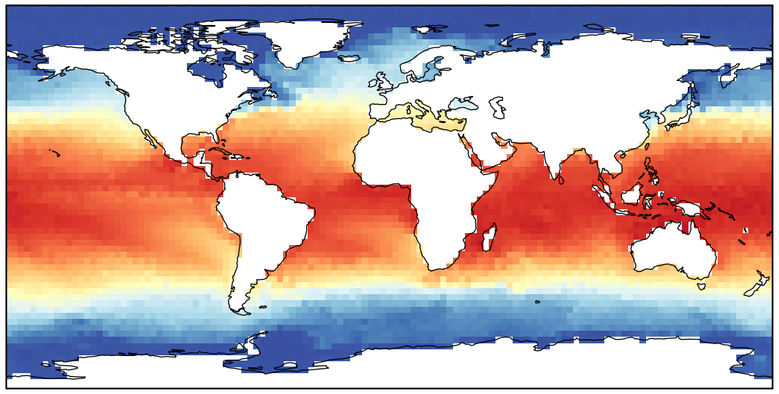


Step 90

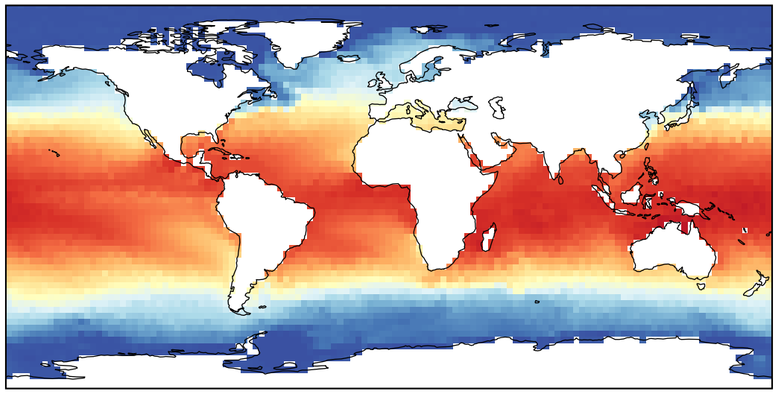


Chain init: 2020-01-11 — Test (°C)

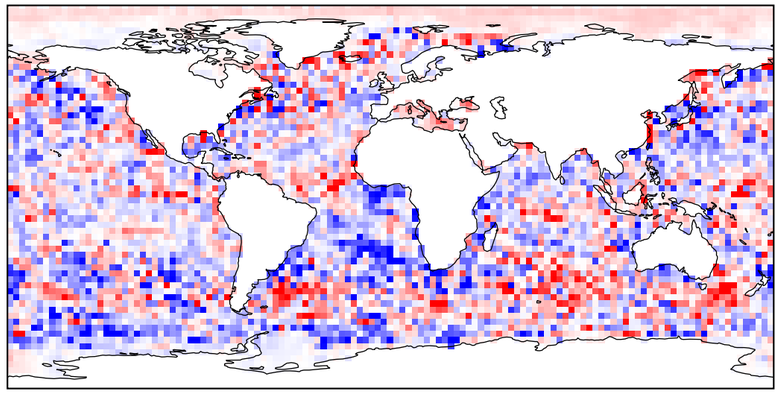
Actual — Step 1 (°C)



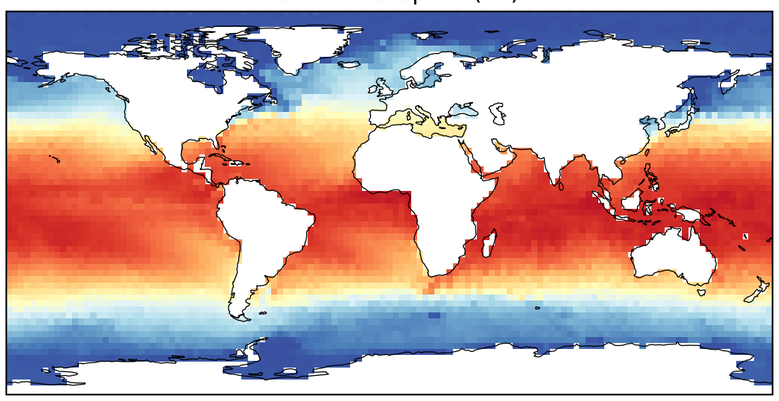
SFNO — Step 1 (°C)



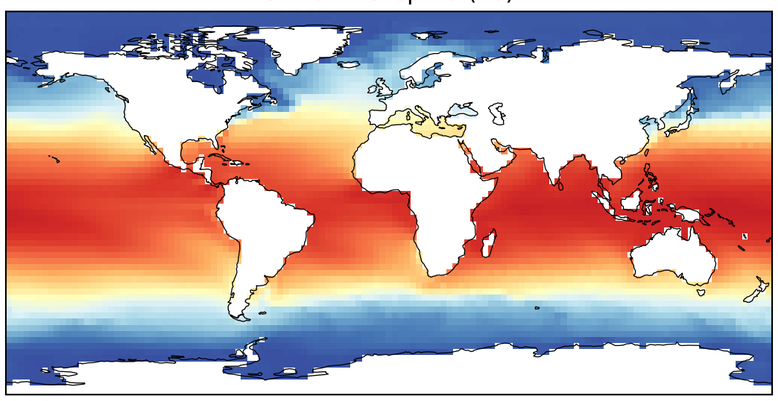
Error — Step 1 (°C)



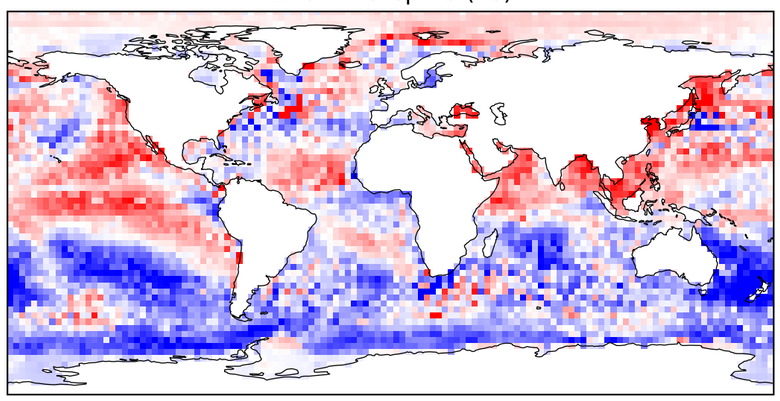
Actual — Step 46 (°C)



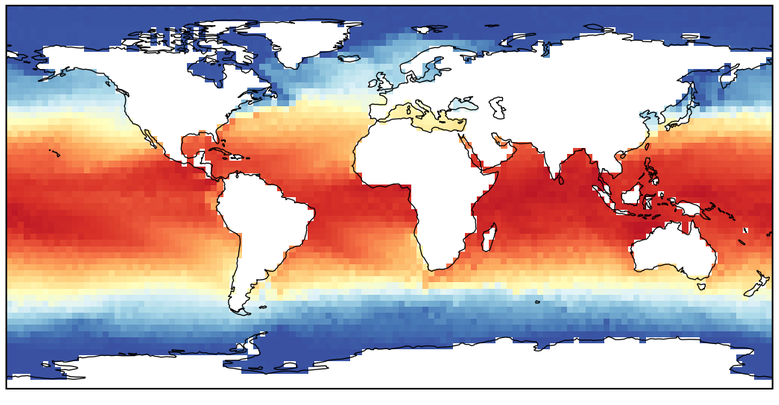
SFNO — Step 46 (°C)



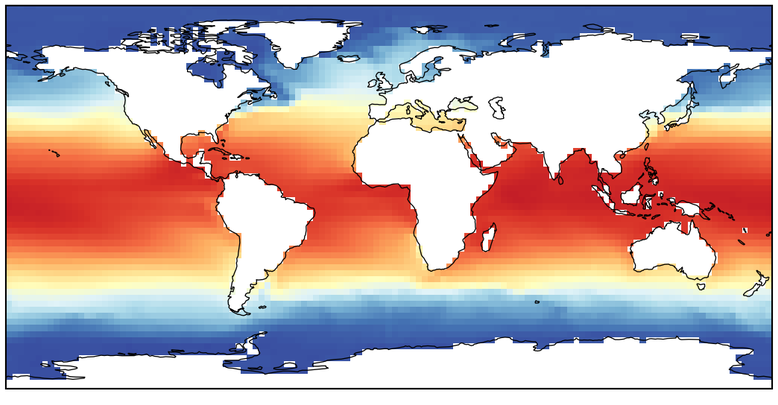
Error — Step 46 (°C)



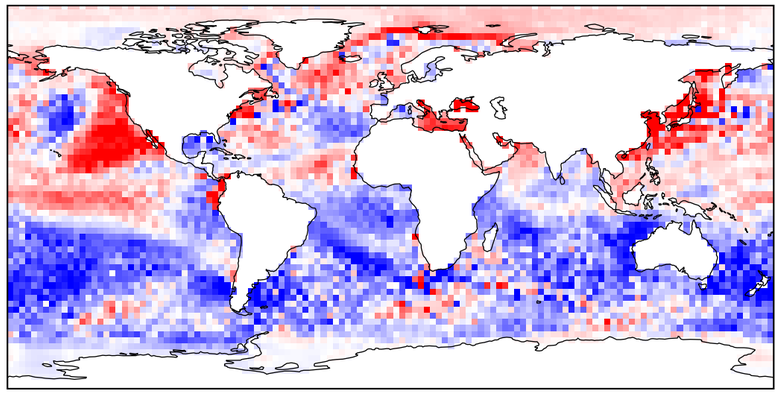
Actual — Step 90 (°C)

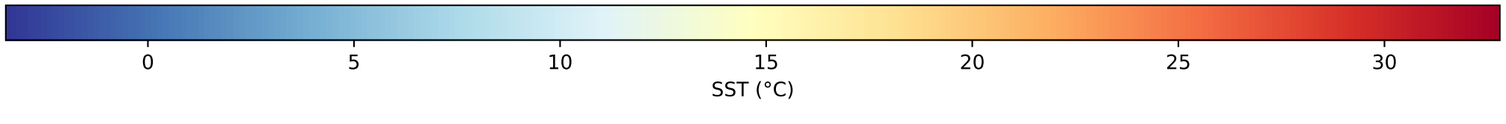


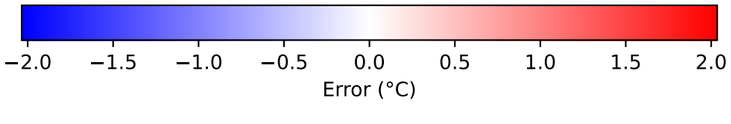
SFNO — Step 90 (°C)



Error — Step 90 (°C)

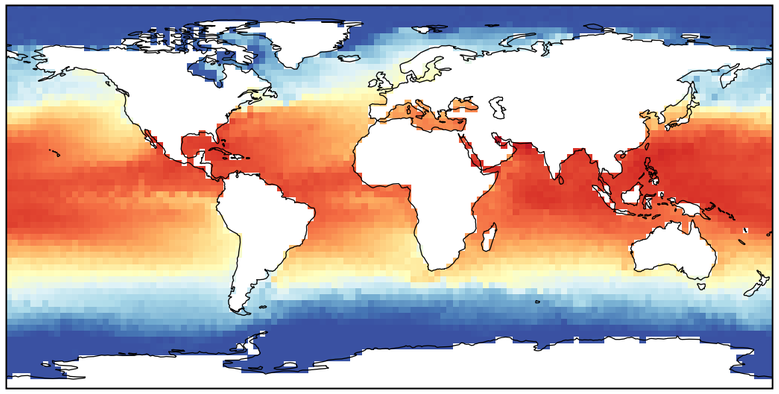




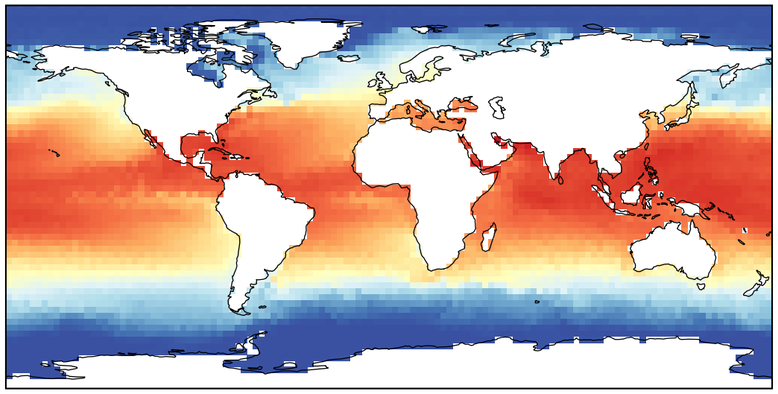


Chain init: 2020-07-09 — Test (°C)

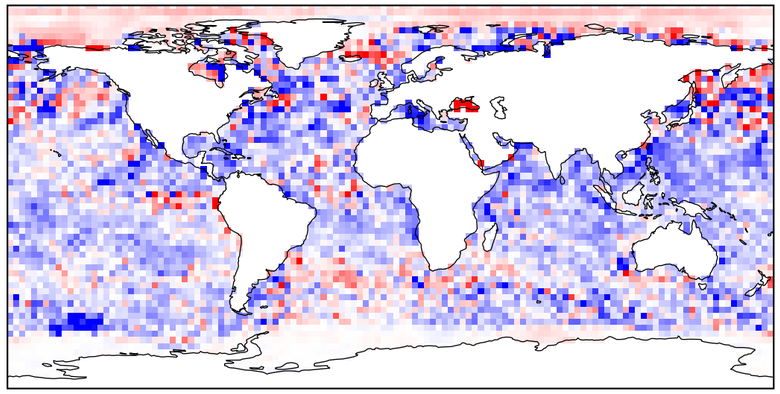
Actual — Step 1 (°C)



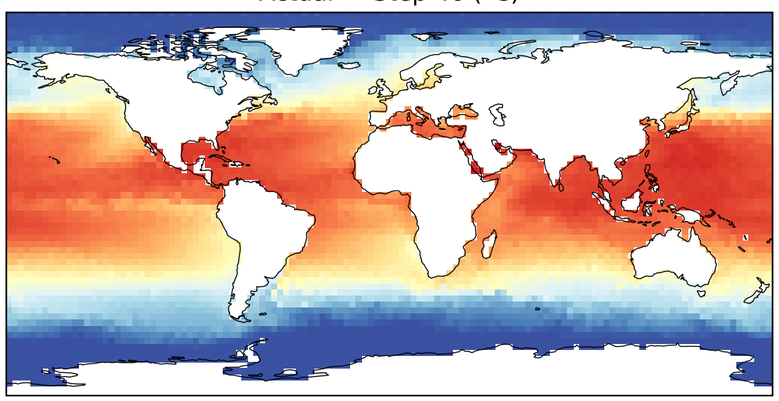
SFNO — Step 1 (°C)



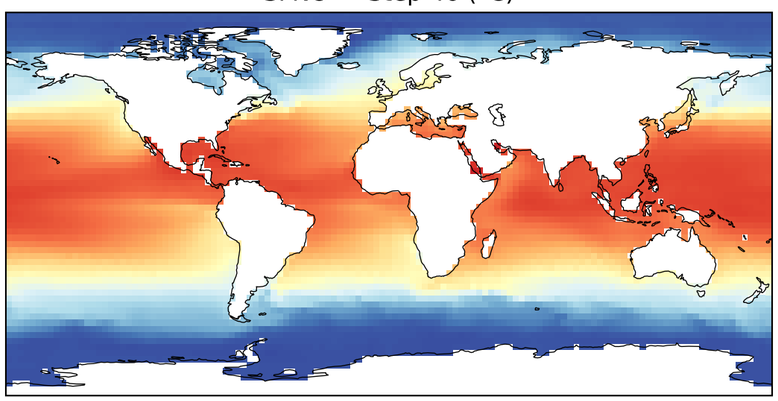
Error — Step 1 (°C)



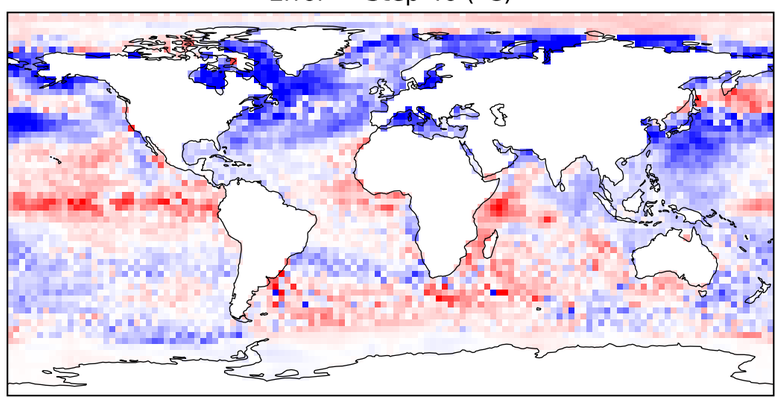
Actual — Step 46 (°C)



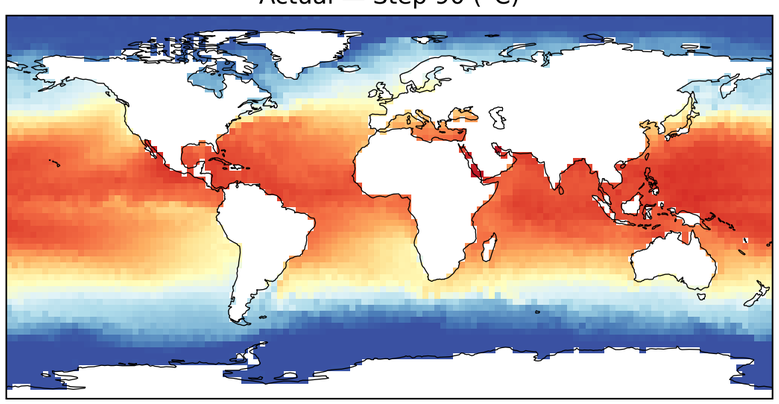
SFNO — Step 46 (°C)



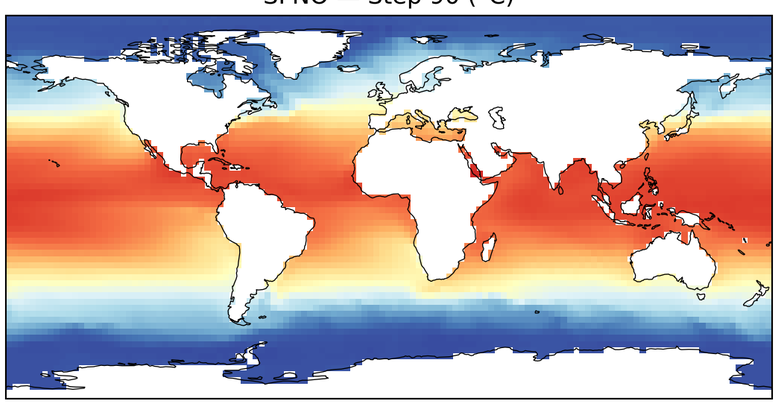
Error — Step 46 (°C)



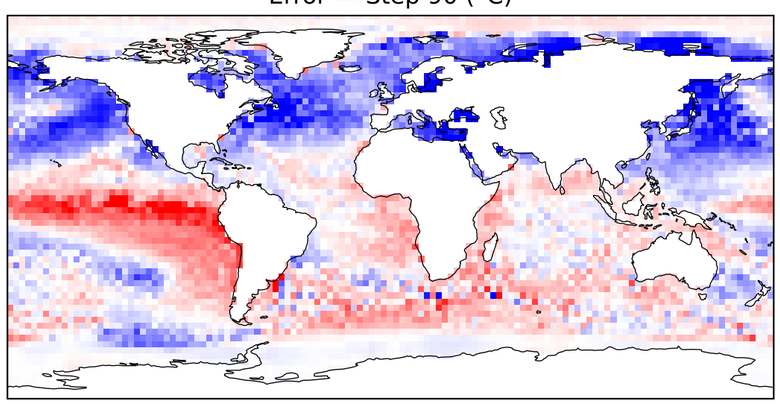
Actual — Step 90 (°C)

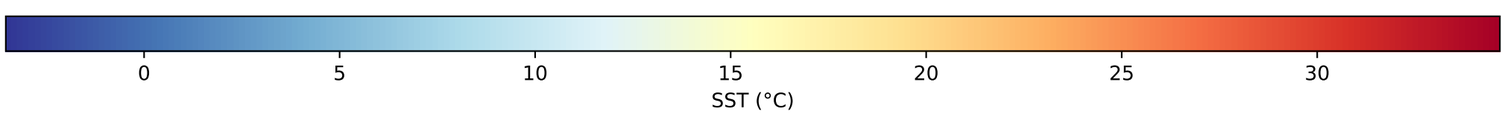


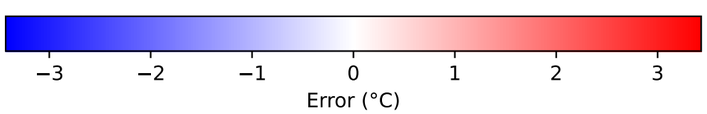
SFNO — Step 90 (°C)



Error — Step 90 (°C)

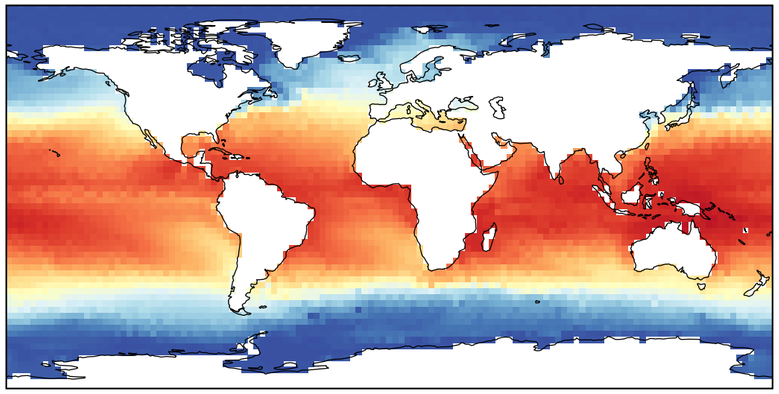




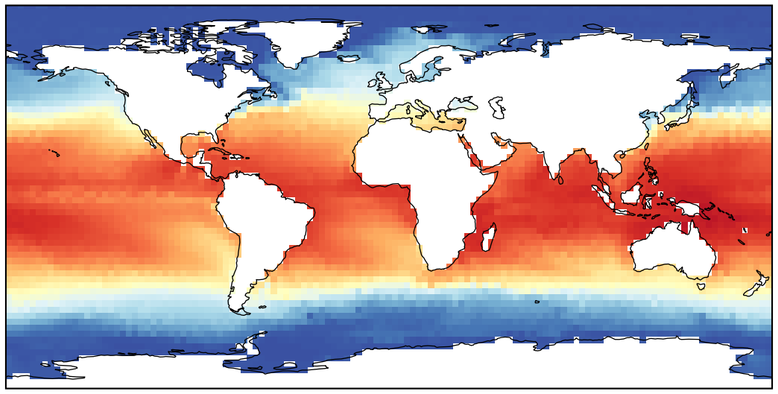


Chain init: 2021-01-06 — Test (°C)

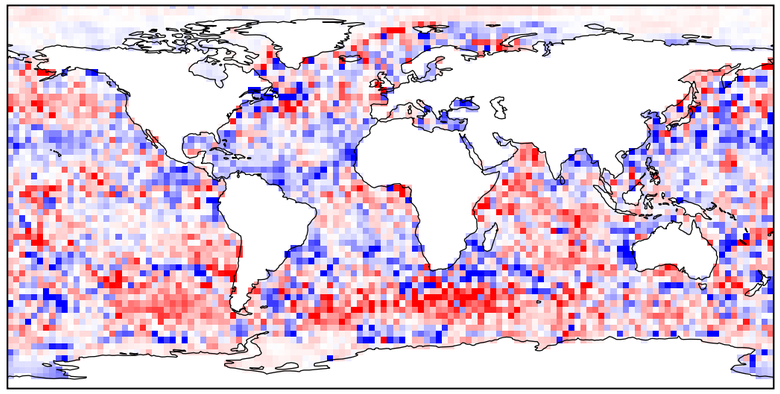
Actual — Step 1 (°C)



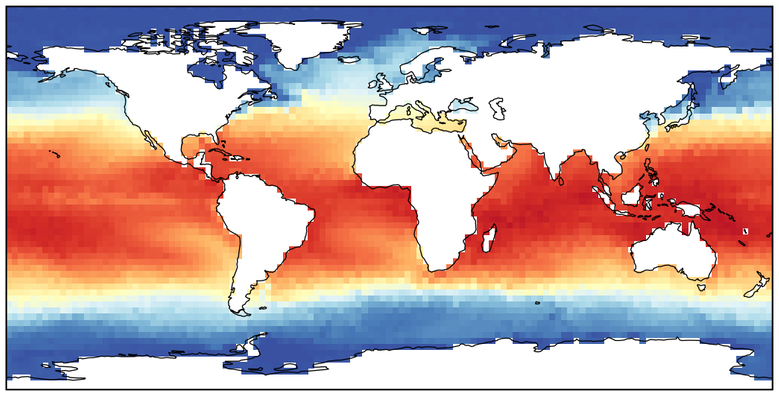
SFNO — Step 1 (°C)



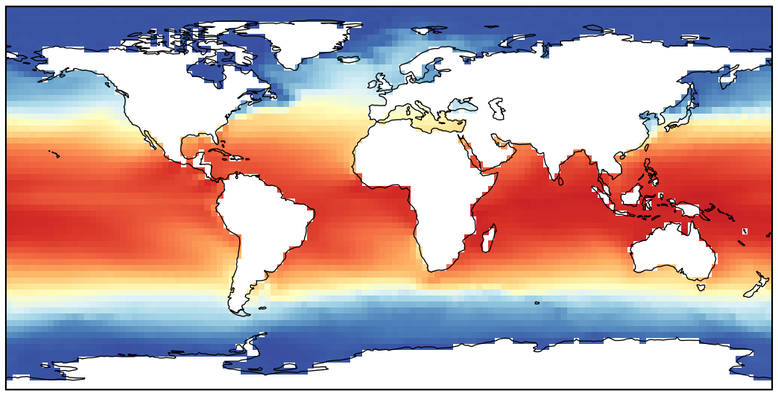
Error — Step 1 (°C)



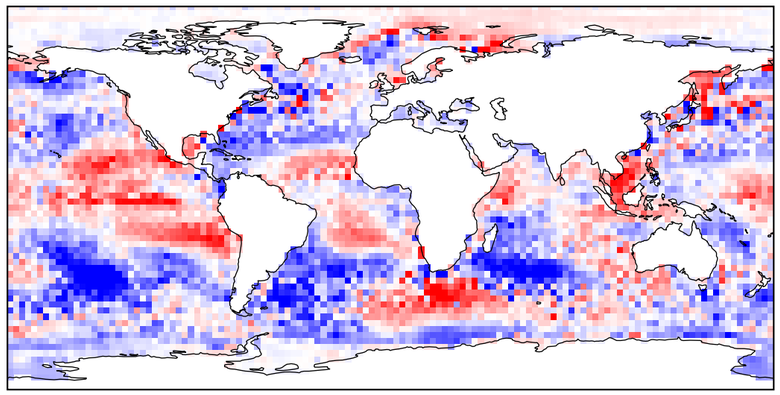
Actual — Step 46 (°C)



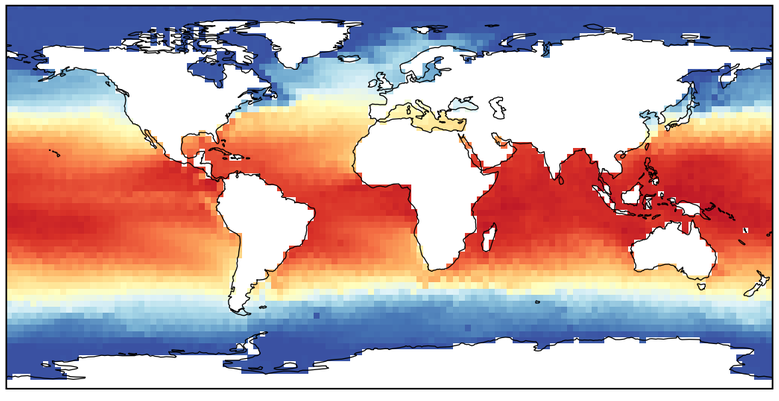
SFNO — Step 46 (°C)



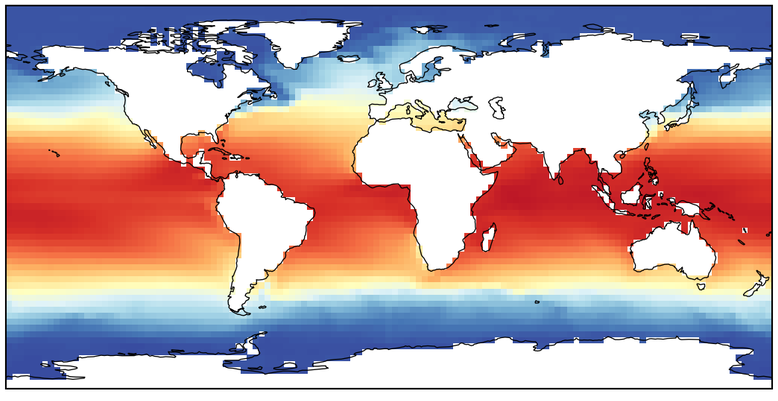
Error — Step 46 (°C)



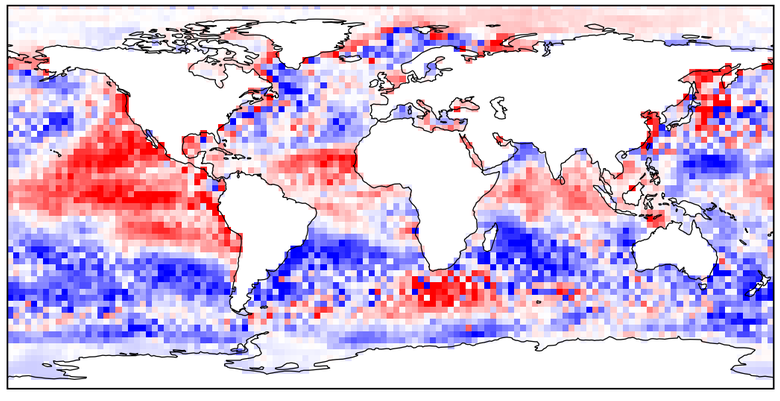
Actual — Step 90 (°C)

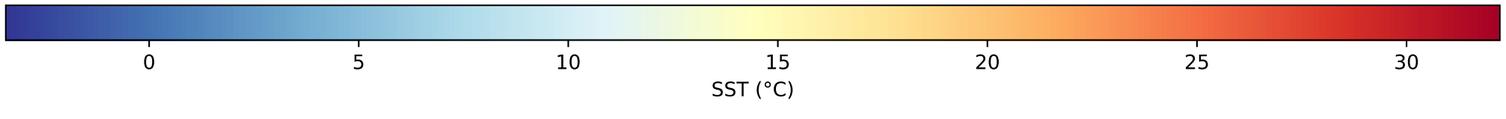


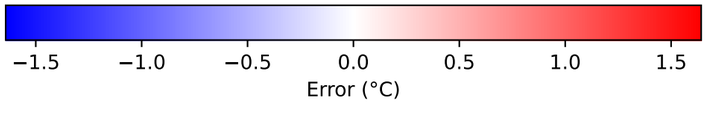
SFNO — Step 90 (°C)



Error — Step 90 (°C)

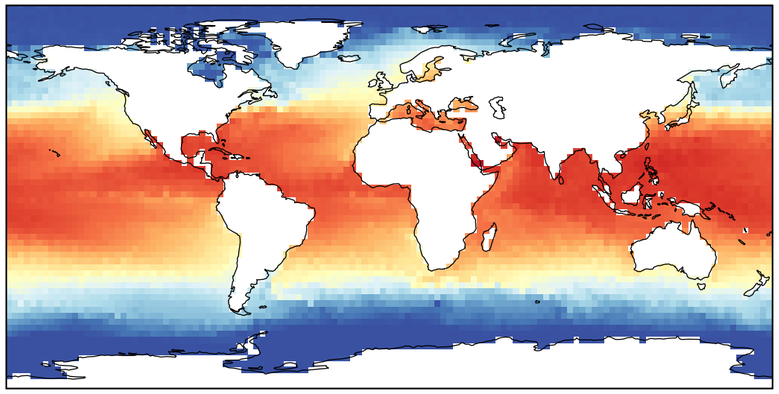




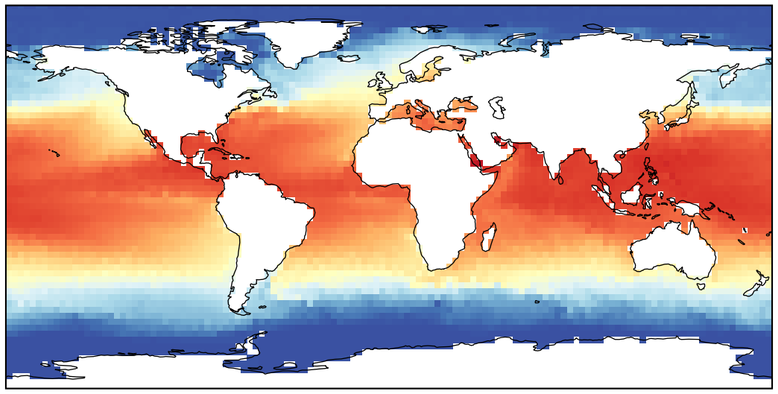


Chain init: 2021-07-05 — Test (°C)

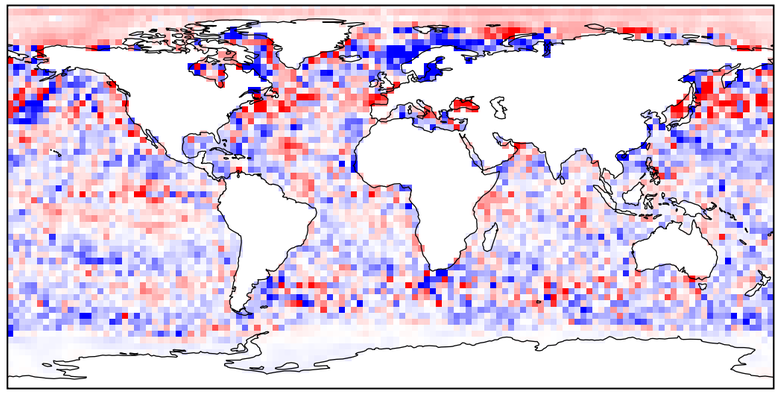
Actual — Step 1 (°C)



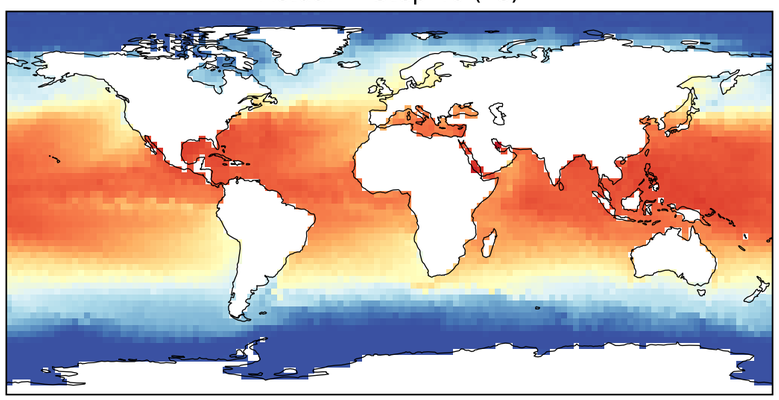
SFNO — Step 1 (°C)



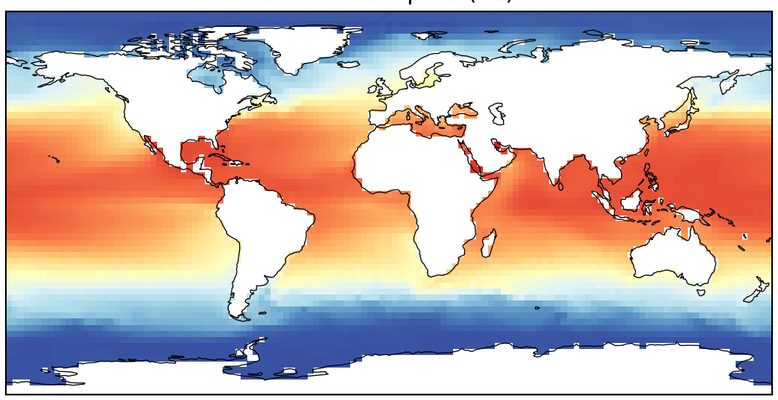
Error — Step 1 (°C)



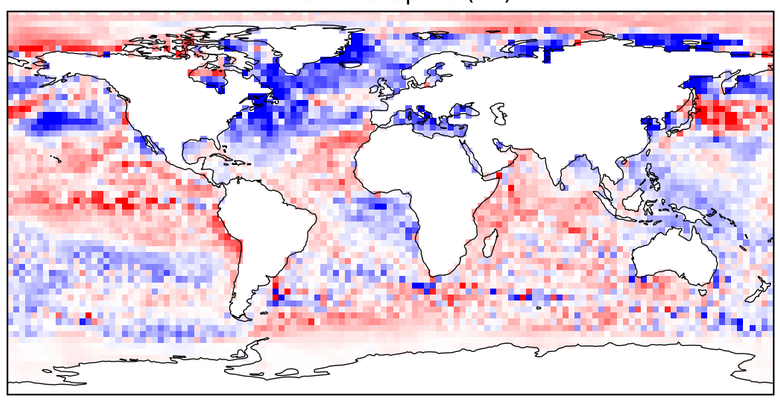
Actual — Step 46 (°C)



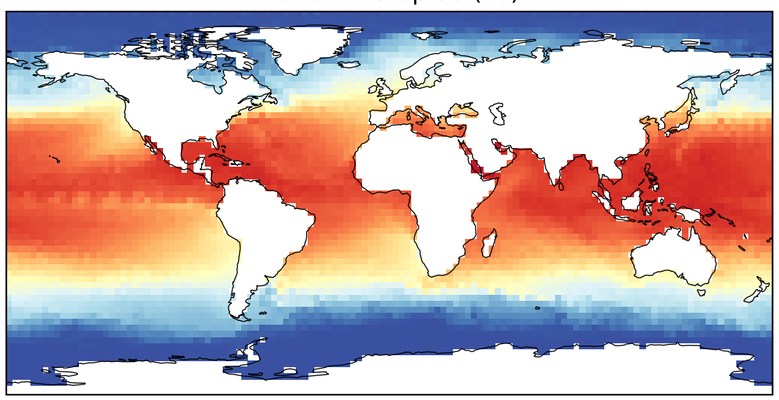
SFNO — Step 46 (°C)



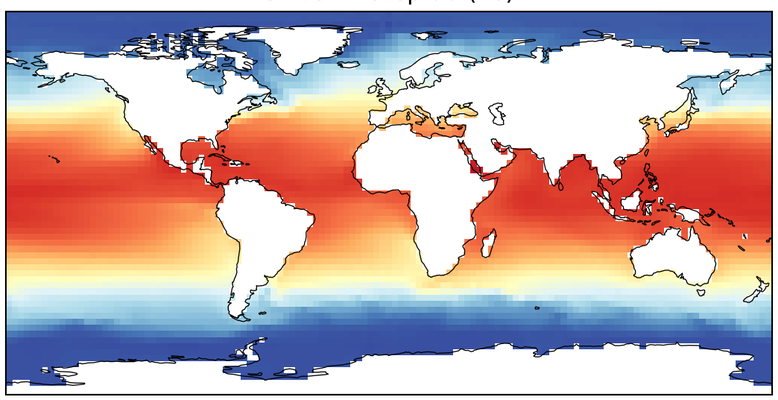
Error — Step 46 (°C)



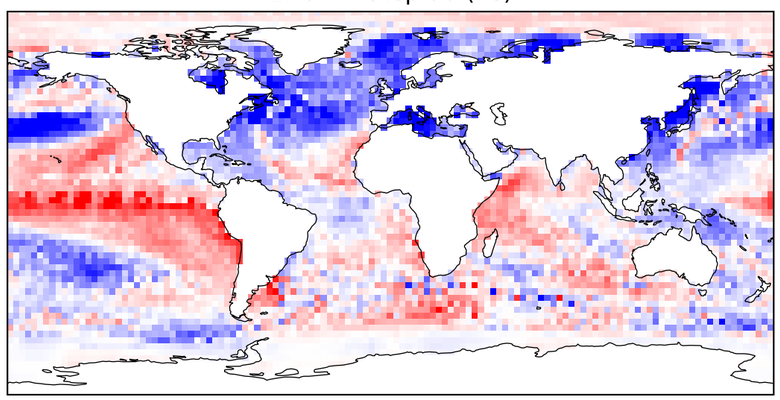
Actual — Step 90 (°C)

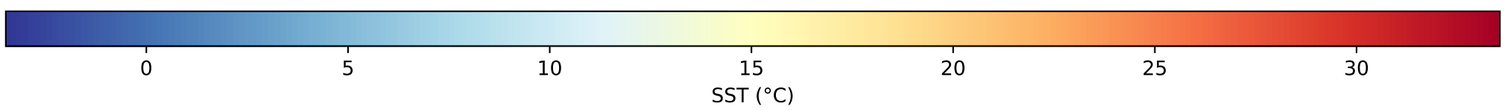


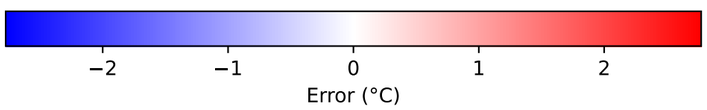
SFNO — Step 90 (°C)



Error — Step 90 (°C)

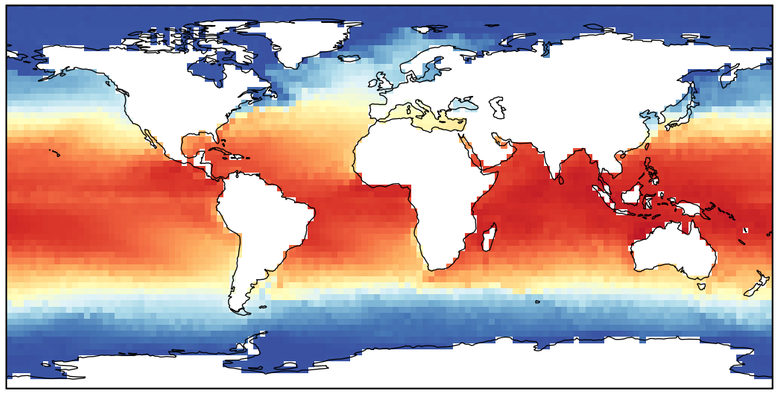




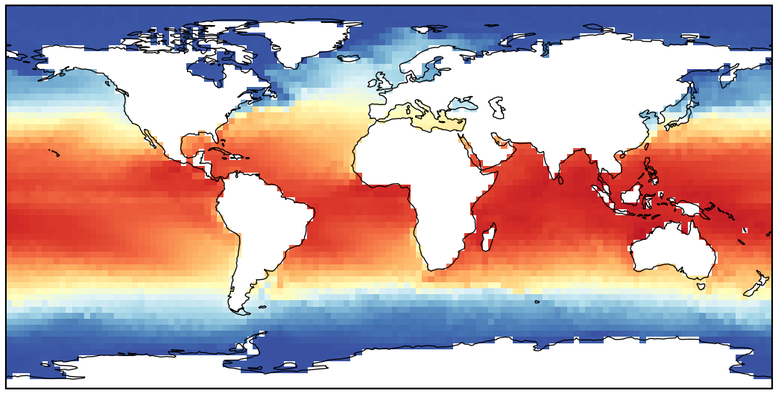


Chain init: 2022-04-01 — Test (°C)

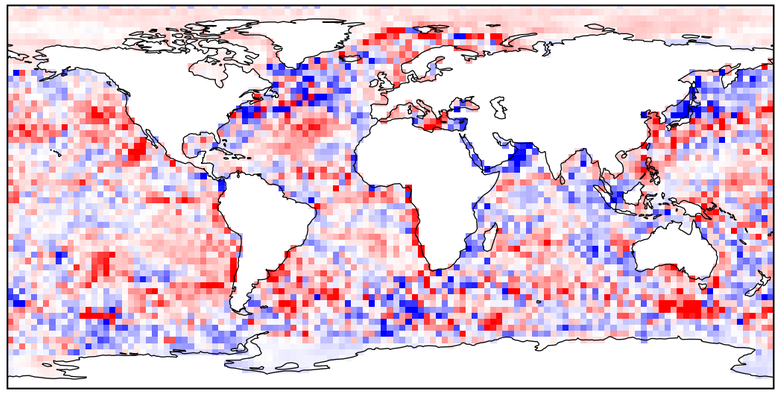
Actual — Step 1 (°C)



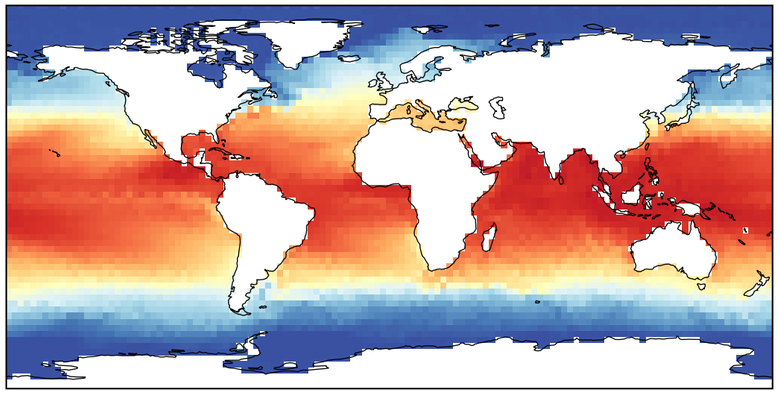
SFNO — Step 1 (°C)



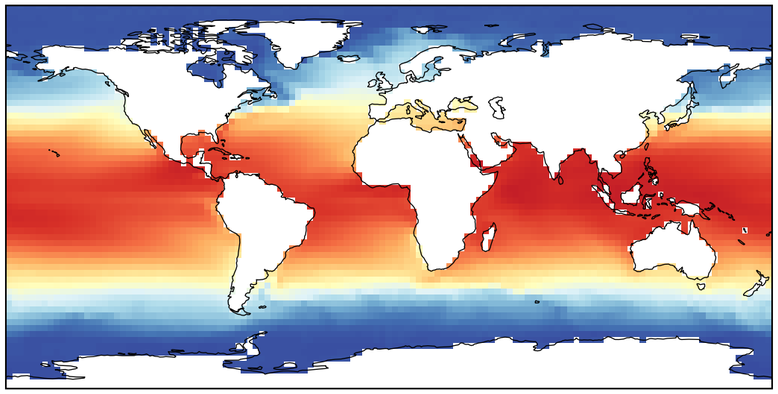
Error — Step 1 (°C)



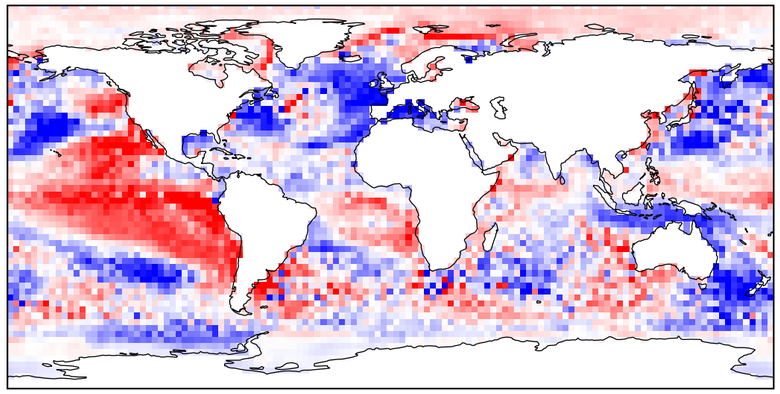
Actual — Step 46 (°C)



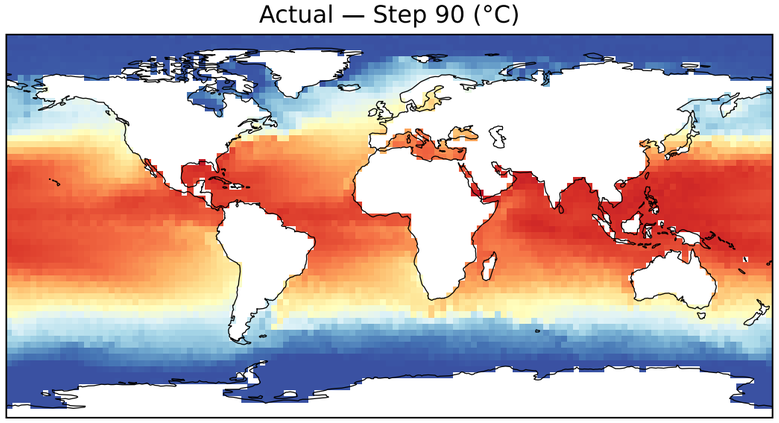
SFNO — Step 46 (°C)



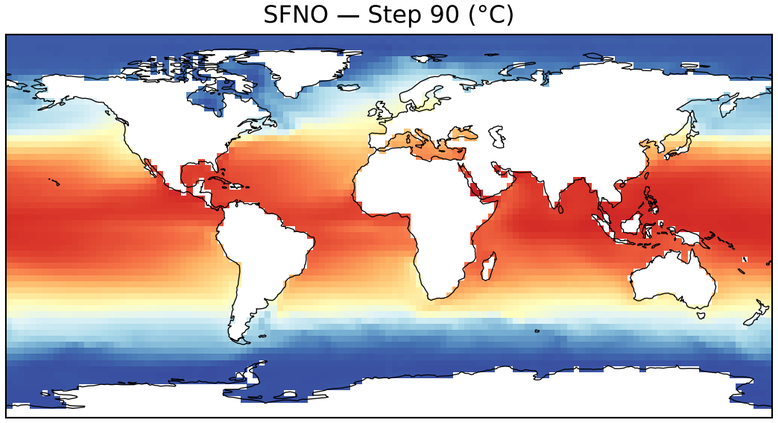
Error — Step 46 (°C)



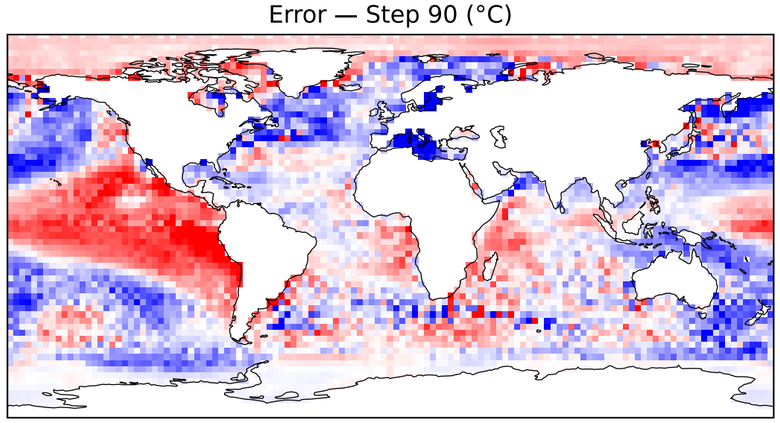
Actual — Step 90 (°C)

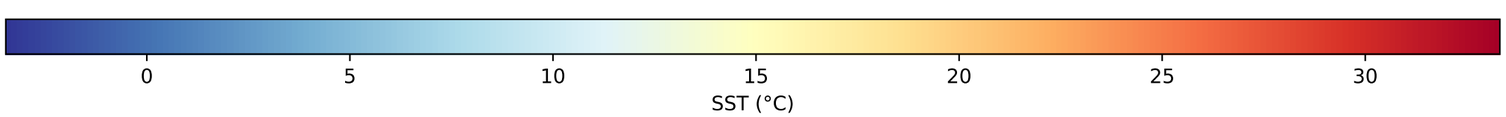


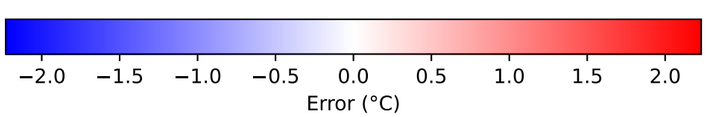
SFNO — Step 90 (°C)



Error — Step 90 (°C)

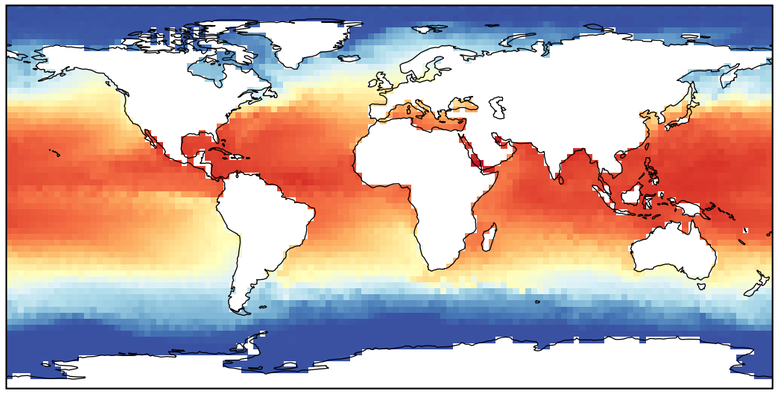




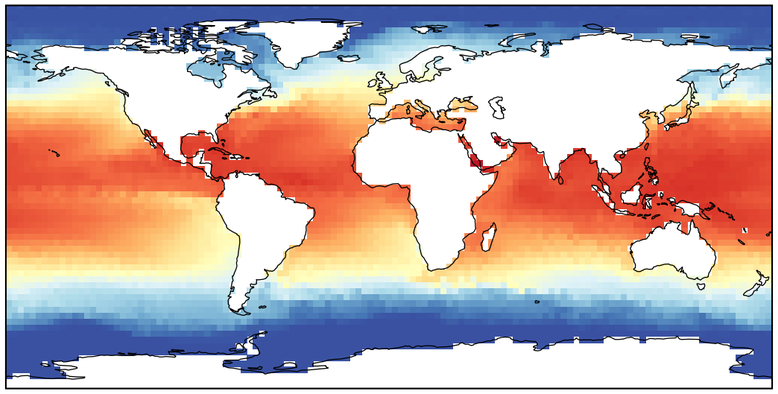


Chain init: 2022-09-28 — Test (°C)

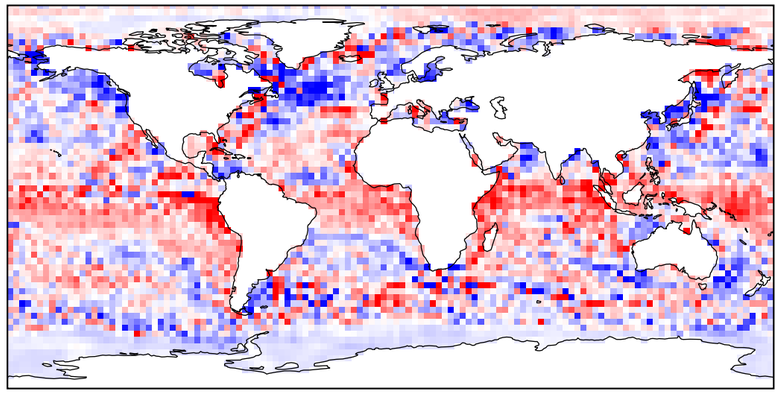
Actual — Step 1 (°C)



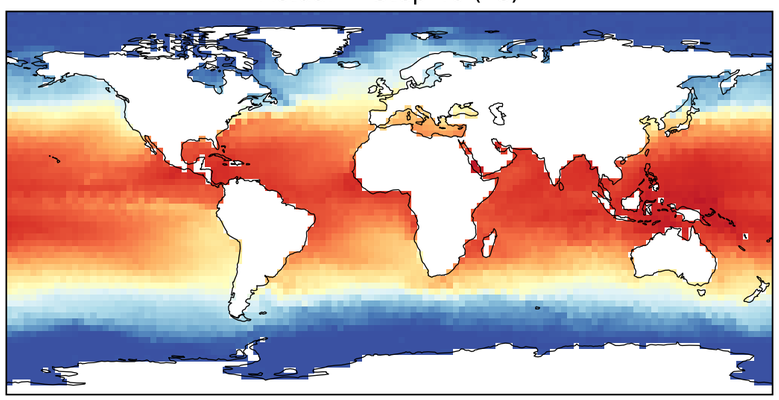
SFNO — Step 1 (°C)



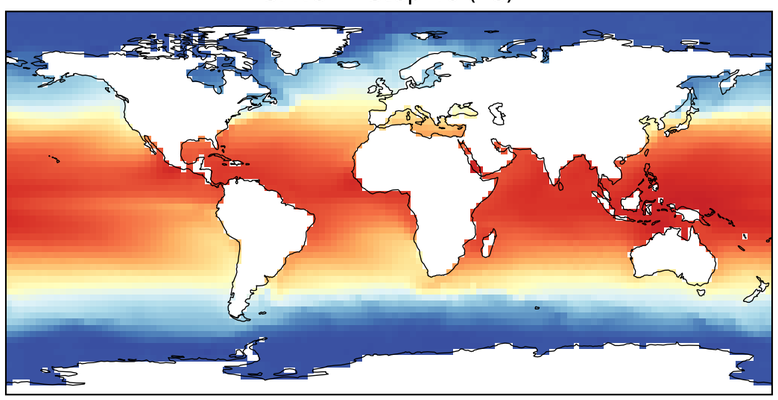
Error — Step 1 (°C)



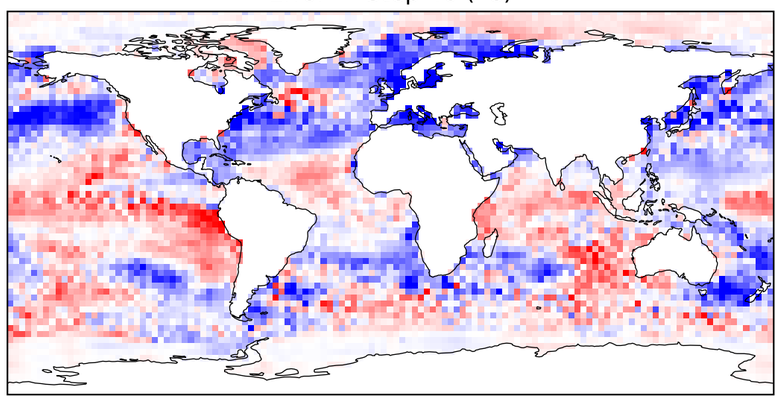
Actual — Step 46 (°C)



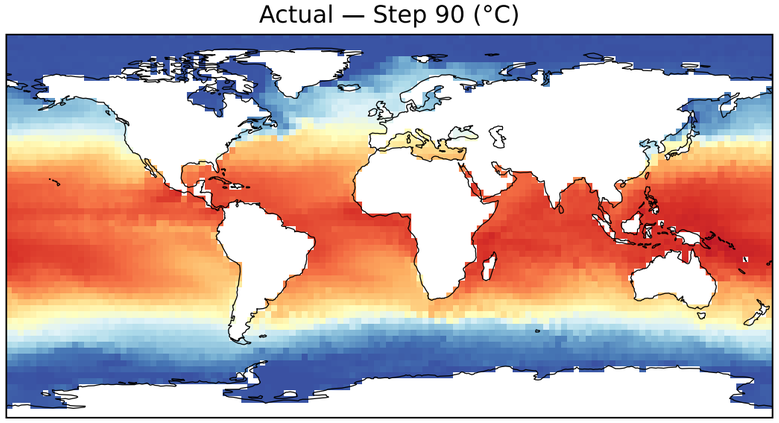
SFNO — Step 46 (°C)



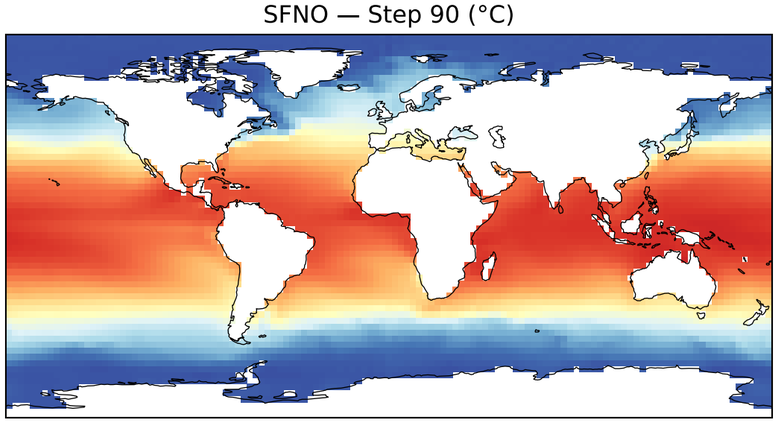
Error — Step 46 (°C)



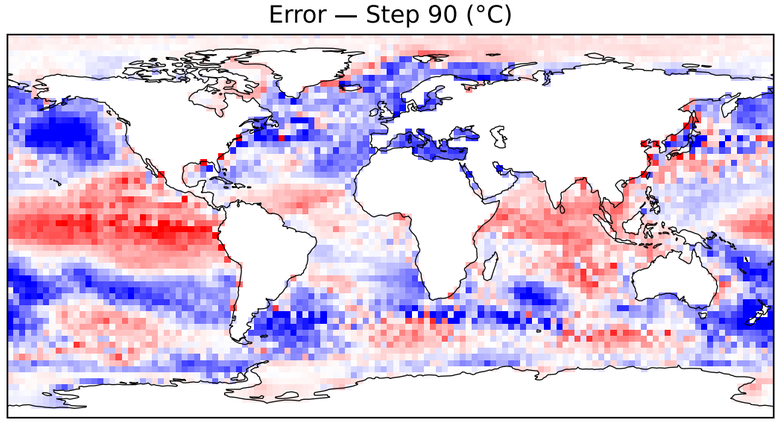
Actual — Step 90 (°C)

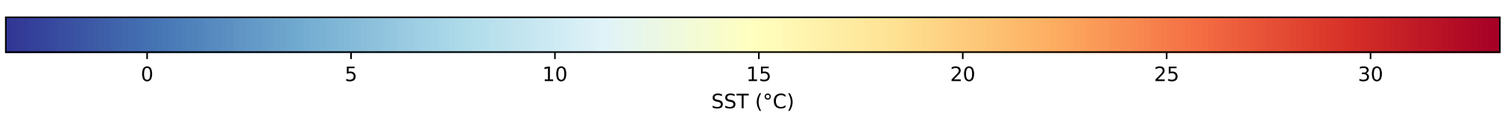


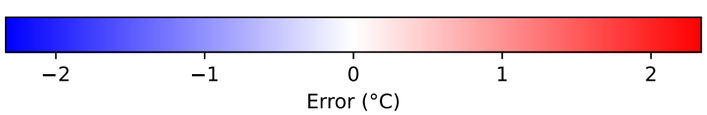
SFNO — Step 90 (°C)



Error — Step 90 (°C)

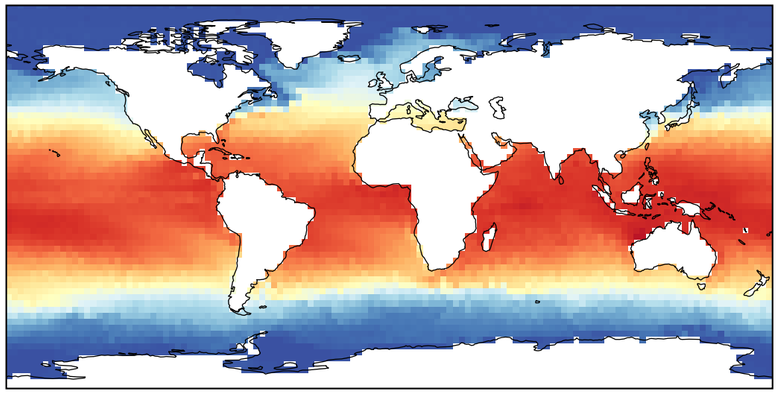




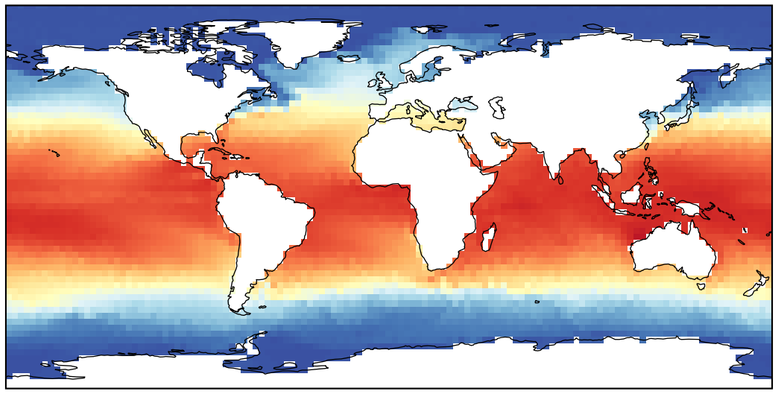


Chain init: 2023-03-27 — Test (°C)

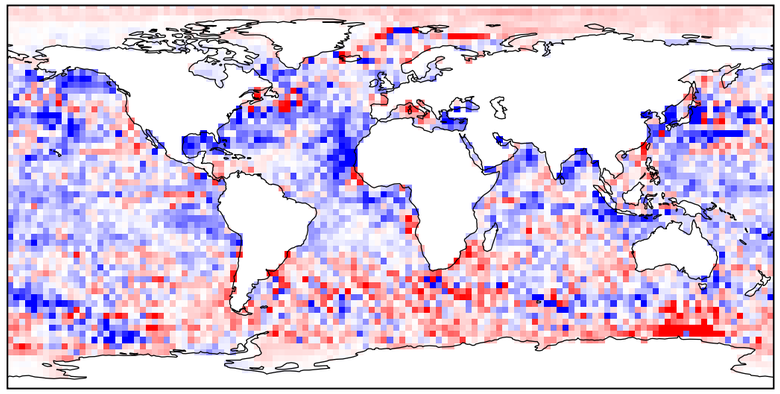
Actual — Step 1 (°C)



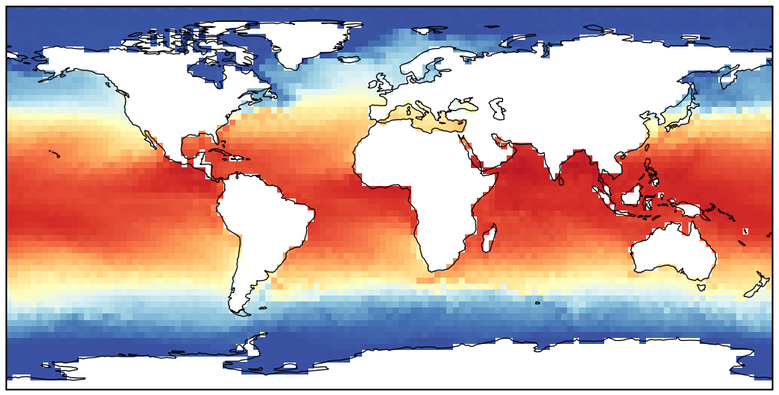
SFNO — Step 1 (°C)



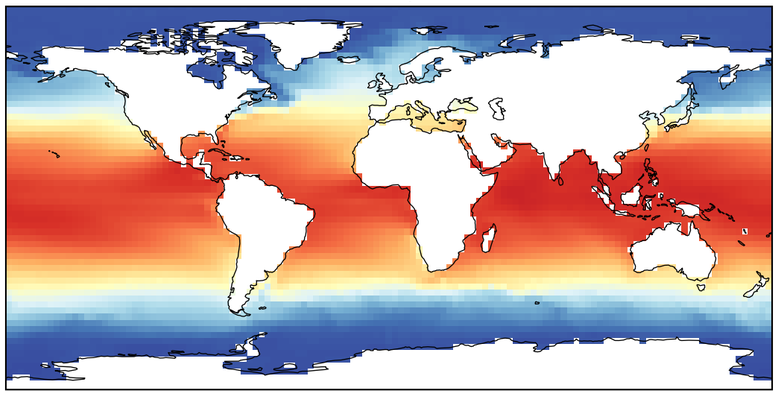
Error — Step 1 (°C)



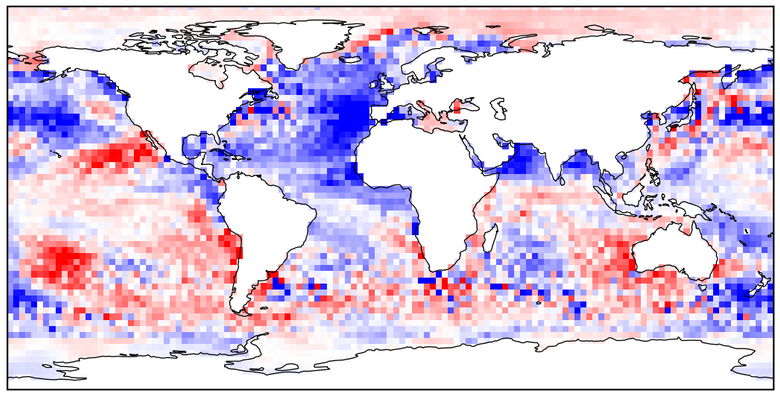
Actual — Step 46 (°C)



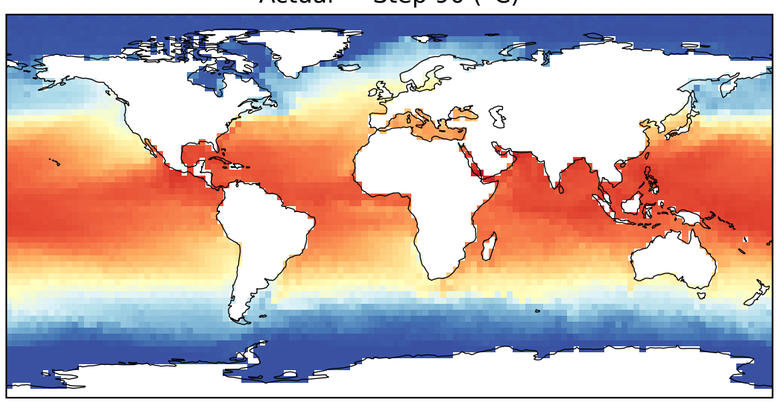
SFNO — Step 46 (°C)



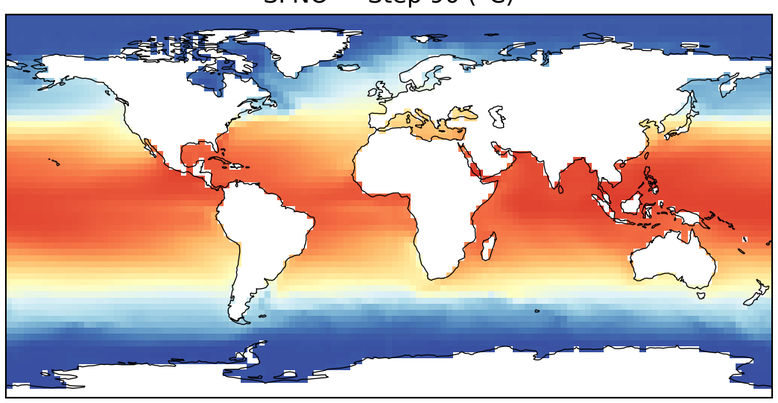
Error — Step 46 (°C)



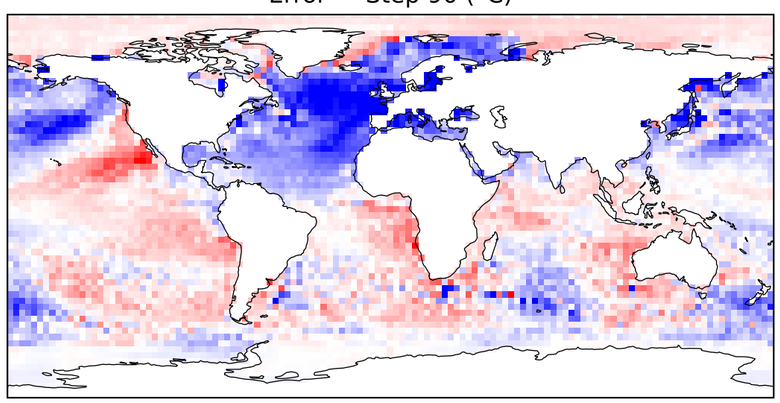
Actual — Step 90 (°C)

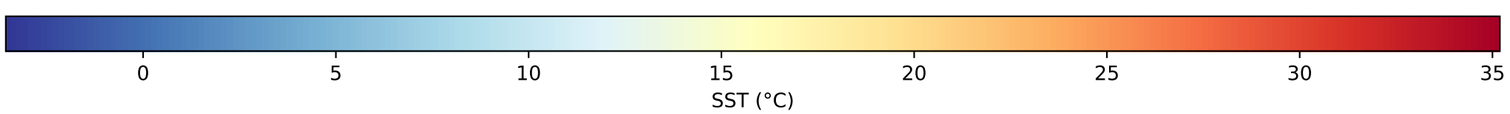


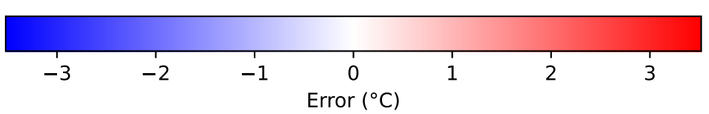
SFNO — Step 90 (°C)



Error — Step 90 (°C)

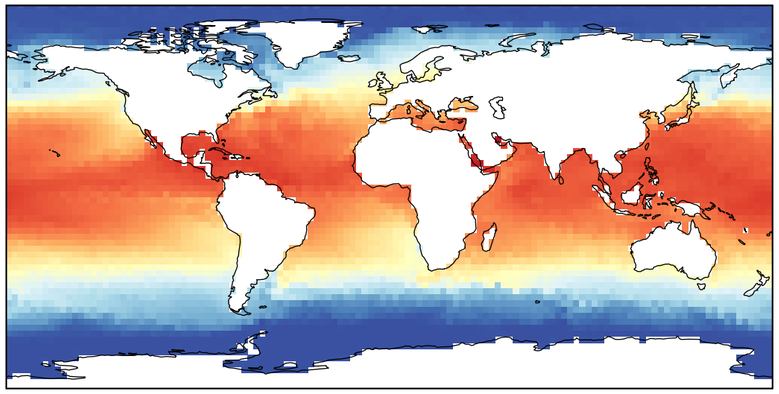




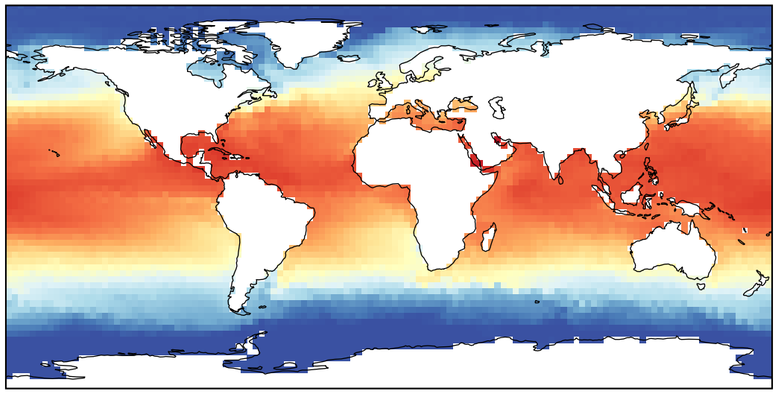


Chain init: 2023-09-23 — Test (°C)

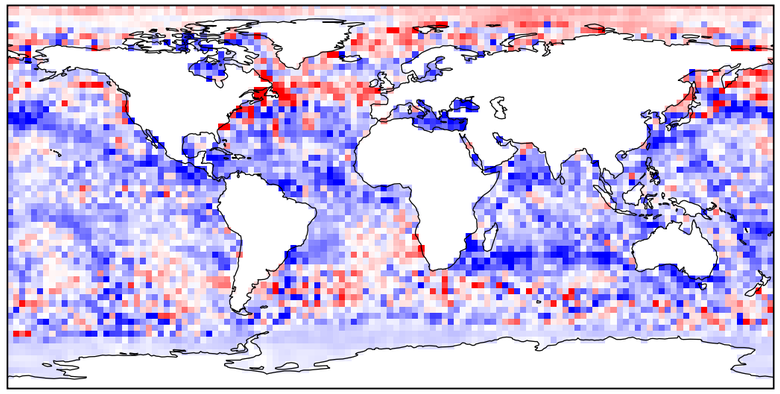
Actual — Step 1 (°C)



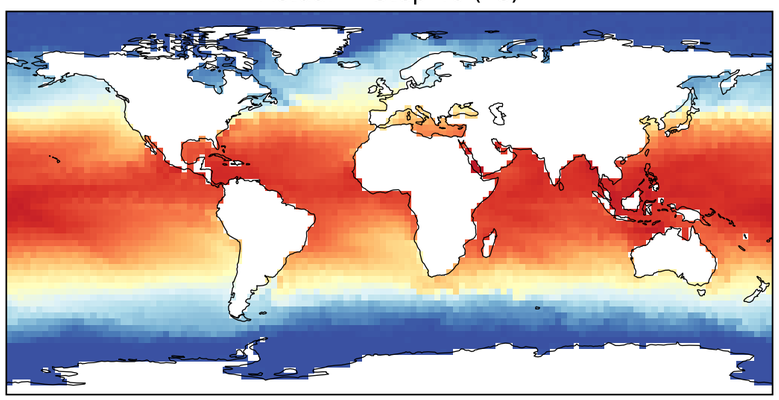
SFNO — Step 1 (°C)



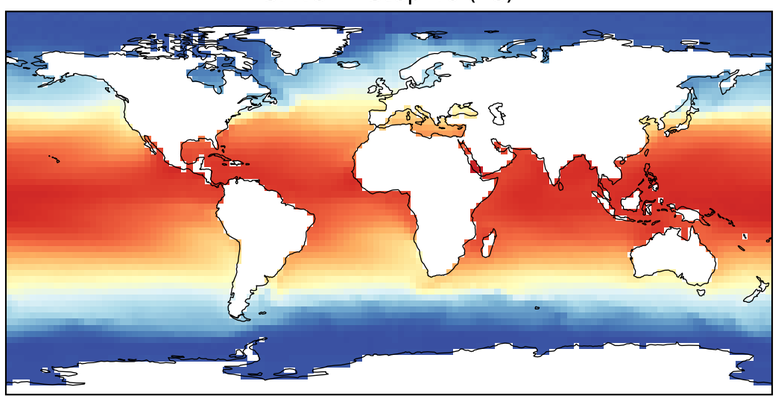
Error — Step 1 (°C)



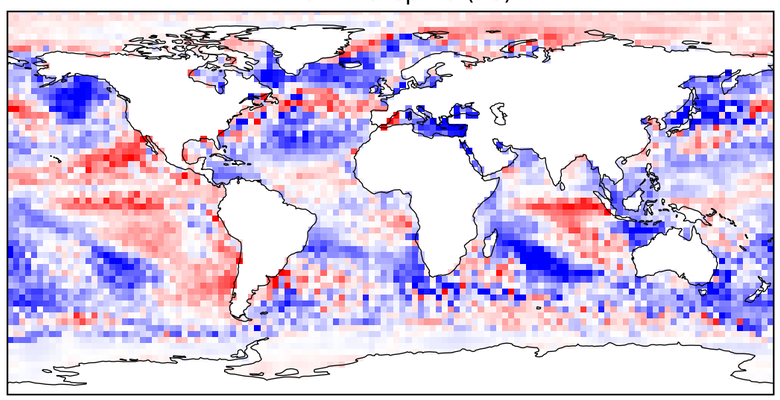
Actual — Step 46 (°C)



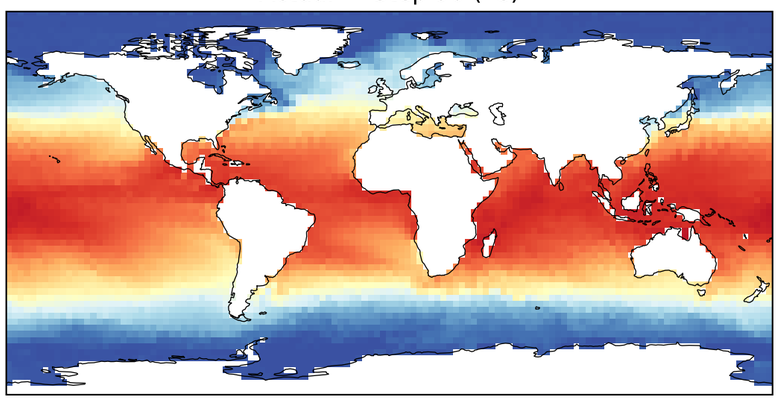
SFNO — Step 46 (°C)



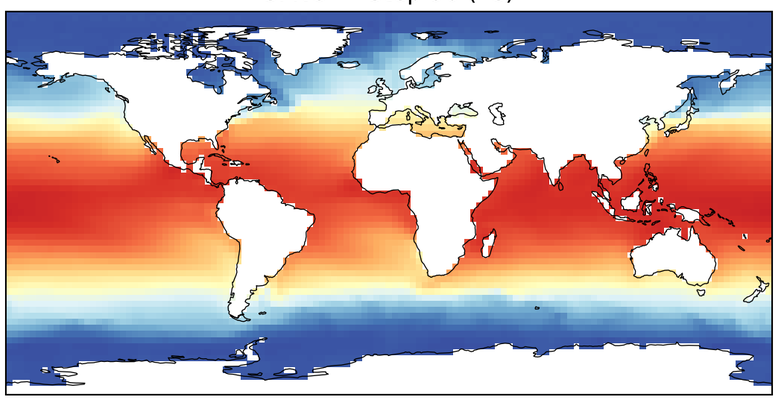
Error — Step 46 (°C)



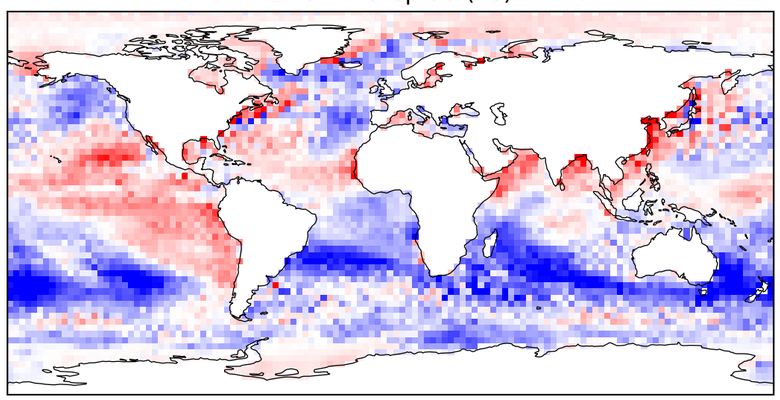
Actual — Step 90 (°C)

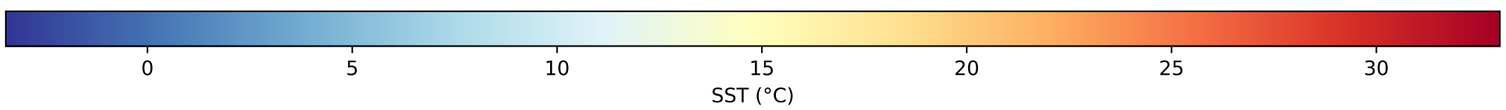


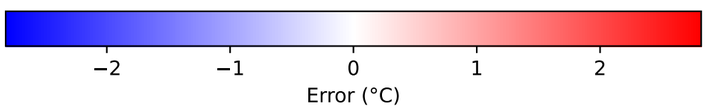
SFNO — Step 90 (°C)



Error — Step 90 (°C)

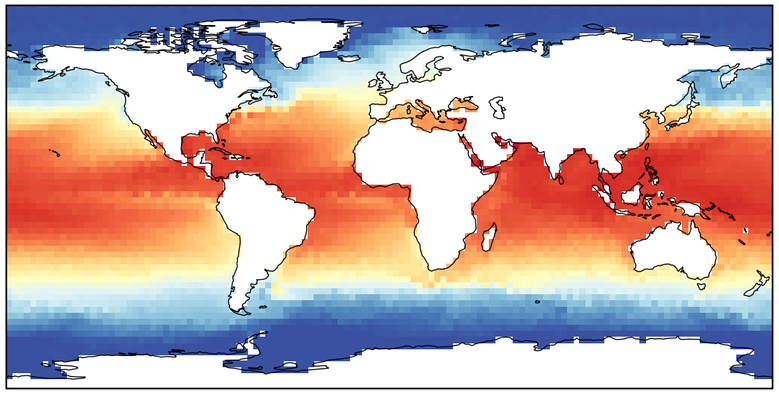




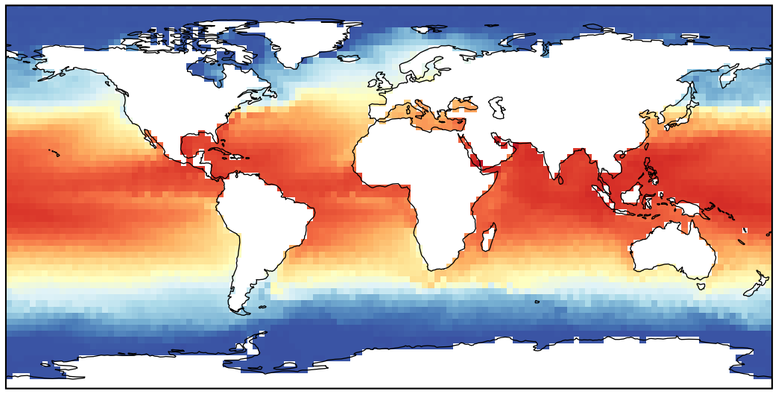


Chain init: 2024-06-19 — Test (°C)

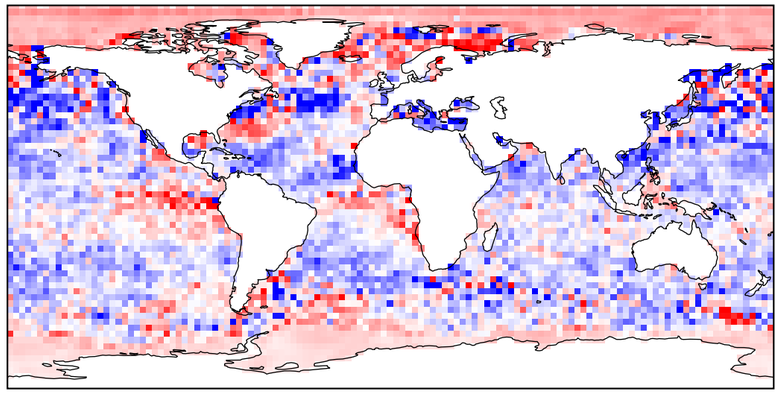
Actual — Step 1 (°C)



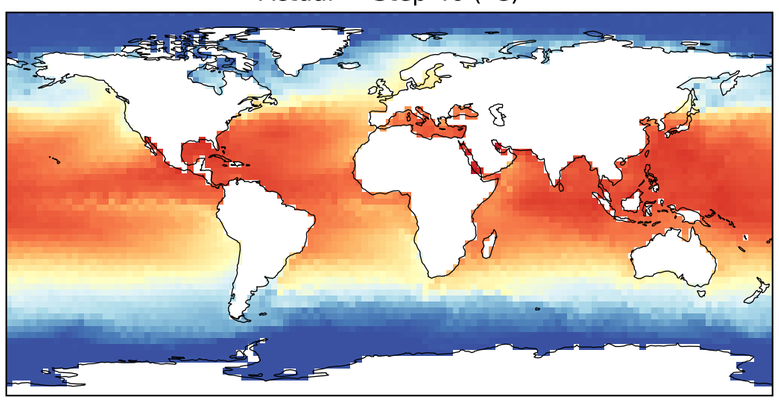
SFNO — Step 1 (°C)



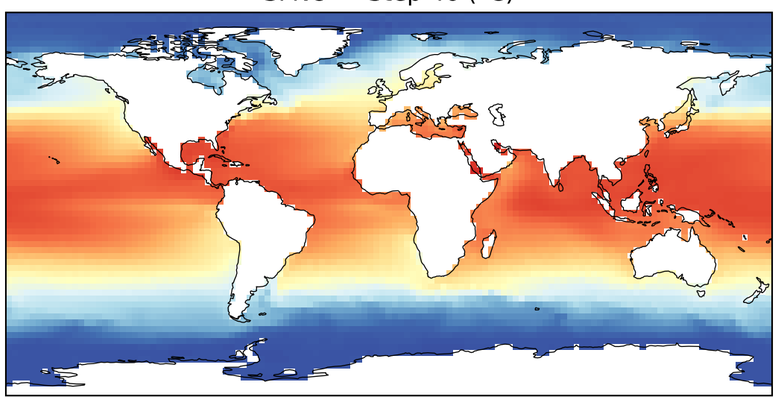
Error — Step 1 (°C)



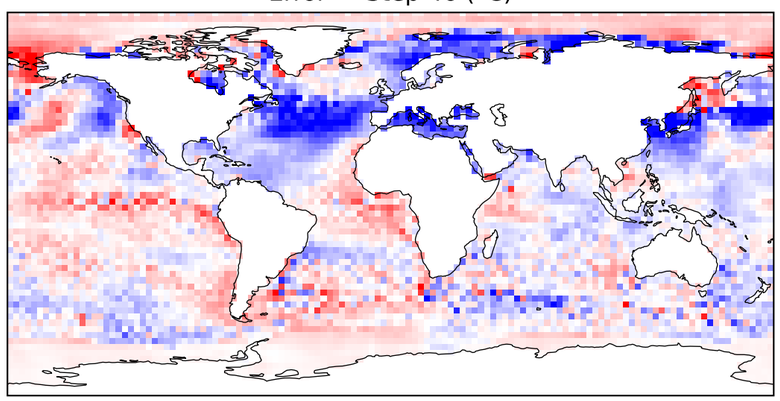
Actual — Step 46 (°C)



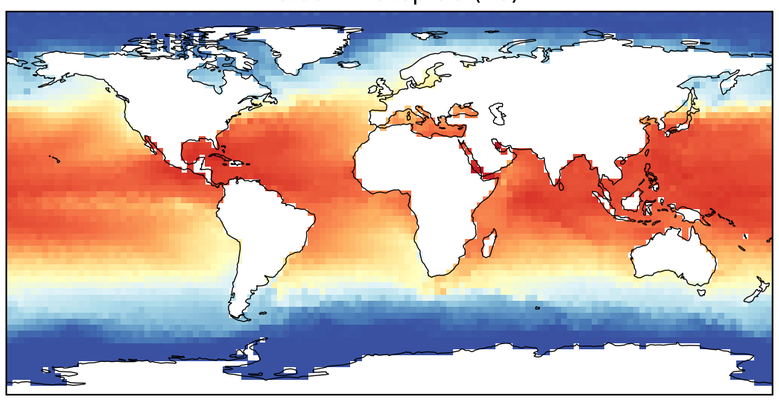
SFNO — Step 46 (°C)



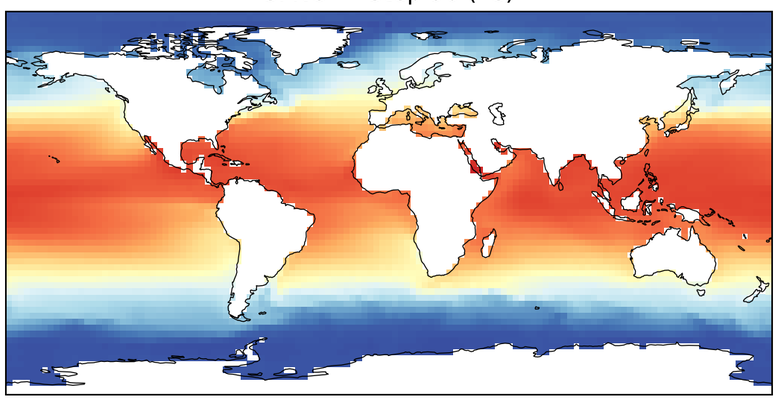
Error — Step 46 (°C)



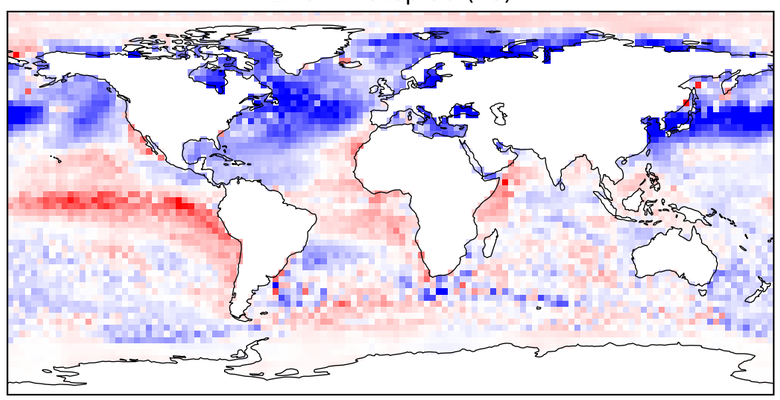
Actual — Step 90 (°C)

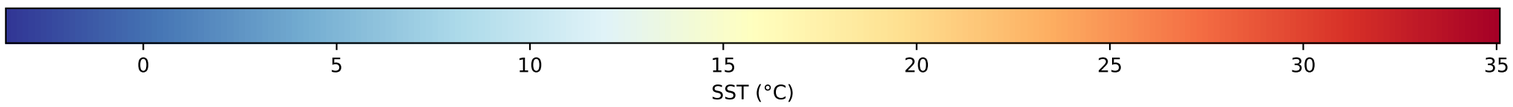


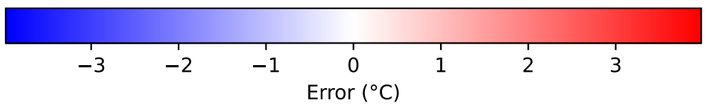
SFNO — Step 90 (°C)



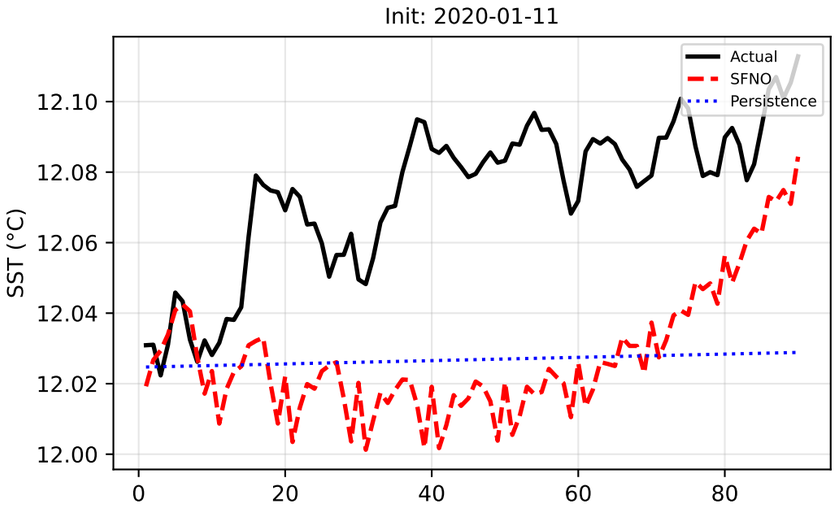
Error — Step 90 (°C)

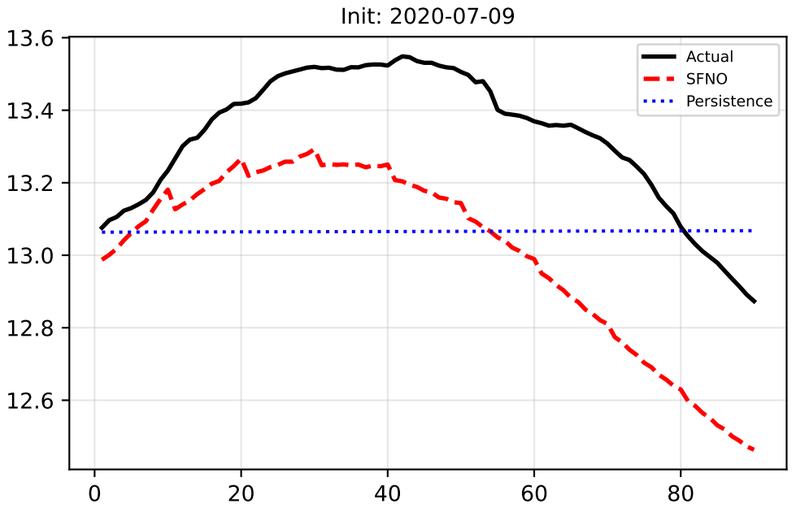


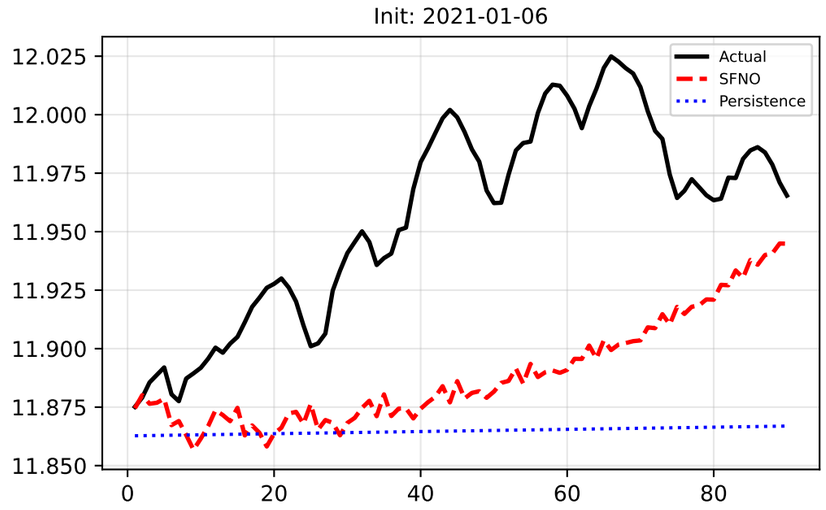


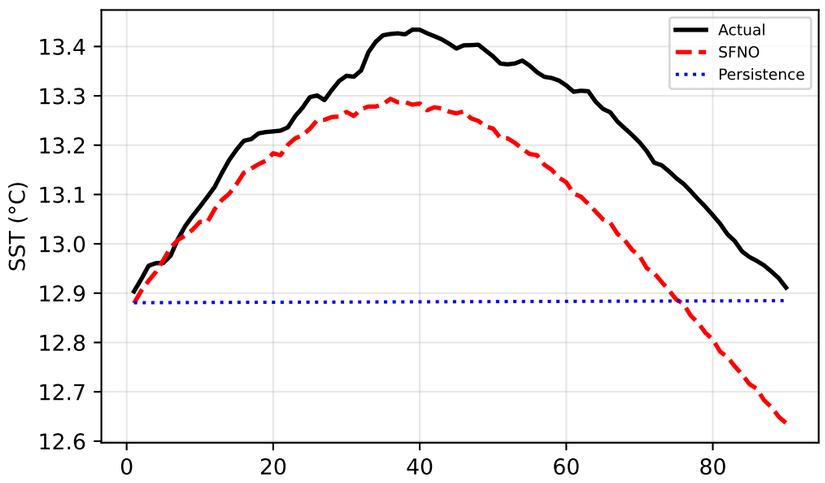


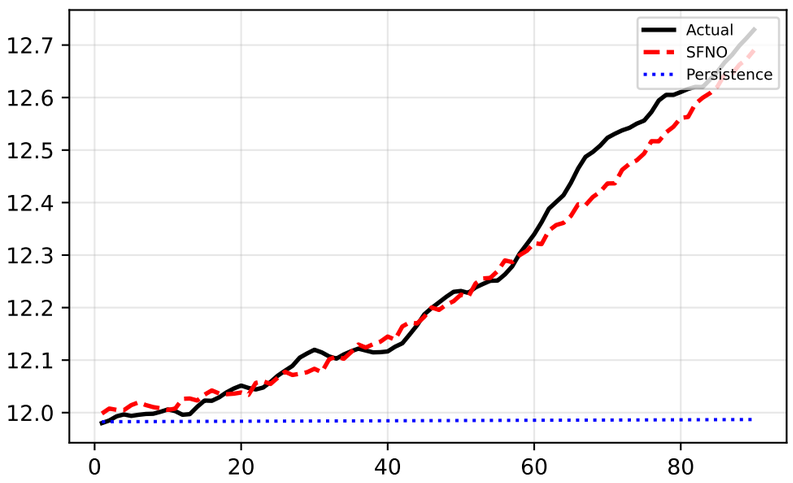
Global-Mean SST — Test (°C)

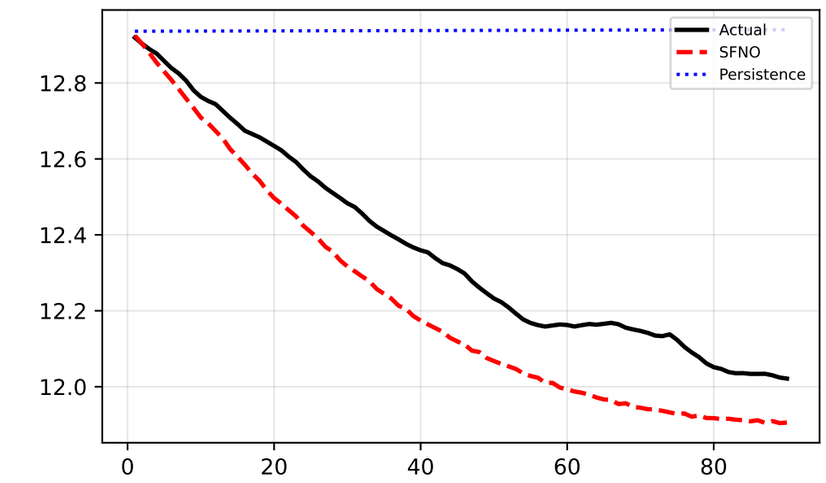


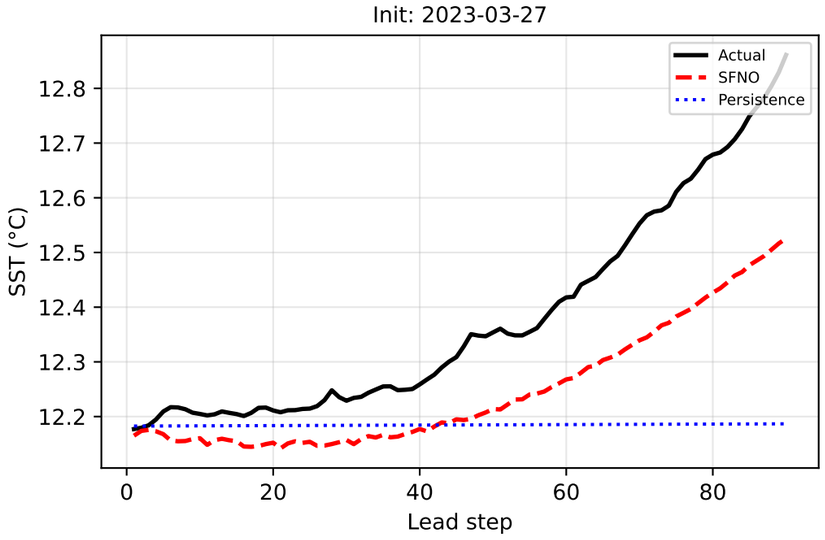


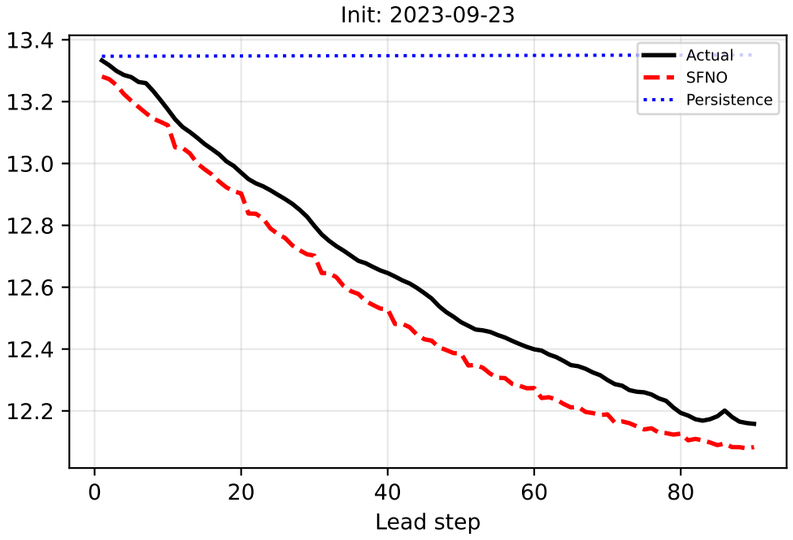


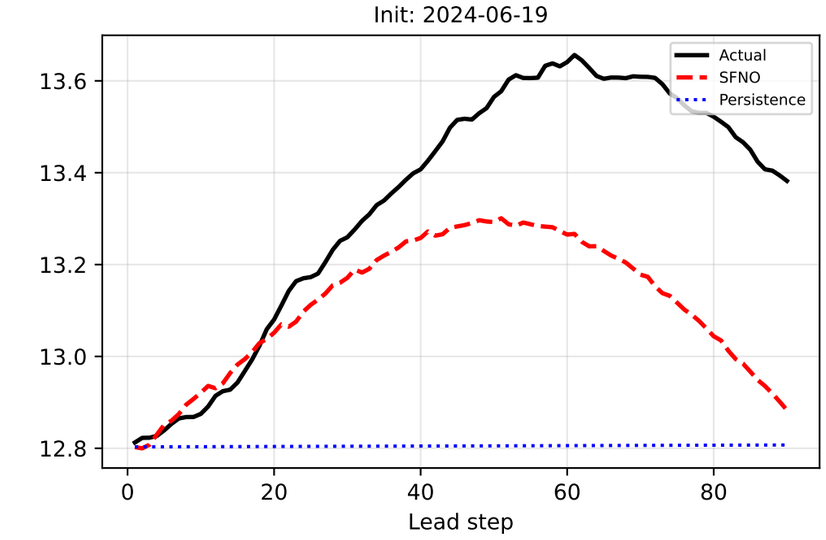






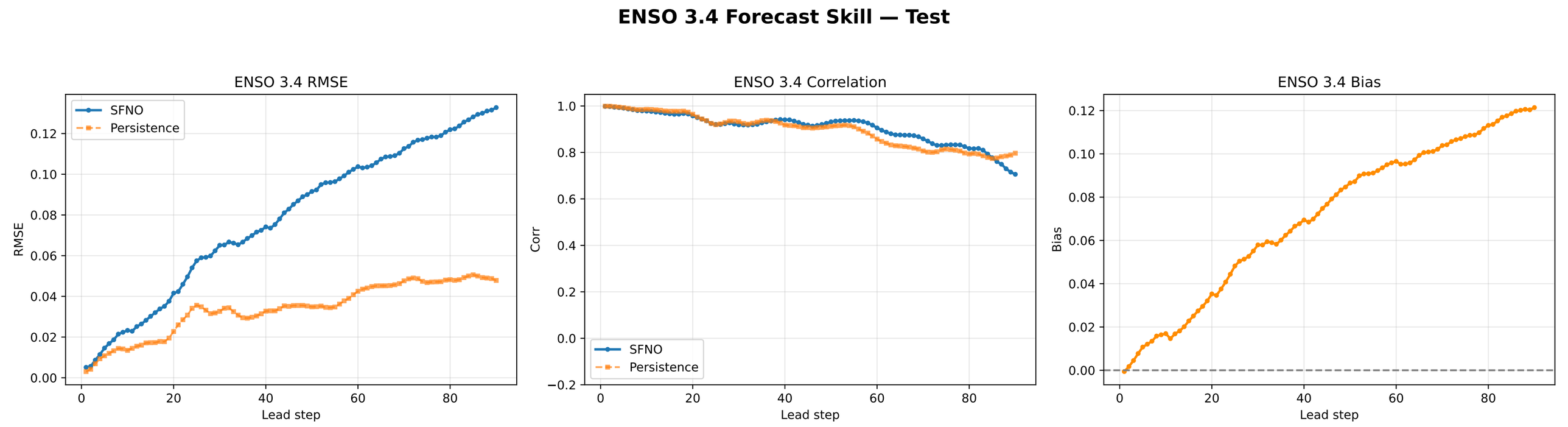




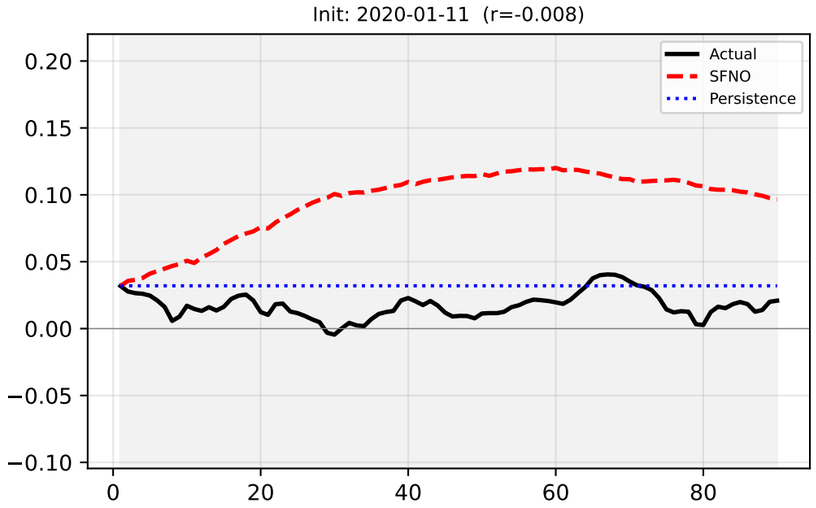


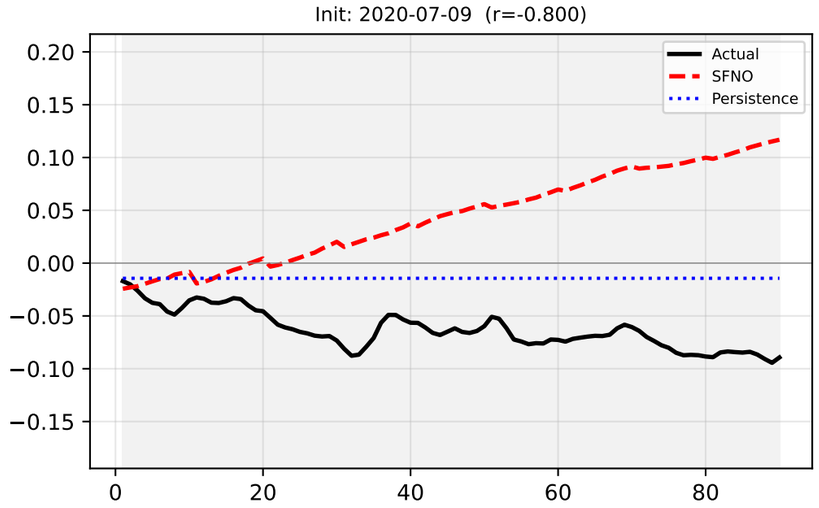
ENSO 3.4 Metrics — Test

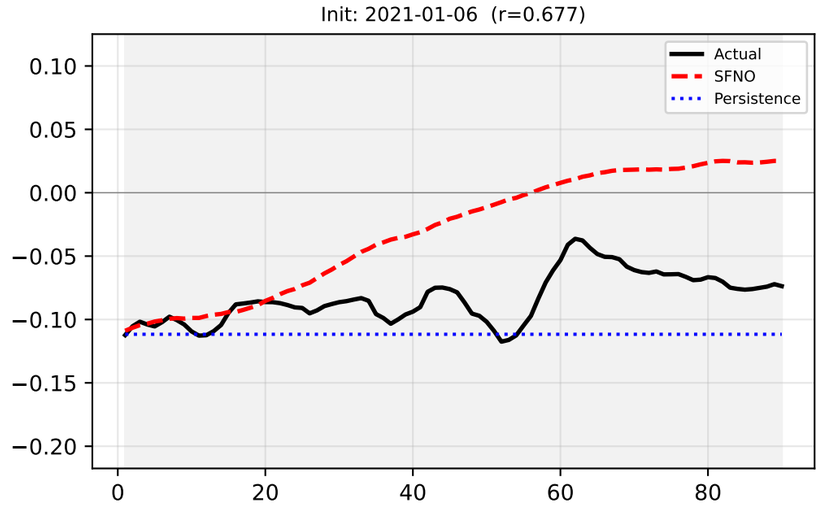
Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0051	0.0031	0.9982	0.9992	-0.0006
2	0.0056	0.0042	0.9974	0.9989	0.0017
3	0.0087	0.0069	0.9947	0.9972	0.0045
4	0.0114	0.0093	0.9931	0.9950	0.0077
5	0.0146	0.0108	0.9907	0.9927	0.0107
6	0.0168	0.0120	0.9870	0.9896	0.0121
7	0.0187	0.0133	0.9840	0.9867	0.0134
8	0.0215	0.0144	0.9804	0.9842	0.0158
9	0.0224	0.0142	0.9790	0.9843	0.0164
10	0.0233	0.0135	0.9779	0.9857	0.0169
11	0.0229	0.0145	0.9763	0.9850	0.0146
12	0.0252	0.0155	0.9737	0.9833	0.0168
13	0.0264	0.0160	0.9718	0.9817	0.0182
14	0.0282	0.0171	0.9685	0.9784	0.0201
15	0.0302	0.0172	0.9664	0.9772	0.0228
16	0.0320	0.0173	0.9642	0.9774	0.0251
17	0.0338	0.0178	0.9647	0.9770	0.0274
18	0.0351	0.0177	0.9675	0.9777	0.0295
19	0.0376	0.0195	0.9654	0.9736	0.0321
20	0.0416	0.0227	0.9576	0.9644	0.0353
21	0.0424	0.0260	0.9499	0.9526	0.0346
22	0.0459	0.0285	0.9433	0.9440	0.0376
23	0.0496	0.0308	0.9362	0.9369	0.0408
24	0.0540	0.0341	0.9246	0.9251	0.0444
25	0.0575	0.0356	0.9205	0.9197	0.0482
26	0.0590	0.0349	0.9204	0.9226	0.0504
27	0.0592	0.0332	0.9238	0.9287	0.0513
28	0.0599	0.0315	0.9272	0.9351	0.0526
29	0.0624	0.0319	0.9223	0.9350	0.0551
30	0.0651	0.0326	0.9188	0.9322	0.0579
31	0.0652	0.0342	0.9182	0.9251	0.0579
32	0.0667	0.0344	0.9177	0.9217	0.0594
33	0.0662	0.0325	0.9192	0.9261	0.0590
34	0.0654	0.0308	0.9211	0.9305	0.0582
35	0.0667	0.0295	0.9274	0.9367	0.0601
36	0.0685	0.0293	0.9318	0.9386	0.0624
37	0.0699	0.0297	0.9354	0.9370	0.0643
38	0.0716	0.0304	0.9395	0.9336	0.0666
39	0.0725	0.0314	0.9418	0.9265	0.0677
40	0.0741	0.0327	0.9407	0.9182	0.0694
41	0.0735	0.0329	0.9404	0.9161	0.0685
42	0.0753	0.0329	0.9351	0.9151	0.0699
43	0.0781	0.0339	0.9292	0.9113	0.0722
44	0.0811	0.0353	0.9208	0.9066	0.0748
45	0.0829	0.0351	0.9177	0.9069	0.0767
46	0.0852	0.0355	0.9135	0.9049	0.0792
47	0.0870	0.0355	0.9164	0.9067	0.0811
48	0.0890	0.0355	0.9203	0.9080	0.0833
49	0.0900	0.0352	0.9255	0.9099	0.0847
50	0.0915	0.0349	0.9318	0.9127	0.0866
51	0.0923	0.0349	0.9351	0.9148	0.0872
52	0.0949	0.0352	0.9362	0.9157	0.0899
53	0.0959	0.0346	0.9369	0.9176	0.0907
54	0.0960	0.0345	0.9371	0.9155	0.0908
55	0.0964	0.0348	0.9380	0.9104	0.0911
56	0.0978	0.0362	0.9358	0.9010	0.0923
57	0.0992	0.0378	0.9325	0.8912	0.0936
58	0.1010	0.0390	0.9266	0.8831	0.0950
59	0.1024	0.0408	0.9175	0.8706	0.0959
60	0.1037	0.0425	0.9056	0.8575	0.0965
61	0.1031	0.0435	0.8955	0.8473	0.0952
62	0.1035	0.0441	0.8878	0.8398	0.0953
63	0.1042	0.0448	0.8806	0.8324	0.0958
64	0.1057	0.0452	0.8757	0.8290	0.0973
65	0.1075	0.0452	0.8754	0.8279	0.0993
66	0.1085	0.0452	0.8742	0.8256	0.1006
67	0.1087	0.0453	0.8743	0.8230	0.1009
68	0.1092	0.0457	0.8725	0.8187	0.1011
69	0.1104	0.0462	0.8678	0.8146	0.1021
70	0.1126	0.0476	0.8584	0.8065	0.1039
71	0.1137	0.0486	0.8489	0.8014	0.1042
72	0.1157	0.0490	0.8380	0.8005	0.1057
73	0.1168	0.0487	0.8305	0.8030	0.1065
74	0.1170	0.0474	0.8304	0.8113	0.1071
75	0.1177	0.0468	0.8319	0.8146	0.1080
76	0.1183	0.0470	0.8332	0.8124	0.1086
77	0.1183	0.0471	0.8328	0.8097	0.1087
78	0.1190	0.0472	0.8324	0.8062	0.1098
79	0.1207	0.0480	0.8252	0.7976	0.1117
80	0.1219	0.0482	0.8169	0.7935	0.1132
81	0.1223	0.0479	0.8159	0.7956	0.1136
82	0.1238	0.0482	0.8173	0.7931	0.1152
83	0.1256	0.0492	0.8099	0.7855	0.1169
84	0.1268	0.0500	0.7932	0.7786	0.1176
85	0.1282	0.0506	0.7771	0.7738	0.1186
86	0.1294	0.0500	0.7607	0.7761	0.1197
87	0.1300	0.0492	0.7491	0.7818	0.1201
88	0.1311	0.0490	0.7298	0.7847	0.1206
89	0.1316	0.0487	0.7154	0.7888	0.1203
90	0.1328	0.0478	0.7059	0.7967	0.1214

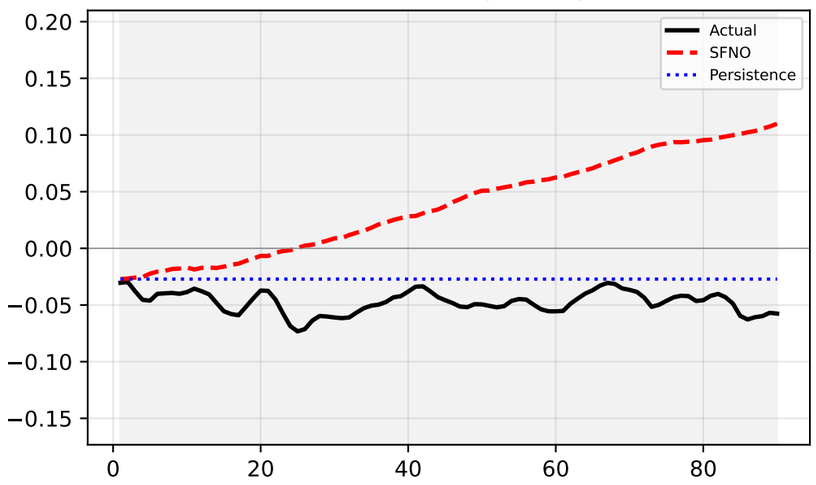

ENSO 3.4 Forecast Skill — Test
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

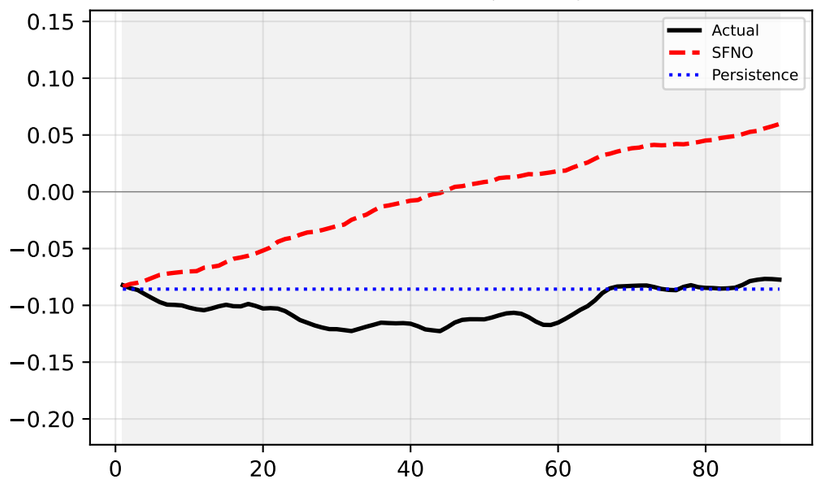
ENSO 3.4 Index — Test

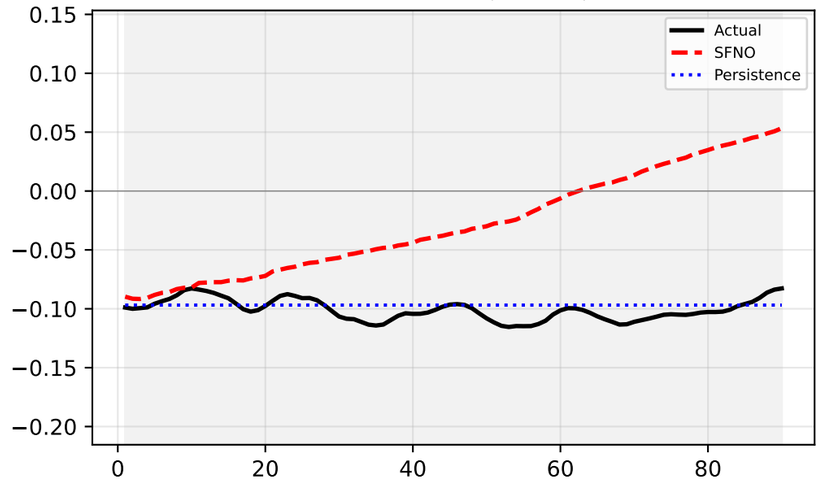


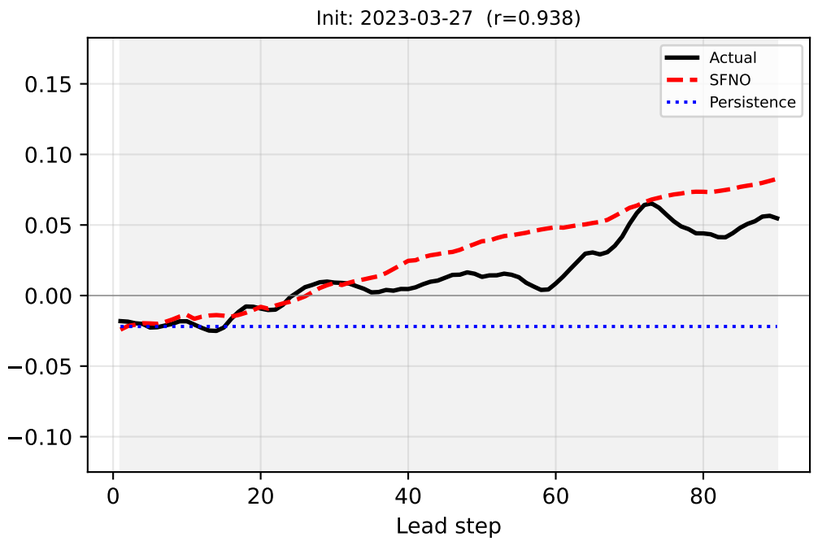


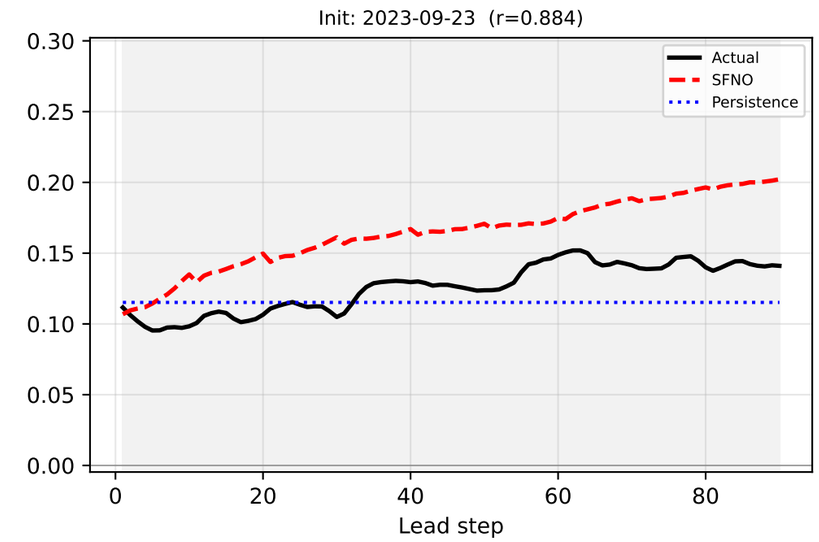


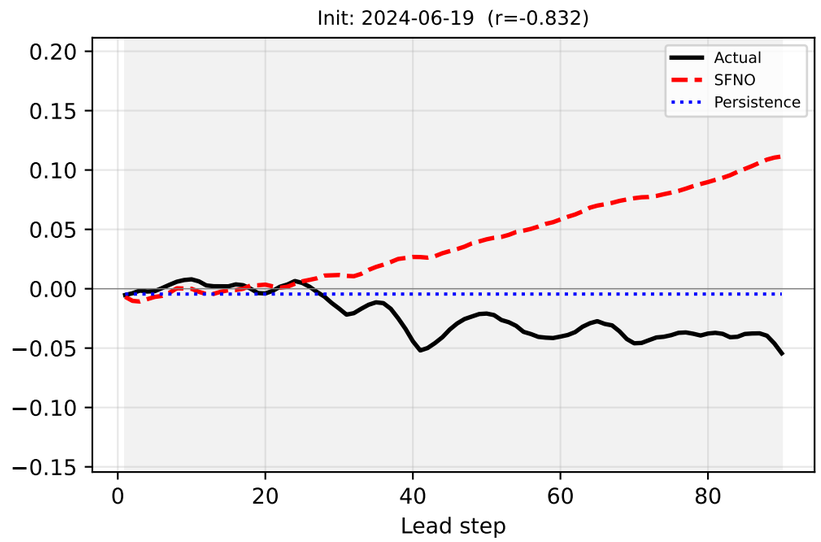












ENSO 3.4 Scatter — Test